第 1 章　蚊子能说话吗？

Can the Mosquito Speak?

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《专家的统治：埃及、技术政治与现代性》（2002）第 1 章完整译稿。以本地原版 PDF 为底本，收录原书 19—53 页正文及 305—315 页的 106 条章末注。页边数字为原书印刷页码；跨页段落合并标出范围。左栏中文、右栏英文，按段对照。上标注号可跳转至章末原注，原注前的向上箭头可返回正文。题名与部分专名为工作译名；校注与作者原注明确区分。

1942 年夏天，两股力量入侵了埃及，各自引发了一场决定性的战斗。其中只有一股力量由人组成，所以只有那一场战斗被人铭记，尽管另一场战斗造成的伤亡更大。在西北海岸，埃尔温・隆美尔的非洲军团从利比亚越过边境，向开罗进军，却在阿拉曼被英国第八集团军挡住。四个月后，英军发动反攻。经过两周的坦克战，英军击溃了德军和意军；其兵力和坦克数量均超过对方两倍。阿拉曼是盟军在第二次世界大战中取得的第一场决定性的陆战胜利，与一个月后苏军在斯大林格勒的胜利一道，似乎扭转了战争局势。由于暴力的规模之大，以及战败部队的混乱，伤亡人数无法统计，但阵亡、受伤或失踪的士兵可能有五万至七万人。[1]而且，在军队开走很久之后，这场战斗仍在夺取生命。阿拉曼标志着地雷首次被用作战争的主要武器。埃及在 20 世纪积存了 2300 万枚尚未清除的地雷，数量居世界之首，其中四分之三源于这场战役。[2]

In the summer of 1942 two forces invaded Egypt, and each provoked a decisive battle. Only one of the two was human, so only that one is remembered, although the casualties in the other battle were greater. On the northwest coast, Erwin Rommel's Afrika Corps crossed the border from Libya and was halted on its march toward Cairo by the British Eighth Army at al-Alamein. Four months later the British counterattacked. After a two-week tank battle they routed the German and Italian forces, whom they outnumbered in men and tanks by more than two to one. Al-Alamein was the Allies' first decisive land victory in World War II and, along with the Soviet victory a month later at Stalingrad, appeared to turn the tide of the war. No count of the casualties was possible given the scale of violence and the disarray among the defeated forces, but somewhere between fifty and seventy thousand soldiers may have been killed, wounded, or missing.[1] Long after the armies moved on, moreover, the battle continued to claim its victims. Al-Alamein marked the first use of land mines as a major weapon of war. It was responsible for three-quarters of the twenty-three million uncleared mines Egypt accumulated in the twentieth century, the largest number of any country in the world.[2]

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与此同时，在这个国家的另一端，又有一个入侵者从苏丹沿尼罗河谷顺流而下：冈比亚按蚊（Anopheles gambiae）。这种蚊子原产于撒哈拉以南非洲，此前在埃及却不为人知。冈比亚按蚊胃内携带着一种恶性的疟疾寄生虫——恶性疟原虫（Plasmodium falciparum）。[3]埃及原本也有其他种类的传疟蚊，但它们传播的是较为温和的疟疾，而且局限于北方的一些小片地区，当地人口已经形成了一定程度的免疫力。对于恶性疟原虫，当地却没有防御能力。1942 年 3 月，横跨埃及与苏丹南部边界的努比亚地区，其村庄首次传出了冈比亚按蚊所致疟疾暴发的报告。疫情 7 月抵达阿斯旺，8 月抵达卢克索，随后继续向北，到达南方最大的城市艾斯尤特。与阿拉曼一样，受害人数既不为人知，也无从确知。据估计，在疫情持续的三年中，可能有 75 万人感染，十万至二十万人死亡。[4]

Meanwhile, at the other end of the country another invader arrived, descending down the Nile valley from Sudan: the *Anopheles gambiae,* a mosquito native to sub-Saharan Africa but previously unknown in Egypt. The gambiae mosquito carried in its stomach the malignant form of the malaria parasite, *Plasmodium falciparum.*[3] Other species of malarial mosquitoes existed in Egypt, but these carried a more benign form of malaria and were confined to small pockets in the north, where the local population had developed a degree of immunity. There were no local defenses against *Plasmodium falciparum.* The first reports of an outbreak of gambiae malaria came in March 1942 from the villages of Nubia, the country lying across Egypt's southern border with Sudan. The epidemic reached Aswan by July and Luxor by August, then continued north to Asyut, the largest city in the south. As at al-Alamein, the number of victims was unknown and unknowable. It was estimated that three-quarters of a million people may have contracted the disease in the three years of the epidemic, and between one and two hundred thousand died.[4]

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我第一次听说 1942 年的疟疾入侵，是 1989 年听一位名叫阿姆・易卜拉欣的人讲起的。他住在卢克索附近的一个村庄，我当时正在那里生活。[5]他那时

I first heard about the 1942 malaria invasion in 1989 from a man named ʿAmm Ibrahim, who lived in a village near Luxor where I was spending time.[5] Then in

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已经八十多岁，是最熟悉村庄历史的讲述者，而疟疾流行的故事总是他叙述中最鲜明的部分。他常说，疟疾夺走了村里三分之一人的生命，活着的健康男子甚至不够抬走死者。人们用骆驼驮着死者，把他们运到墓地。

his eighties, he was the most informed narrator of the history of the village, and the story of the malaria epidemic was always the most vivid part of his narrative. It killed one-third of the village, he used to say, and there were not enough healthy men left alive even to carry the dead. People were hauled to their graves on the back of a camel.

战争与疫情还同威胁这个国家的第三种力量相互作用，即战时严重的粮食短缺。这种短缺本身有着复杂的成因。1933 年，世纪之交建在阿斯旺、横跨尼罗河的大坝被加高，至此，一个始于 19 世纪中叶、由大坝、拦河闸与运河组成的网络建成，使全国大部分农田改用全年灌溉。[6]如今，尼罗河谷只有五分之一的土地仍由河流一年一度的洪水灌溉；过去，洪水沉积一层淤泥与养分，使土壤保持肥力。其余五分之四的土地则需要化学肥料。[7]到 20 世纪 30 年代末，埃及农民每年使用 60 万吨肥料——主要是新型人工合成硝酸盐——单位耕地面积的使用量居世界之首。[8]化学品制造商组成的国际卡特尔，将埃及市场的 80% 划给了一个由德国法本工业集团（I. G. Farben）领衔的联合体；该集团的一家公司发明了合成硝酸铵的工艺。[9]战争爆发切断了这些供应。

〔校注：本章引用的统计和医学判断均属于 2002 年底本的历史叙述。化学名称中的明显疑点保留原文，并在下文相应位置说明。〕

The war and the epidemic interacted with a third threat to the country, a severe wartime shortage of food. The shortage had complex causes of its own. In 1933 the dam across the river Nile at Aswan, built at the turn of the century, had been increased in height, completing a network of dams, barrages, and canals begun in the mid-nineteenth century that converted most of the country's agricultural land to year-round irrigation.[6] Only one-fifth of the Nile valley was now irrigated by the river's annual flood, which in the past had fertilized the soil by depositing a layer of silt and nutrients. The other four-fifths required chemical fertilizers.[7] By the end of the 1930s Egyptian farmers were using 600,000 tons of fertilizer a year—mostly the new artificial nitrates—at the highest rate per cultivated area in the world.[8] An international cartel among chemical manufacturers had assigned 80 percent of the Egyptian market to a consortium led by the German business group I. G. Farben, one of whose companies had invented the process for synthesizing ammonium nitrate.[9] These supplies were cut off by the outbreak of war.

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肥料匮乏使小麦及其他作物的产量下降了多达四分之一。政府实行粮食配给，以供应城市和英国军队，又实行肥料配给与种植面积管制，迫使土地所有者把全国一半的棉田改种粮食。[10]然而，在最南部，主要的商品作物是甘蔗而非棉花，甘蔗并未受到管制。战争期间，甘蔗种植园主把种植面积扩大了多达 30%，加剧了冈比亚按蚊疟疾肆虐地区主食的短缺，也增加了蚊虫的孳生地。[11]疫情第二年的伤亡要高得多，因为许多家庭上一年病得无法收获粮食，此后又因饥荒和营养不良而衰弱。糖业庄园工人的伤亡率最高。[12]在卢克索以南几英里处一个规模最大的甘蔗种植园中，经理估计有 80% 至 90% 的人感染了疟疾；附近阿尔曼特镇的医生则报告，每天有八十至九十人死亡。[13]

The lack of fertilizer caused the yield of wheat and other crops to drop by as much as a quarter. The government introduced food rationing to supply the cities and the British troops, and introduced fertilizer rationing and acreage controls to force landowners to switch half the country's cotton fields to the cultivation of food.[10] In the far south, however, the main commercial crop was sugarcane rather than cotton, for which no controls were introduced. The owners of the cane plantations extended the crop's acreage by as much as 30 percent during the war, exacerbating the shortage of staple foods in the region hit by gambiae malaria (and increasing the breeding grounds for mosquitoes).[11] In the second year of the malaria epidemic casualties were much higher, since many households had been too sick to harvest the previous year's food crop and were weakened by famine and malnutrition. The highest casualty rates were recorded among the workers on the sugar estates.[12] At one of the largest cane plantations, a few miles south of Luxor, the manager estimated that malaria affected 80 to 90 percent of the people, and the doctor in the nearby town of Armant reported eighty to ninety deaths a day.[13]

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共同造成 1942—1944 年灾难的各种因素，代表着 20 世纪一些最强有力的转变。首先是河流筑坝。1898—1902 年阿斯旺最初那座拦河坝的兴建，帮助开启了全球工程规模空前扩大的时代。截断大河水流的计划，将成为这个世纪规模最大的建设项目。大坝改变资源在空间与时间上、在整个社区与生态系统之间的分配，其范围和方式独一无二。它们许下的不仅是农业发展或技术进步的承诺。对于许多后殖民政府而言，这种重新安排自然与社会环境的

The elements combining to cause the disaster of 1942–44 represented some of the most powerful transformations of the twentieth century. First, there was the damming of the river. The building of the original barrage at Aswan in 1898–1902 helped inaugurate around the world an era of engineering on a new scale. Schemes to block the flow of large rivers were to become the century's largest construction projects. Dams were unique in the scope and manner in which they altered the distri-

能力，成为展示现代国家作为技术—经济力量之实力的手段。[14]其次是合成化学品。人工硝酸盐的制造带来的转变，比筑坝还要巨大。从 1925 年那个基本没有合成物的世界出发，以硝酸盐为首的新化学品产量以惊人的速度增长。在美国，产量每十年就增长十倍。到 20 世纪 80 年代，投入生产的合成化学品达四百万种，其中六万种被普遍使用。这种转变在细胞和有机体层面的影响，足以与大坝在国家层面的影响相匹敌。[15]第三是疟疾。它借助灌溉计划、人口流动和农业变化，成为世界上最致命的传染病。恶性疟原虫引起的病例仅占临床疟疾病例的 30%，却造成了高达 90% 的死亡。其传播如此广泛，以至于它每年夺走多少生命，连精确到百万位都无法达成共识。[16]最后是战争。阿拉曼被铭记为第一场大规模机械化冲突：德国坦克以新型战术与反坦克炮、飞机协同，迎战体型更大的格兰特和谢尔曼坦克。然而，战线过于狭窄，德意两军的机械装备又缺乏燃料和弹药，因此这场为期两周的战斗以近距离交战展开，仿佛第一次世界大战中的一场战斗。它集中体现了人与机器之间一种新出现的致命互动。

bution of resources across space and time, among entire communities and ecosystems. They offered more than just a promise of agricultural development or technical progress. For many postcolonial governments, this ability to rearrange the natural and social environment became a means to demonstrate the strength of the modern state as a techno-economic power.[14] Second, there were the synthetic chemicals. The manufacture of artificial nitrates introduced a transformation even greater than the building of dams. From the largely synthetic-free world of 1925, the production of new chemicals, led by nitrates, grew at a phenomenal rate. In the United States output increased tenfold in each decade. By the 1980s there were four million synthetic chemicals in production, sixty thousand of which were in common use. This transformation had an impact at the level of the cell and the organism to rival that of dams at the national level.[15] Third, there was malaria, which took advantage of irrigation schemes, population movements, and changes in agriculture to become the world's most deadly infectious disease. *Plasmodium falciparum* represented only 30 percent of clinical malaria cases but was responsible for up to 90 percent of the deaths. It was so widespread that no one could agree even to the nearest million how many lives it was taking each year.[16] Finally, there was the war. Al-Alamein was remembered as the first great mechanized conflict, in which the German panzers, used in new kinds of tactical combination with antitank guns and aircraft, engaged the larger Grant and Sherman tanks. Yet the battle front was so narrow, and the German and Italian machinery so short of fuel and ammunition, that the two-week battle was fought at close quarters, like a battle of World War I. It epitomized a new and lethal interaction of man and machine.

大坝、血液传播的寄生虫、合成化学品、机械化战争与人为饥荒，同时出现并相互作用。环境转变带来疾病、工业化学受军事需要塑造，或战争伴随饥荒，这些现象并不令人惊讶。然而，它们的相互作用提出了一个难题：坦克、寄生虫与合成硝酸盐究竟如何彼此影响？怎样的解释才能把它们联系起来？

Dams, blood-borne parasites, synthetic chemicals, mechanized war, and man-made famine coincided and interacted. It is not surprising to find disease brought by environmental transformation, industrial chemistry shaped by military needs, or war accompanied by famine. Nevertheless, their interaction presents a challenge. How exactly did tanks and parasites and synthetic nitrates affect one another? What kind of explanation can bring them together?

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战争与疫情在若干层面相互作用。战事爆发时，英国重新在埃及实施戒严，此前埃及在摆脱始于 1882 年的殖民占领后，已经享有将近二十年的部分独立。当局审查有关疟疾疫情的报道，希望把疫情限制在南方。英国人已在准备一旦隆美尔突破阿拉曼便撤出开罗，不愿从北方调拨人员和资源去对付来自南方的入侵者。这帮助了冈比亚按蚊的推进。英方还面临奎宁短缺，这是当时唯一的抗疟治疗药物；就在努比亚报告冈比亚按蚊疟疾的同一个月，日本占领了爪哇，切断了荷兰金鸡纳树种植园的供货，而欧洲的奎宁正是从这些树木中取得的。[17]于是，埃及卫生部只能自行发动抗疟运动。灭蚊队针对的是疾病媒介——蚊子——而非寄生虫本身：他们在积水池中撒布 Malariol，即掺有铺展剂的柴油。油在水面形成一层薄膜，阻止蚊幼虫孵化。不过，Malariol 经常不翼而飞，因为灌溉水泵可用其中的柴油作为燃

The war and the epidemic interacted on several levels. At the outbreak of hostilities Britain had reimposed martial law on Egypt, after the country had enjoyed almost two decades of partial independence from the colonial occupation established in 1882. The authorities censored reporting of the malaria epidemic, hoping to contain it in the south. Already preparing to evacuate Cairo in case Rommel broke through at al-Alamein, the British were unwilling to divert men and resources from the north to meet the invader from the south. This helped the gambiae mosquito advance. The British also faced a shortage of quinine, the only treatment against the infection, for in the same month that gambiae malaria was reported in Nubia the Japanese had occupied Java, cutting off the Dutch cinchona plantations whose trees supplied the drug to Europe.[17] So the Egyptian Ministry

料，而战争使燃料难以获得。灭蚊队后来改用巴黎绿，即砷粉与醋酸铜的混合物，原先用作绘画颜料。事实证明，它是一种更可靠的杀幼虫剂，或者至少更不容易被挪作他用。

〔校注：底本写的是阻止幼虫‘孵化’（larvae from hatching）。按蚊生命周期为卵、幼虫、蛹、成虫，严格说孵化的是卵；此处保留作者措辞。〕

of Health was left to launch its own antimalaria campaign. Its eradication teams attacked the disease vector—the mosquito—rather than the parasite itself, spreading Malariol, diesel oil mixed with a spreading agent, on pools of standing water. The oil formed a film on the water surface, which prevented the mosquito larvae from hatching. Malariol tended to go missing, however, since irrigation pumps could use the diesel oil as fuel, which the war had made it difficult to obtain. The eradication teams later replaced it with Paris green, a mixture of arsenic powder and copper acetate used originally as a painters' pigment, which proved a more reliable larvicide, or at least one less liable to be taken over for other purposes.

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甚至可能是战争本身带来了疫情。按蚊的活动范围只有两英里，因此，要抵达埃及，它自己也需要传播媒介。一种看法认为，它必定是乘飞机而来；蚊子搭乘飞机并非罕见。德军的空袭与潜艇袭击使地中海不再安全，因此英国人开辟了一条经西非和苏丹飞往开罗的补给航线。但战事也可能使蚊子得以乘船抵达埃及。战争增加了与苏丹之间的河运，阿斯旺大坝的修建与加高又沿途为昆虫创造了新的孳生地。一旦进入埃及，蚊子便乘船、火车和汽车继续北上。为阻止其移动，人们用一种新技术——除虫菊喷雾——处理这些交通工具。这项技术是在此前十年中，为应对南非东海岸纳塔尔省一次严重的疟疾暴发而开发的；该省同上埃及一样，是一个甘蔗产区。除虫菊粉由除虫菊的干花制成，有时也被焚烧以熏杀屋内害虫；人们将其与绿皂及甘油混合，再用脚踏固定的手压泵从喷嘴喷出，形成细密的毒雾，杀死成蚊。[18]

The war may even have brought the epidemic. The anopheles mosquito has a range of only two miles, so to reach Egypt it needed vectors of its own. One view was that it must have arrived by airplane, a mode of travel not unusual for mosquitoes. German air and submarine attacks had made the Mediterranean unsafe, so the British were flying a new supply route to Cairo via West Africa and Sudan. But the hostilities may also have enabled the mosquito to reach Egypt by boat. The war had increased river traffic with Sudan, and the building and raising of the Aswan Dam had created new breeding places for the insect along the route. Once in Egypt, the mosquito continued to travel north, moving by boat, train, and motorcar. To prevent its movement, these vehicles were treated by a new technique, the pyrethrum spray, developed over the previous decade to combat a major outbreak of malaria in Natal Province on the east coast of South Africa—like Upper Egypt, a region producing sugarcane. Pyrethrum powder, made of the dried flowers of the pyrethrum variety of chrysanthemum and sometimes burned to fumigate houses against insects, was mixed with green soap and glycerine and then forced through the spray nozzle of a stirrup pump, making a fine poisonous mist that killed the adult mosquitoes.[18]

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疾病往往随人群流动方式的改变而移动，现代战争又迫使大批人口在既有贸易与迁移网络之外另寻路径。然而，这种昆虫在利用新型交通工具与运输路线之后，还需要通过进入新的地域和人群来立足。战争与交通的模式必须同其他变化交织在一起，尤其是水流系统的变化。就在冈比亚按蚊开始从赤道非洲沿上尼罗河谷北移的那些年里，它也跨越了大西洋，到达巴西海岸。在巴西和上尼罗河地区，蚊子都利用了新近的灌溉工程和用水模式的改变。在尼罗河，英国人通过在英埃占领下的苏丹增建蓄水库，扩大了阿斯旺水利工程对河流的控制。1925 年，位于苏丹首都喀土穆以南二百英里的森纳尔，建成了横跨青尼罗河的大坝；1937 年，在喀土穆上游三十英里的杰贝勒奥利亚，又建成了横跨白尼罗河的大坝。工程完成后，地方病发病水平上升的报告随之出现，包括疟疾和血吸虫病。后者是一种由水生螺传播的寄生虫感染，最终波及整个埃及；其治疗后来又引入了另一种地方性感染——丙型肝炎，这可能是世界上由医疗干预造成的最大规模血液传播病原体传播事件。[19]河流控制工程相互连通，使蚊子能够跨越障碍，从一个地区抵达另一个地区。随之发

Disease often moves with the changing movements of people, and modern war causes large numbers to find routes outside existing networks of trade and migration. But having taken advantage of new kinds of transport and traffic routes, the insect also needed ways to establish itself by colonizing new territory and populations. The patterns of war and transportation had to intersect with other developments, in particular changing hydraulics. In the same years that the gambiae mosquito began to move north from equatorial Africa along the Upper Nile valley, it also crossed the Atlantic to the coast of Brazil. In both Brazil and the Upper Nile the mosquito took advantage of recent irrigation works and changed patterns of water use. In the case of the Nile, the British had extended the control of the river at Aswan by constructing further storage reservoirs in the Anglo-Egyptian-occupied Sudan. Dams were completed across the Blue Nile at Sennar, two hundred miles south of Khartoum, the Sudanese capital, in 1925, and across the White Nile at Jabal Aulia, thirty miles above Khartoum, in 1937. These projects were followed by reports of

展的常年灌溉农业创造了大量孳生地，其间生活着更为稠密的人类宿主；由于许多地区不再泛滥，人们往往住得离水边更近。修建灌溉工程的工程师没有考虑到，螺或蚊子可能利用他们的工程迁移，某些寄生虫可能随这些宿主移动，或者由此产生毁灭性的后果。然而，英国人在 1942 年的一份内部报告中承认，要恢复埃及人口的健康，最可靠的办法就是拆除大坝，回到盆地灌溉。[20]

new levels of endemic disease, including schistosomiasis (a parasitic worm infection carried by an aquatic snail that would eventually affect all of Egypt, and whose treatment later introduced another endemic infection, hepatitis C, in possibly the world's largest transmission of blood-borne pathogens from medical intervention) as well as malaria.[19] The linking together of the river control projects enabled the mosquito to jump barriers from one region to the next. The accompanying cultivation based on perennial irrigation created many breeding places among a thicker population of human hosts that often lived much closer to the water now that flooding no longer occurred in many areas. The engineers who built the irrigation works had not considered the possibility that snails or mosquitoes would make use of their work to move, or that certain parasites would travel with these hosts, or that devastating consequences would ensue. In a private report in 1942, however, the British acknowledged that the surest way to restore the health of the Egyptian population would be to destroy the dams and return to basin irrigation.[20]

灌溉工程还产生了其他意料之外的影响。筑坝改变了河流流量的分布和时间节律，也改变了水温与水的化学性质。这些变化影响河床与河岸，改变了河流环境的特征。依赖河流涨落平衡的微生物和植物消失了，另一些更具侵占性的物种却利用了这一变化。菹草（Potamogeton crispus）是侵入性最强的水生植物之一，它开始形成大片草岛，河水把成团的草带往下游。一位埃及疟疾专家证实，冈比亚按蚊又利用了菹草，借它把蚊幼虫从一个孳生区运往另一个孳生区。[21]

The irrigation works led to other unexpected effects. The damming of the river altered the distribution and timing of its flow, as well as the temperature and chemistry of the water. This affected the riverbed and banks, altering the character of the riverine environment. Microorganisms and plants dependent on the balance of the river's ebb and flood disappeared, while other, more aggressive species took advantage of the change. Curly pondweed, or *Potamogeton crispus,* one of the most invasive aquatic plants, began to form large islands of weed, which the river's current carried in clumps downstream. An Egyptian malaria expert established that the *Anopheles gambiae* in turn made use of the pondweed, which transported the larvae of the mosquito from one breeding area to the next.[21]

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如果说冈比亚按蚊得益于尼罗河流量及化学性质的变化，那么需要人体来繁殖的寄生虫也得以从中获益。作为一种形成孢子的寄生虫，疟原虫并不是为了杀死人类受害者而来；它进入人体，只是为了完成自己不寻常的生命周期。雌蚊叮咬时把幼小的孢子转移到人体内，它们在受害者的肝细胞中停留约一周。随后，每个孢子裂开，向血流中释放多达四万个后代；这些后代以血细胞中的血红蛋白为食，继续增殖，其中一些分别形成雄性和雌性的形态。这种爆发式繁殖并不是为了杀死受害者，而是要确保另一只蚊子叮咬时，能把一定数量的孢子再次吸入昆虫胃内，使它们在那里受精，完成繁殖周期。然而，新入侵者带到埃及南部的这种恶性寄生虫，会使受害者的红细胞格外黏稠，堵塞动脉，使身体缺氧。多数受害者在经历严重发热后活了下来，这保证了寄生虫仍有可栖居的宿主。但如果脑部或其他重要器官缺氧，这个并非自愿接纳它的宿主就可能死亡。

〔校注：底本以 spores 泛称疟原虫的不同阶段，并称其堵塞 arteries（动脉），这是简化的生物学叙述。CDC 的生命周期说明区分子孢子、裂殖子和配子体；雌蚊吸入的是配子体。本文保留原文，而不把这些不同阶段静默改写为同一种规范术语。〕

If the gambiae mosquito benefited from the changes in the flow and chemistry of the Nile, its parasite, needing human bodies for reproduction, was also able to take advantage. As a spore-forming parasite, the plasmodium did not set out to kill its human victims, but entered their bodies merely to complete its unusual life cycle. Transferred by the bite of the female mosquito, the young spores take up residence for about a week in the cells of the victim's liver. Each spore then bursts apart and releases into the bloodstream up to forty thousand offspring, which feed off the blood's cell hemoglobin and multiply into further offspring, some of which assume separate male and female forms. The explosive reproduction is not intended to kill the victim, but to ensure that with the bite of another mosquito a number of spores are ingested back into the stomach of the insect, where they fertilize and complete the reproductive cycle. However, the malignant form of the parasite brought by the new invader to southern Egypt makes the red blood cells of its victims particularly sticky, clogging the arteries and depriving the body of oxygen. Most victims survive after a severe fever, which ensures that the parasite still has hosts in which to live. But if the brain or

another vital organ is deprived of oxygen, the unwilling host can die.

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在上埃及，疟原虫遇到的人群没有能够打断其感染周期的免疫反应，因为它是新来的。它也遇到了一群身体已被制糖业改变的人。从 20 世纪 20 年代起，新近独立的埃及政府首次能够保护本国制造业，尤其是制糖业——这个国家历史最久、规模最大的现代工业。20 世纪 30 年代和 40 年代，对全球市场实行的价格保护与灌溉工程相结合，支持了种植面积的扩张。常年灌溉和甘蔗种植降低了土壤肥力，也缩减了可用于粮食生产的土地。当战争中断人工肥料供应时，这些因素共同使埃及南部居民更容易受到疟原虫侵害。与营养不良的南部居民形成对照的是，政府官员、医务人员、灭蚊队员，以及那些从开罗来到南方开展慈善救济的富裕妇女，无一在疫情中丧生。[22]此外，巴西的报告表明，甘蔗汁会加重疟疾的影响，而甘蔗种植园的工人会在劳作现场折断甘蔗，咀嚼吸汁。[23]因此，寄生虫在多个层面发现，糖使它所遇到的身体更难抵抗感染。疫情的化学作用同时发生在国家层面和细胞层面。

In Upper Egypt the plasmodium found a population with no immune response to interrupt its infection cycle, for it was a new arrival. It also found a population whose bodies had been transformed by the sugar industry. From the 1920s Egypt's newly independent government was for the first time able to protect local manufacturing, in particular sugar production, the country's oldest and largest modern industry. Price protection against the global market in the 1930s and 1940s combined with the irrigation work supported the extension of cultivation. Perennial irrigation and cane cultivation reduced the fertility of the soil and the land available for food production. When the war interrupted the supply of artificial fertilizer, these factors combined to make the people of southern Egypt far more vulnerable to the plasmodium parasite. In contrast to the badly nourished residents of southern Egypt, none of the government officials, medical workers, or eradication teams, nor the wealthy women from Cairo who launched a charity relief operation in the south, lost their lives in the epidemic.[22] Furthermore, reports from Brazil indicate that sugarcane juice, which those working on sugar plantations consumed on site by breaking and chewing the cane, can worsen the effects of malaria.[23] Thus on several levels the parasite found that sugar had left the bodies it encountered less able to resist infection. The chemistry of the epidemic operated at the level of the nation and of the cell.

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造成营养不良的肥料短缺，也体现了多层面力量之间的相互作用。德国硝酸盐供应被战争切断后，之所以找不到替代性的化肥来源，还存在一个更大的原因。天然硝酸铵的供应只存在于世界上的一个地方——智利阿塔卡马沙漠，而在那里经营的美国加工公司只能向埃及提供少量产品。它们同人工硝酸盐制造商一样，正把肥料工厂用于一个更迫切的目的。硝酸铵是两种化学上相似、社会上却不同的过程的主要原料；这两种过程都与生死攸关：给作物施肥，以及制造烈性炸药。欧洲和美国已经把肥料厂改为生产战时弹药。埃及农业缺乏硝酸盐，进而发生使大量人口营养不足的粮食危机，并不只是因为失去了某一个特定的供应来源。硝酸盐的化学力量参与塑造了事件的进程。

〔校注：底本此处确写 ammonium nitrate（硝酸铵）。USGS《Mineral Resources in Deserts》把阿塔卡马历史上用于肥料与炸药的天然矿物明确列为 sodium nitrate（硝酸钠）。此处化学名称疑误，译文保留底本。〕

The fertilizer shortage that contributed to malnutrition also represented the interaction of forces on several levels. After German nitrate supplies were cut off by the war, there was a larger reason why alternative sources of chemical fertilizer could not be found. Natural supplies of ammonium nitrate were available in only one place in the world, the Atacama Desert in Chile, and the U.S. processing companies operating there could supply Egypt with only small amounts. Along with the manufacturers of artificial nitrates, they were using their fertilizer factories for a more urgent purpose. Ammonium nitrate provided the main ingredient for two chemically similar but socially different processes, each concerned with life and death: the fertilizing of crops and the making of high explosives. Europe and America had converted their fertilizer plants to the manufacture of wartime ammunition. The lack of nitrates for Egyptian agriculture, and the consequent food crisis that left much of the population undernourished, was due not only to the loss of a particular source of supplies. The chemical powers of the nitrates contributed to the course of events.

最后，战争还提供了战胜疫情的方法，因为除虫菊喷雾和巴黎绿并不足够。疫情经历第一个冬天后，政府宣布冈比亚按蚊已经被根除，但 1943—1944 年又出现了第二轮、且严重得多的暴发。问题的一部分在于，灭蚊运动受当时公共卫生观念的影响，认为积水有害健康，因而把注意力集中在村庄边缘常见的大型积水池上；但这种特定的蚊子却愿意在极小的沟渠、灌溉水道，以及修筑铁路路堤留下的取

Finally, war provided the method used to defeat the epidemic, for pyrethrum sprays and Paris green were not enough. After the first winter of the epidemic the government declared the gambiae mosquito eradicated, but in 1943–44 there was a second and much more severe outbreak. Part of the problem was that the eradication campaign, influenced by current public health concerns about the unhealthiness of stagnant water, had

土坑中孳生。人们没有把这些地方同疾病联系起来，常常将其忽略。[24]

concentrated on the large standing pools often found on the edges of villages, whereas this particular mosquito was willing to breed in the smallest ditches and irrigation channels and in the borrow pits left by the construction of railway embankments, which were not associated with disease and were often overlooked.[24]

埃及政府得以向一种新型跨国世俗组织——非营利法人——求助。世纪初，美国在加勒比地区的军事扩张，尤其是巴拿马运河的修建，推动了大规模灭蚊工作，因为蚊子同时传播疟疾和黄热病。（组织修建苏伊士运河的费迪南·德·雷赛布，也是第一个尝试开凿贯穿巴拿马地峡运河的人；但 1889 年，他被迫放弃了这项已进行十年的工程，部分原因正是这两种疾病造成的死亡。）1915 年，也就是运河建成后的第二年，新成立的洛克菲勒基金会接手美国陆军的灭蚊运动，启动了一项研究和控制这两种蚊媒疾病的全球计划。因此，蚊子的全球移动塑造了一种跨国法人慈善事业。

The Egyptian government was able to turn for help to a new form of transnational secular body, the nonprofit corporation. Early in the century, U.S. military expansion in the Caribbean, in particular the building of the Panama Canal, had encouraged extensive efforts to control mosquitoes, which carried both malaria and yellow fever. (Ferdinand de Lesseps, the man who organized the construction of the Suez Canal, had been the first to attempt to dig a canal across the Isthmus of Panama, but in 1889 was forced to abandon the ten-year effort, in part because of deaths from these two diseases.) In 1915, the year after the canal's completion, the newly established Rockefeller Foundation took over the mosquito campaign from the U.S. army and launched a worldwide program to study and control the two mosquito-borne diseases. Thus the global movements of the mosquito gave shape to a transnational corporate philanthropy.

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相较于疟疾，黄热病是更迫在眉睫的威胁，因为它可能借巴拿马运河进入太平洋。洛克菲勒基金会在巴西设立项目，以消除南美洲沿海地区的黄热病。[25]运动的负责人弗雷德·索珀医生开发了以现代战争为基础的根除方法：身穿制服、手持喷雾枪的人组成“旅”，执行搜索与歼灭任务。疾病将通过物理消灭敌方物种来战胜，而不是通过改善社会条件或医疗干预。详细的地图和索引卡记录着待搜查房屋的位置、每只蚊子的发现情况，以及喷洒或倾倒化学品的路线与时间。纽约的洛克菲勒基金会总部把注意力放在黄热病上，对冈比亚按蚊已抵达巴西的报告并不感兴趣。不过，索珀认为，这种新来且分布范围相对有限的冈比亚按蚊，为展示他的技术方法提供了理想机会。他于 1938 年组织了一场运动，到 20 世纪 40 年代初根除了冈比亚按蚊传播的疟疾。这次成功使索珀成为世界上最有影响力的疟疾专家，能够重塑战术，并制定彻底根除一个物种的新方法；此后又过了五十年，这些方法才受到严肃挑战。[26]

Yellow fever was a more immediate concern than malaria, for it threatened to use the Panama Canal to cross into the Pacific. Rockefeller set up a program in Brazil to eliminate the disease from the coastal areas of South America.[25] The leader of the campaign, Dr. Fred Soper, developed eradication methods based on modern warfare, in which “brigades” of uniformed men armed with spray guns went on search and destroy missions. Disease was to be defeated not by improved social conditions or medical intervention but by the physical elimination of the enemy species. Detailed maps and index cards recorded the location of houses to be searched, the discovery of each mosquito, and the routes and timing of missions to spray or dump the chemicals. With its focus on yellow fever, the Rockefeller Foundation headquarters in New York was not interested in reports that the gambiae mosquito had reached Brazil. However, Soper saw in the arrival of the new and relatively well contained *Anopheles gambiae* an ideal opportunity to demonstrate his technical methods. He organized a campaign in 1938 that eradicated gambiae malaria by the early 1940s. The success made Soper the world's most influential malaria expert, able to reshape tactics and lay down the new methods of total species eradication, methods that were not seriously challenged for another fifty years.[26]

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1942 年 11 月，为配合英军在阿拉曼的攻势，美国投入了地中海战场，在法属北非登陆。疾病再次成为一个关切，但这次是斑疹伤寒——它曾在第一次世界大战中夺走数万名士兵的生命。为了研究保护军队的方法，华盛顿设立了斑疹伤寒委员会，总部设在开罗。洛克菲勒基金会的弗雷德·索珀被借调到该委员会，并被派往埃及。与巴西的情形一样，他为另一项任务而来，却恰逢冈比亚按蚊疟疾暴发。[27]美国慈善权力与军事权力交织的网络，再一次把索珀与冈比亚按蚊带到了一起。他拟定了一项军事

In November 1942, in coordination with the British offensive at al-Alamein, the United States entered the Mediterranean war, landing troops in French North Africa. Disease was again a concern, but in this case it was typhus, which had killed tens of thousands of soldiers in World War I. To develop ways to protect its forces, Washington set up a Typhus Commission with its headquarters in Cairo. Fred Soper of the Rockefeller Foundation was seconded to the commission and sent to Egypt. As in Brazil, his arrival on an unrelated mission

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式的灭蚊计划，但英国当局反对美国势力借此进入埃及政治，迫使政府搁置计划。1943—1944 年疫情再起时，英国人开始担心，它会威胁北方的人口中心与军队集结地。他们同意埃及方面任命“一种疟疾独裁官”，组织抗疫运动——“独裁官”一词在当时很流行。[28]他们不得不放弃此前的反对，允许埃及政府在索珀指导下采用巴西的根除模式。[29]1945 年 2 月，埃及灭蚊队终于在卢克索附近、阿姆 • 易卜拉欣所在村庄以南几公里处，发现并消灭了最后一批冈比亚按蚊幼虫。

coincided with an outbreak of gambiae malaria.[27] The intersecting networks of U.S. philanthropic and military power had once again brought Soper and the gambiae mosquito together. He drew up a plan for a military-style eradication campaign, but the British authorities, objecting to this introduction of an American influence into Egyptian politics, forced the government to shelve the plan. When the epidemic reemerged in 1943–44, the British began to fear that it threatened the population centers and troop concentrations of the north. They agreed that the Egyptians should appoint “a sort of malaria dictator” to organize a campaign against the disease (the word “dictator” was in vogue in those days).[28] They were forced to drop their earlier objections and allow the Egyptian government to follow the Brazilian model of eradication, advised by Soper.[29] The Egyptian teams finally identified and destroyed the last gambiae larvae in February 1945, a few kilometers to the south of ʿAmm Ibrahim’s village near Luxor.

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埃及这一连串事件似乎构成了一个三角形，由战争、疾病与农业相互联结而成。地中海的战争把注意力和资源从南方袭来的疫情上移开；蚊子利用战时运输，把疫情带了过来。昆虫的移动还得到战前灌溉工程及其造成的生态转变的帮助。灌溉工程为工业作物提供了用水，却也使农业依赖人工肥料。施入土壤的硝酸铵是制造炸药的主要原料，因而被转用于战争需要。没有肥料，田地生产的粮食减少，于是蚊子携带的寄生虫遇到了营养不良的人类宿主，每天夺走数百人的生命。

The chain of events in Egypt seems to create a triangle, formed by the interconnection of war, disease, and agriculture. War in the Mediterranean diverted attention and resources from an epidemic arriving from the south, brought by mosquitoes that took advantage of wartime traffic. The insect also moved with the aid of the prewar irrigation projects and the ecological transformations those brought about. The irrigation works made water available for industrial crops but left agriculture dependent upon artificial fertilizers. The ammonium nitrate used on the soil was the main ingredient in the manufacture of explosives and was diverted for the needs of war. Deprived of fertilizer the fields produced less food, so the parasite carried by the mosquito found its human hosts malnourished and killed them at the rate of hundreds a day.

事实上，这条链不止是一个三角形。战争、疫情与饥荒之间的联系，取决于河流、大坝、肥料、食物网之间的联系，以及我们还将看到的若干其他环节和互动。引人注意的是，这些不同要素的性质究竟以怎样的方式相互作用。它们并非只是在社会层面彼此影响的独立历史事件。它们之间的联结涉及水力、化学、军事、政治、病因与机械。没有任何研究这一时期埃及的人描述过这种互动。有关军事战术、灌溉方法、英埃关系、水利工程、寄生虫、制糖业和农民的研究都有，却没有叙述认真对待这些要素如何相互作用。仿佛这些要素在某种意义上不可通约。它们似乎涉及截然不同的力量、行动者、要素、空间尺度和时间性。[30]它们相互塑造，但其异质性却抵抗着解释。

The chain is in fact more than a triangle. The connections between a war, an epidemic, and a famine depended upon connections between rivers, dams, fertilizers, food webs, and, as we will see, several additional links and interactions. What seems remarkable is the way the properties of these various elements interacted. They were not just separate historical events affecting one another at the social level. The linkages among them were hydraulic, chemical, military, political, etiological, and mechanical. No one writing about Egypt in this period describes this interaction. There are studies of military tactics, irrigation methods, Anglo-Egyptian relations, hydraulic engineering, parasites, the sugar industry, and peasants. But there are no accounts that take seriously how these elements interact. It is as if the elements are somehow incommensurable. They seem to involve very different forces, agents, elements, spatial scales, and temporalities.[30] They shape one another, yet their heterogeneity offers a resistance to explanation.

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这种抵抗或许与自然世界和社会世界的混杂有关。化学与生物过程，无疑同军事和政治力量属于不同层次。每种过程与力量都有自己的科学，确定适用于其分析的行动者、时间线、地理空间尺度和互动方式。这往往使它们各自孤立于所属学科之中。

The resistance may have something to do with the mixing of natural and social worlds. Chemical and biological processes are surely of a different order than military and political forces. Each of these processes and forces has its own science, which identifies the agents,

对于某一门科学或某一种技术专长的任务而言，这种隔离也许是合适的；但只要开始追问我所描述的那些互动，其局限便立即凸显。既然这些互动如我所说，属于现代时期一些最深刻的转变，这就给社会科学提出了一个问题。社会理论并未发展出能够处理这些互动、回应 20 世纪技术科学转变的分析方式，而是在很大程度上仍然受困于 19 世纪的方法和劳动分工。

time lines, geo-spatial scales, and modes of interaction appropriate to its analysis. This tends to leave each of them isolated in their separate sciences. The isolation may be appropriate for the task of a particular science or technical expertise, but its limitations are striking as soon as one begins to ask about the kinds of interactions I have described. Since those interactions belong, as I suggested, to some of the most profound transformations of the modern era, this presents a problem for social science. Instead of developing the kinds of analysis that might address these interactions, responding to the techno-scientific transformations of the twentieth century, social theory is still largely trapped in the methods and divisions of labor of the nineteenth century.

社会解释有两个特征与这一问题相关。首先，社会理论通常通过把特定案例与一种更大的模式或过程联系起来而运作。像埃及这样的地方所发生的事件，被解释为某种更普遍现象的地方性发生，或通常情形之外的例外，或一般可能性范围内的一种特殊变体。在某些社会科学中，这一目标十分明确，通过方法规则与写作风格表现出来。在另一些学科中，它虽是隐含的，却仍在发挥作用。例如在历史研究中，叙述可能聚焦于特定情境，但其结构和意义却来自它同其他更一般性案例之间的暗含比较。此类叙述中的一般案例，不可避免地是欧洲或西方的历史；欧洲以外发生的具体事件，则被解释为欧洲历史的复制品、相对于该历史模式的变体，或它的替代形式。[31]例如在埃及研究中，我一直描述的那些事件，会被纳入各种更宏大的叙事：民族及其发展的故事、新社会阶级与其他民族行动者的成长，以及现代国家的兴起；这些又常被放到资本主义发展、欧洲扩张，或现代性的全球历史之中。故事之所以采取某种形态，是因为它被嵌入一套用于讲述每个地方的支配性叙事——理性化、技术与社会进步、生产的增长与转变，以及西方文化和权力普遍化的故事。对这样一种普遍性骨架的预设，是社会理论得以成立的基础。当然，把特定事件置于普遍框架之中的解释方式的发展，恰好伴随着西方权力、财富与技术知识非常切实的扩张。问题不在于这种扩张是否发生，而在于它同社会理论赖以建立的基础之间是什么关系。社会理论所向往的普遍性，是在西方的特殊历史内部奠基、并由这段历史表达的一个范畴。

There are two characteristics of social explanation relevant to this problem. First, social theory typically operates by relating particular cases to a larger pattern or process. Events in a place like Egypt are explained as the local occurrence of something more general, or an exception to what generally occurs, or a particular variation in the general range of possibilities. In some of the social sciences this aim is quite explicit, expressed in rules of method and styles of writing. In others it is implicit but still at work, for example in historical scholarship, in which the narrative may focus on a specific context but draws its structure and relevance from an implied comparison with other, more general cases. Inevitably the generic case in such accounts is the history of Europe or the West, and the particulars of what happened outside Europe are explained as replicas of Europe's history, or variations from that historical pattern, or alternatives to it.[31] In studies of Egypt, for example, events like those I have been describing fit into a variety of larger narratives: the story of the nation and its development, the growth of new social classes and other national actors, and the rise of the modern state, often placed in the context of the development of capitalism, the expansion of Europe, or the global history of modernity. The story takes it shape from the way it fits into a sovereign narrative told about every place, the story of rationalization, technological and social progress, the growth and transformation of production, and the universalization of the culture and power of the West. This assumption of a universal armature is the foundation that makes social theory possible. The development of forms of explanation placing particular events into a universal framework coincided, of course, with a quite palpable expansion of Western power, wealth, and technical knowledge. The issue is not whether such expansion occurred, but its relationship to the grounds on which social theory is built. The universal to which social theory aspires is a category founded within and expressed by the particular history of the West.

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社会解释的第二个特征由第一个特征而来：所有行动者都是人。民族史、现代性史、资本主义史的主角，都是人。故事围绕人类行动者的行动和意图而书写。这必然如此，因为正是人类行动者的意向性或理性，赋予解释以逻辑，并使特定案例能够作为某种一般现象的实例而纳入其中。社会理论试图辨

The second feature of social explanation follows from the first: all the actors are human. The protagonists of the history of the nation, of modernity, of capitalism, are people. Human beings are the agents around whose actions and intentions the story is written. This is necessarily the case, for it is the intentionality or rationality

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认的事件之一般性或普遍性，恰恰表现为这种人类理性、技术知识或集体意识的扩散。相比之下，尽管尼罗河是跨国的，按蚊也相当具有全球性，但它们的一般性不同于资本主义、民族观念或现代科学的一般性。尼罗河不被视为一种抽象，蚊子也不被体验为普遍性的表达。

of human agents that gives the explanation its logic and enables particular cases to fit as instances of something general. The general or universal aspect of events that social theory attempts to identify occurs precisely as the spread of this human reason, technical knowledge, or collective consciousness. By contrast, although the river Nile is transnational, and anopheles mosquitoes are quite global, their generality is not the same as that of capitalism, the idea of the nation, or modern science. The Nile is not considered an abstraction, nor is the mosquito experienced as an expression of the universal.

社会理论这两个特征带来的结果是，在解释事件时，人们总是预先知道谁是主角。埃米尔·涂尔干曾经比较过：自然对理解构成阻力，而社会却轻而易举地得到了解释。“研究物质自然的科学家十分敏锐地意识到，自然对他构成了阻力，而克服这种阻力何其困难；社会学家却似乎置身于一个在他眼中完全透明的领域，连最晦暗的问题都能如此轻易地得到解决。”[32]这种轻易、这种透明，究竟是什么？它部分来自人们早已决定了谁才算是行动者。社会分析未必忽略疾病、农业、化学品或技术，而是把它们当作外在之物——自然、工具、障碍、资源——其作用本质上是被动的。即便偶尔赋予它们较为独立的力量，人类能动性与非人要素之间仍存在一道根本分界。

The result of these two features of social theory is that in the explanation of events one always knows in advance who the protagonists are. Emile Durkheim once described the resistance that nature offers to understanding compared to the ease with which society is explained. “While the scientist who studies physical nature is very keenly aware of the resistance it offers him, and which he has so much difficulty in overcoming, the sociologist seems to move in a sphere perfectly transparent to his view, so great is the ease with which the most obscure questions are resolved.”[32] What is this ease, this transparency? It arises in part from having already decided who counts as an agent. It is not that social analysis necessarily ignores disease, agriculture, chemicals, or technology, but that these are externals—nature, tools, obstacles, resources—whose role is essentially passive. Even on the occasions when they are given a more independent force, there is still a fundamental divide between human agency and the nonhuman elements.

社会科学始终建立在一种范畴性区分之上：一边是人类意图与目的的观念性，另一边是这些意图与目的作用于其上、并反过来可能影响它们的对象世界。几乎没有空间去考察，两者如何在多种组合中共同涌现，或所谓的人类能动性如何通过试图转引其他形式的能量或逻辑、或依附于它们而获得力量。任何以人类计划和意图的普遍化力量为基础的解释，都无法探究：人、意向性和抽象之所以可能，是否一面依赖着非人要素，一面又忽视着它们。这些要素显得只是物质性的、次要的和外在的。

Social science is always founded upon a categorical distinction between the ideality of human intentions and purposes and the object world upon which these work, and which in turn may affect them. There is little room to examine the ways they emerge together in a variety of combinations, or how so-called human agency draws its force by attempting to divert or attach itself to other kinds of energy or logic. No explanation grounded in the universalizing force of human projects and intentions can explore whether the very possibility of the human, of intentionality, of abstraction depends on, at the same time as it overlooks, nonhuman elements. These appear merely physical, secondary, and external.

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如果说战时埃及的事件之网对解释构成某种抵抗，部分原因或许在于，它包含了多种并非纯粹属于人的能动作用：按蚊、恶性疟原虫、硝酸铵的化学性质、谢尔曼坦克的 75 毫米炮、河流的水力，以及稍后还将介绍的一两种力量。它们不只是同人类行动者的活动相互作用。它们使一个世界成为可能，而这个世界不知怎的又显得像是人类理性与规划的产物。它们塑造着各种社会过程，有时遵循人类计划，但同样经常不遵循，或至少不完全遵循。我们需要追问：理性、规划、专门知识与利润的各种形式，究竟是如何从这种效果中产生的？

If the web of events in wartime Egypt offers a certain resistance to explanation, part of the reason may be that it includes a variety of agencies that are not exclusively human: the anopheles mosquito, the falciparum parasite, the chemical properties of ammonium nitrate, the 75mm guns of the Sherman tank, the hydraulic force of the river, and one or two more to be introduced shortly. These do not just interact with the activities of human agents. They make possible a world that somehow seems the outcome of human rationality and programming. They shape a variety of social processes, sometimes according to human plans, but just as often not, or at least not quite. How is it, we need to ask, that forms of rationality, planning, expertise, and profit arise from this effect?

社会理论通常把人类行动置于中心，并把外部世界当作这种行动的场所，而非能动性与力量之形式的来源；但这一规则有一个重要例外，见于马克思的著作。对马克思而言，个别资本家不应被理解为凭自身而行动的主体，而应被理解为使资本力量人格化的人。驱动资本主义历史运动的“动力”，不是人的意图，而是通过商品交换，尤其是劳动力交换而实现的价值增殖。马克思写道，当这种通过交换实现的资本增殖成为一个货币所有者的主观目的时，他便成了资本家。此时，他“作为资本家起作用，也就是说，作为人格化的、具有意识和意志的资本起作用”。[33]因此，马克思把资本理解为一种双重之物。它产生于货币流通、技术过程的发展，以及特定的商品交换和权力关系模式。然而，这些物质过程又取得了一种准人的形式。通过交换，对象的力量获得了意识和意志。大多数借鉴马克思的分析，都很快越过了这一观念。某些历史行动者能够使资本力量人格化，这种能力很容易被视为理所当然。人们也许会争论，究竟哪些行动者承担了这一角色，他们完成得有多成功。但资本被人格化，究竟意味着什么？非人的事物或过程，究竟如何同人的意识组成这种混合体？如果资本的力量取决于它同时既是人的又是非人的，那么这样理解资本意味着什么？正如德里达所说，马克思是“最早思考技术的思想家之一”，也是最早着手处理人—机器、资本—意识、自动性—意志这些混合体的人。[34]他的写作把握到，人类意识是一种人为制成的构成体，尽管最终他总想把对意识的批判建立在一些绝对区分之上：实在与抽象、在场与表象、对象与价值、劳动与观念。我们必须探究这些区分表面上的稳定性。非人与人、或实在与抽象之间这种含混的关系，是如何构成的？在现代政治中，两者之间不可化约的交换或张力，又是如何被化解成如此简单的对立的？

In social theory there is an important exception to the rule that human action is put at the center and the external world is treated as an arena for such action rather than the source of forms of agency and power. It is found in the work of Marx. For Marx, individual capitalists are to be understood not as agents in their own right, but as those who personify the power of capital. The “main-spring” that powers the movement of capitalist history is not human intention but the expansion of value through the exchange of commodities, in particular the exchange of labor power. An individual possessor of money becomes a capitalist, Marx writes, when this expansion of capital through exchange becomes his subjective aim. He then “functions as a capitalist, that is, as capital personified and endowed with consciousness and a will.”[33] Thus Marx understands capital as something twofold. It arises from the circulation of money, the development of technical processes, and particular patterns of commodity exchange and power relation. Yet these material processes acquire a quasi-human form. Through exchange, the powers of objects take on a consciousness and a will. Most analyses that draw on Marx move quite quickly over this idea. The ability of certain historical actors to personify the force of capital is easily taken for granted. There might be disagreement over which particular actors achieve this role, and how successfully they accomplish it. But what does it mean for capital to become personified? How exactly do nonhuman things or processes form this hybrid with the consciousness of humans? What does it mean to think of capital as something whose power depends on being simultaneously human and nonhuman? Marx, as Derrida says, was “one of the first thinkers of technics,” the first to grapple with the hybrids of man-machine, capital-consciousness, automatism-will.[34] His writing grasped that human consciousness is an artifactual body, even if in the end he always wanted to ground his critique of consciousness in absolute distinctions between real and abstract, presence and representation, object and value, labor and ideas. These are distinctions whose apparent stability we must explore. How is the ambivalent relation between the nonhuman and the human, or the real and the abstract, constituted? How are the irreducible exchanges or tensions between the two resolved in modern politics into so simple an opposition?

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要开始这项工作，我们需要找到一位资本家，一位能在我们的故事中充当人格化资本的人。幸好确有这样一位资本家，而且是位大资本家。巧的是，在前文提到的阿尔曼特种植园——每天有八九十名工人死于疟疾的地方——他就有一座大宅。想必读者不会惊讶，他正是那座种植园的主人。艾哈迈德·阿布德还控制着顺流而下五百码处的加工厂，以及埃及另外十八座大型糖厂；这些工厂共同构成了全国的制糖业。他是埃及商界与政界最有权势的人物之一。[35]他在格拉斯哥大学接受工程师训练，第一次世界大战前曾在奥斯曼帝国治下的伊拉克从事灌溉工程，战争期间则在叙利亚与巴勒斯坦的铁路系统工作。1924 年，他获得了一份疏浚和维护政府出资新建灌溉渠的合同，由此在埃及开始经商；随着公共

To begin this task, we need to find a capitalist, someone who can function in our story as capital personified. Fortunately there is such a capitalist available, and a big one. As it happens he had a large house on the same plantation at Armant mentioned above, where eighty or ninety workers a day were dying of malaria. He was, it will be no surprise to learn, the plantation’s owner. Ahmad ʿAbbud also controlled the processing mill five hundred yards down the river, together with the eighteen other large sugar factories in Egypt that made up the country’s sugar industry, and he was one of the most powerful figures in Egyptian business and politics.[35] Trained as an engineer at the University of Glasgow, he had worked on irrigation schemes in Ottoman

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灌溉系统的扩展，他的财富也不断增长。[36]他的建筑公司参与了 1929—1933 年阿斯旺大坝的加高工程，以及其他大型国家项目。同其他少数成功企业家一样，他借助这些利润丰厚的政府合同与特许权，进入航运、公共交通、房地产、贸易和银行等其他商业领域。他购入阿尔曼特那座六千费丹的甘蔗种植园，加入埃及新兴的大地主阶层；1939 年，他又控制了埃及糖业公司。这是全国历史最久、规模最大的工业企业，在国家保护下垄断着甘蔗原料加工以及糖的销售和出口。[37]到第二次世界大战爆发时，他的商业帝国将总部搬入开罗第一座高层建筑——十八层的伊莫比利亚大楼。此时，阿布德控制着全国仅有的两三个家族企业集团之一，它们正争相垄断本国金融、贸易、运输和工业中的大片领域。战后，国际报刊会将他列为世界十大富豪之一。[38]

〔校注：本章的 acre 按作者原注 67 专指 Egyptian acre，即 feddan。译文统一作‘费丹（埃及亩）’，不换算为英制英亩。〕

Iraq before World War I and on the railway system in Syria and Palestine during the war. He started business in Egypt in 1924 by obtaining a contract to dredge and maintain the new government-financed irrigation canals, his wealth expanding with the expansion of the public irrigation system.[36] His construction company worked on increasing the height of the Aswan Dam in 1929–33 and other large state projects. Like a handful of other successful entrepreneurs, he used these lucrative government contracts and concessions to move into other business sectors, including shipping, public transport, real estate, trade, and banking. He joined Egypt's new class of big landowners by acquiring the six-thousand-acre sugar plantation at Armant, and in 1939 took control of the Egyptian Sugar Company, the country's oldest and largest industrial venture, which enjoyed a state-protected monopoly over the processing of raw cane and the sale and export of sugar.[37] By the outbreak of World War II, as his business empire moved its headquarters into Cairo's first high-rise structure, the eighteen-story Immobilia Building, ʿAbbud controlled one of just two or three family-based entrepreneurial groups competing to monopolize large sectors of the country's finance, trade, transportation, and industry. Following the war, the international press was to rank him as one of the ten richest men in the world.[38]

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阿布德帝国的成长，依赖于他不断建立并重组政治与社会权力的回路。1942 年 2 月，英国强迫埃及任命了一个由华夫脱党领导的政府；该党早先通过谈判争取到了埃及对英国的部分独立，看起来也是英国最可靠的战时盟友。阿布德通过与瓦基勒家族的商业往来，重新建立了早年同华夫脱党的关系，开始资助其活动，并帮助自己的盟友与商业伙伴掌握权力。瓦基勒家族既经营棉花贸易，又拥有土地；其女儿宰娜卜嫁给了华夫脱党籍首相。该党执政三个月后，阿布德商业伙伴的兄弟阿卜杜勒•瓦希德•瓦基勒被任命为卫生部长，恰逢该部收到冈比亚按蚊疟疾抵达埃及的第一批消息。[39]与此同时，阿布德促成了华夫脱党新任财政部长的免职；后者曾试图实行食糖配给，并起诉瓦基勒家族在战时走私棉纺织品。[40]部长被免职，使阿布德的糖业垄断企业能够放手同英国谈判一笔厚利交易。1943 年 2 月，谈判接近完成时，英国大使到阿布德和他出生于苏格兰的妻子杰迈玛家中访问，在糖业庄园那座宽敞的红瓦别墅住了六个星期。这是一次期待已久的旅行，却变成了“噩梦”：大使患上危险的发热病，阿布德也两度险些丧命，先是遭遇飞机失事，随后又在驾车时因马匹惊奔，被甩撞到砖墙上。[41]大使离开后，阿布德达成了那笔令不满的英国人在私下称作“强盗买卖”的交易，同意以可观的利润，把公司“剩余”的糖库存卖给英国军事当局——而此时糖业庄园正发生饥荒。交易还使阿布德获得了稀缺肥料，用于甘蔗种植园，尽管这些肥料官方原本指定供其他作物使用。[42]

The growth of ʿAbbud's empire depended upon his making and remaking circuits of political and social power. In February 1942 the British forced the appointment of an Egyptian government led by the Wafd party, which had earlier negotiated the country's partial independence from Britain and seemed its most reliable wartime ally. Through a business association with the Wakil family, cotton merchants and landowners whose daughter Zaynab was married to the Wafdist prime minister, ʿAbbud reestablished earlier ties with the Wafd, began to finance its activities, and helped put his allies and business associates in control. Three months after the party took power, ʿAbd al-Wahid al-Wakil, a brother of ʿAbbud's business partner, was made minister of health, just as the ministry received the first news of the arrival of gambiae malaria.[39] At the same time ʿAbbud secured the dismissal of the Wafd's new minister of finance, who had tried to introduce sugar rationing (and to prosecute the Wakils for the wartime smuggling of cotton textiles).[40] The minister's removal gave ʿAbbud's sugar monopoly a free hand to negotiate a lucrative deal with the British. As the negotiations neared completion in February 1943, the British ambassador spent six weeks visiting ʿAbbud and his Scottish-born wife Jemima at their spacious red-tiled villa on the sugar estate (a long anticipated trip that became a "nightmare" after the ambassador suffered a dangerous fever and ʿAbbud was twice almost killed, first in a plane crash and then in a driving accident when his horse bolted and threw him against a brick wall).[41] Following the stay ʿAbbud concluded his "thief's bargain," as the disgruntled British called it in private, agreeing to sell the British military authorities his company's "surplus" stock of sugar (at a

moment of famine on the sugar estates) for a considerable profit. The deal also gave ʿAbbud scarce supplies of fertilizer, officially earmarked for other crops, to use on the sugar plantations.[42]

此后的几个月，疟疾疫情再度蔓延，阿布德则展开了一项大胆计划，要利用食糖交易和扩大甘蔗种植所得的利润，把自己变成一位英国官员再次借用当时流行用语所说的“一种商业独裁者”。[43]阿布德运作着要接管埃及另一个主要企业集团——米斯尔集团——的银行和附属企业。该集团在纺织制造、棉花贸易、航空运输、电影、保险及其他领域占据支配地位。[44]与此同时，他重启了 20 世纪 30 年代的一项计划：在新近加高的阿斯旺大坝安装水力涡轮机，建设埃及最大的工业工程。他提议利用所发的电制造肥料，满足全国的全部需求。

Over the following months, as the malaria epidemic took hold again, ʿAbbud embarked on an audacious scheme to use his profits from the sugar deal and the expansion of cane cultivation to make himself into what a British official, invoking once again the nomenclature of the day, called “a kind of commercial dictator.”[43] ʿAbbud maneuvered to take over the bank and the affiliated enterprises of the country’s other major business conglomerate, the Misr group, which had a dominant position in textile manufacture, the cotton trade, air transport, cinema, insurance, and other fields.[44] At the same time he revived a plan from the 1930s for Egypt’s largest industrial scheme, to install hydroelectric turbines in the dam at Aswan that had recently been made higher. He proposed to use the electricity to manufacture and supply the country’s entire demand for fertilizer.

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两次世界大战之间，竞争中的企业集团为争夺埃及政治中罗伯特·维塔利斯所谓“租金回路”的支配地位，展开了日益激烈的斗争。所谓租金回路，指的是凭借对流经这个国家的经济资源享有特权控制而获取的利润。然而，控制租金的流转，又依赖于控制那些具有其他相互关联的流转形式的资源。这包括复杂的家族权力与殖民关系网络，其重要性我已经有所提示。但从 20 世纪 20 年代到 50 年代，这些斗争的中心，是试图支配、或至少从中获利的一种特定回路——尼罗河水的流动。[45]阿斯旺大坝提供了机会，使一系列其他流转——水力、电力、政治、化学和农业的流转——能够重新组织，并集中到更少的人手中。阿布德与其对手争夺在阿斯旺兴建水电站的高利润权利，要把河流的水力转化为驱动工业的动力，再由工业供给农业。大量电力将被用于把氮转化为人工肥料。正如冲积淤泥曾由尼罗河洪水携带并沉积下来一样，未来，合成化学品将装袋从阿斯旺的硝酸盐工厂运出，撒向全国田野，恢复土壤已经丧失的一小部分肥力。尼罗河洪水的复杂流动——被引入蓄水盆地、留存数周以让淤泥和养分沉降、再被排回河道——将被重新组织和转化为更狭窄的流动：水穿过涡轮，电力以高电压沿输电线传送，电能转入硝酸盐，肥料袋运往乡间，氨从土壤转入甘蔗和棉株的蛋白质。控制租金回路的政治斗争，就是一场建立并控制这些相互联结的回路的斗争。而蚊子也正是经由同样这些回路——大坝、灌溉、甘蔗种植——进入埃及的。

The interwar years had seen a growing struggle among the rival business groups for the dominant position in what Robert Vitalis calls the “rent circuits” of Egyptian politics, meaning the profits to be made from privileged control of the economic resources circulating through the country. Controlling the circulation of rents, however, was dependent upon the control of resources that had other, interconnected forms of circulation. These included the complex networks of family power and colonial affiliation whose significance I have already suggested. But at the center of these struggles from the 1920s to the 1950s was an effort to command, or at least turn to profit, one particular kind of circuit, the flow of the waters of the Nile.[45] The Aswan Dam offered the opportunity to reorganize and concentrate into fewer hands a series of further circulations—hydraulic, electrical, political, chemical, and agricultural. ʿAbbud and his rivals competed for the lucrative rights to build a hydroelectric power station at Aswan and convert the force of the river’s flow into the power to drive industry, which in turn would fuel agriculture. Large quantities of electric power would be used to convert nitrogen into artificial fertilizer. Just as alluvial silt had once been carried and deposited by the floodwaters of the Nile, synthetic chemicals would in future be transported in sacks from the nitrate plant at Aswan and deposited across the country’s fields to restore a little of the lost fertility of the soil. The complex flows of the Nile flood, channeled into storage basins, held for several weeks to allow silt and nutrients to settle, and released again into the river, were to be reorganized and transformed into the narrower flows of waters through turbine wheels, high voltages along transmission cables, electrical energy into nitrates, fertilizer sacks across the countryside, and ammonia from the soil into the proteins of cane and cotton plants. The political struggle to control rent circuits was a battle to build and control these interconnected circuits. And it was through these same circuits—dams, irrigation, sugar cultivation—that the mosquito had entered Egypt.

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p. 33–34 我们是否应该这样解释阿布德的权势和财富：他能够使资本“人格化”，并成为资本自我再生产和扩张力量的自觉代表？相比另一种解释——阿布德的成功仅仅源于他善于算计，能够比对手算得更精，并获取越来越大的利润——这种说法似乎更好。后一种解释把成功全归于阿布德本人。它甚至没有追问：究竟是哪些安排——法律、财产、政治经济、工程、灌溉，以及更多安排——使这种计算成为可能？又有哪些能动作用维持着这些安排？前一种解释至少承认了另一种力量，即资本流通的力量。资本能够流通，并通过与其他力量结合而蜕变为其他形式：从货币变为财产和劳动，从财产和劳动变为甘蔗，从甘蔗变为加工后的糖或酒精，再重新变回货币；它所利用的资源和安排，并非仅由阿布德提供。[46]然而，显然，在阿布德的成功中发挥作用的流转，不只有资本的运动与形态变化。他试图把租金回路转为己用的斗争，同时也是发展并引导一整套相互联结回路的努力：水、电、硝酸盐、军方征购、甘蔗、加工糖、棉花，以及其他若干回路。

Should we explain ʿAbbud’s power and wealth in terms of his ability to “personify” capital and become the conscious representative of its power to reproduce and expand? This seems preferable to the alternative of saying that ʿAbbud’s success resulted simply from his skill as a calculating agent who was able to out-calculate his rivals and make an ever larger profit. The latter explanation attributes all the success to ʿAbbud himself. It does not even ask what arrangements (of law, property, political economy, engineering, irrigation, and much more) made such calculation possible, or what agencies kept those arrangements in place. The former at least gives some credit to another power, the circulating force of capital. Capital can circulate and, by combining with further forces, go through metamorphoses into other forms—from money into property and labor, property and labor into sugarcane, sugarcane into processed sugar or al-cohol, and back again into money—using resources and arrangements that do not come from ʿAbbud alone.[46] Clearly, however, the movements and metamorphoses of capital were not the only circulations at work in ʿAbbud’s success. His struggle to divert rent circuits to his own advantage was at the same time an effort to develop and direct a whole series of interconnected circuits: water, electricity, nitrates, military requisitions, cane, processed sugar, cotton, and several others.

p. 34 同样明显的是，认为这些流转与力量被“人格化”，或由特定个人的行动加以代表，这个想法过于简单。个人有时也许能够控制某些要素，甚至声称在社会世界中代表着这些要素。但没有哪个人能够完全驾驭它们，或让世界服从自己的意图。更多时候出现的是一系列诉求、亲和关系与互动，它们全部超出了相关人类行动者的把握或意图。人类能动性与意图，是这些互动的局部而不完整的产物。我们将会看到，这种不完整性意味着：不存在一条单一界线，可以把人与非人分开，或把意图和计划同它们所指向的对象世界分开。

Clearly, too, the idea that these circulations and forces are “personified,” or represented by the actions of particular individuals, is too simple. Individuals may at times secure control of certain elements, and they may even claim to represent those elements in the social world. But no individual masters them, or submits the world to their intentions. More often there occurs a series of claims, affinities, and interactions, all of which exceed the grasp or intention of the human agents involved. Human agency and intention are partial and incomplete products of these interactions. This incompleteness, as we will see, means that no single line divides the human from the nonhuman, or intentions and plans from the object-world to which they refer.

但为什么要坚持纳入所有这些额外的能动作用、流转与力量？社会科学的任务，难道不应像一切科学一样，是进行简化，辨认数量有限、更加具有决定性的行动者吗？为什么不接受一种较为简单却更有解释力的叙事，一种能够描绘全局、甚至辨认出某些模式或作出预测的叙事？对此有一个老答案：如果世界本就是复杂且未定的，存在许多发挥作用的行动者与力量，那么对它的准确描绘也必然是复杂且未定的。[47]但我在这里想提出的答案，涉及专门知识与理性、解释与简化在 20 世纪政治中的作用。政治本身就在努力简化世界，试图通过把世界归结为简单的力量与对立，为自己取得专门知识的权力。

But why insist on all these additional agencies, circulations, and forces? Surely the task of social science, like all science, is to simplify, to identify a limited number of more decisive agents. Why not accept a simpler but more powerful story, one that can depict the big picture and even identify certain patterns or predictions? There is an old answer to this question: that if the world is a complicated and indeterminate place, with many agencies and forces at work, then an accurate picture of that world will be a complex and indeterminate one.[47] But the answer I want to propose here has to do with the role of expertise and reason, explanation and simplification, in the politics of the twentieth century. Politics itself was working to simplify the world, attempting to gain for itself the powers of expertise by resolving it into simple forces and oppositions.

p. 34–35 因此，这并不是要引入自然决定论或水力决定论，来替代现代技术创新或资本主义扩张的决定论。如果社会和经济网络同世界最长河流之一不断变化的

This is not, therefore, a question of introducing a natural or hydraulic determinism to replace the determinism of modern technological innovation or capitalist ex-

生态相联结，这并不意味着要把社会结果归因于自然变化。早在阿斯旺大坝之前，早在 19 世纪所有那些灌溉工程之前，这条河就既是技术与社会现象，也是自然现象。其河水由机械、人、动物与水力的相互作用加以引导、储存、提升、分配和排放。埃及政府水库主管威廉·威尔科克斯对尼罗河水力的研究，被用于确定阿斯旺大坝的工程设计。他认为，旧有灌溉系统把洪水依次引入数百个相互连通的田间盆地，留存一定时间，再依次排回河道；这样的灌溉机制，比取而代之的庞大却单一的拦河坝与水库更为复杂。[48]旧方法制造出一种地理，它的自然性并不比人造性更多，也并不更少。毋宁说，它始终兼具两者。

pansion. If social and economic networks were connected with the changing ecology of one of the world's longest rivers, this does not mean attributing social outcomes to changes in nature. Long before the Aswan Dam, before all the irrigation work of the nineteenth century, the river was already as much a technical and social phenomenon as a natural one. Its waters were channeled, stored, raised, distributed, and drained by the interaction of mechanical, human, animal, and hydraulic power. William Willcocks, the director of reservoirs for the Egyptian government, whose studies of Nile hydraulics were used to determine the engineering of the Aswan Dam, considered the old system of channeling floodwaters in sequence into hundreds of interconnected field basins, holding them for a certain period, and releasing them in sequence again into the river a more complex irrigation mechanism than the enormous but singular barrage and reservoir that replaced it.[48] The old methods had manufactured a geography that was no more natural than it was human, and no less. Rather, it was always both.

自然不是正在发生的变化的原因，而是其结果。正是 20 世纪技术与工程建设的规模，产生了一种把尼罗河纯然体验为自然力量的新方式。一位欧洲作家在参观阿斯旺大坝后受到启发，出版了第一部通俗的尼罗河叙述，并称之为这条河生命的“传记”。埃米尔·路德维希写道：“1924 年底，当我第一次看到阿斯旺大坝时，它的象征意义如此强烈地扑面而来，我仿佛能从河道上这个关键点，向前向后理解整条尼罗河。一股强大的自然力量已被人类的才智驯服，使沙漠能够结出果实；百岁的浮士德也曾尝试这项成就，视之为人服务于同胞所能达到的最高境界。”[49]这里提到浮士德十分贴切。歌德这部关于以殖民方式改造自然的伟大小说，受到他与圣西门主义者谈话的启发。这些人是工程的世俗祭司，他们在 19 世纪前往埃及，开创了后来随着阿斯旺大坝而完成并转变的灌溉工程。20 世纪工程的新规模——阿斯旺大坝是全世界最早也最引人注目的例子之一——把圣西门主义者那种奇异的宗教变成了日常信念：“人类才智”如今能够支配自然的“强大力量”。工程师在制造大坝的同时，也制造了自然。

〔校注：底本将《浮士德》称作 novel（小说），这里照译。魏玛古典基金会作品目录将其列为 Drama（戏剧），并称为悲剧。〕

Nature was not the cause of the changes taking place. It was the outcome. The very scale of the technical and engineering works of the twentieth century produced a new experience of the river Nile as exclusively a force of nature. A visit to the Aswan Dam inspired a European writer to publish the first popular account of the river, which he called a "biography" of its life. "When, at the end of 1924, I first saw the Great Dam at Aswân," wrote Emil Ludwig, "its symbolic significance burst upon me with such force that I seemed to comprehend the River Nile forwards and backwards from this crucial point in its course. A mighty element had been tamed by human ingenuity so that the desert should bring forth fruit, an achievement which the centenarian Faust had attempted as the highest attainable to man in the service of his fellow-men."[49] The reference to Faust is quite appropriate. Goethe's great novel of the colonizing transformation of nature was inspired by conversations with Saint-Simonians, secular priests of engineering who had traveled to Egypt in the nineteenth century and initiated the irrigation projects completed, and transformed, with the Aswan Dam. The new scale of twentieth-century engineering, of which the Aswan Dam was among the first and most dramatic examples anywhere in the world, turned the bizarre religion of the Saint-Simonians into an everyday belief: that "human ingenuity" could now dominate the "mighty elements" of nature. In manufacturing the dam, the engineers also manufactured nature.

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这项新建设的若干特征，帮助产生了一种效果：世界被分成两边，一边是人类的专门知识，另一边是自然。首先，河流控制机制集中到了一个地点。旧有的灌溉机制分布在整个河谷，由数百条渠道、排水沟、堤坝、盆地、闸门、水泵和水车，以及河流自身的河道构成，并利用蒸汽、动物、水力与人力。在描述这些安排时，很难说自然力量在哪里结束、技术在哪里开始，也很难在才智与自然之间画出界线。相比之下，阿斯旺大坝把全部工程集中到一个地点，提供了一个观察位置；路德维希这样的作家可以站

Several features of the new construction helped produce the effect of a world divided into human expertise on one side and nature on the other. First, there was the concentration of the river control mechanisms at one site. The old irrigation mechanisms were distributed along the length of the valley, formed out of hundreds of canals, drains, dikes, basins, sluices, pumps, and water wheels, as well as the channel of the river itself, and drew upon steam, animal, hydraulic, and human power. It would have been difficult in describing these arrange-

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在那里，突然“理解”这条河是一种被人驯服的自然力量。其次，工程的集中要求资本相应集中。最初的大坝造价为 244 万英镑，而水库蓄水后，为加固坝基立即又花了 28 万英镑。[50]要组织这笔支出，并在后来为其辩护，就需要一系列建议书、计划、财务报表、政治备忘录、年度报告与报刊报道；所有这些都以不同方式描述、列举、计算和争论大坝的建设。围绕旧有水利系统的论证与计算，则分布在广阔得多的地域。因此，随着水力集中于大坝，核算、计算、描述和知识也发生了重大的重组与集中。正是通过这些及其他重组过程，世界逐渐被简化为：一边仿佛是自然，另一边则是人类计算与专门知识。

ments to say where natural forces ended and technology began, or to draw a line between ingenuity and nature. In contrast, the dam at Aswan gathered all the engineering into one location, providing an observation point where writers like Ludwig could stand and suddenly “comprehend” the river as a force of nature tamed by man. Second, the concentration of engineering required a parallel concentration of capital. Building the original dam cost £2,440,000 sterling, and a further £280,000 was spent to strengthen the base immediately after the reservoir was filled.[50] To organize, and later justify, this expenditure required a series of proposals, plans, financial statements, political memoranda, annual reports, and newspaper accounts, all of which in different ways described, enumerated, calculated, and argued about the building of the dam. The arguments and calculations accompanying the old hydraulic system had been distributed over a much wider territory. Thus a significant reorganization and concentration of accounting, calculation, description, and knowledge accompanied the concentration of hydraulic power in the dam. These and other reorganizations were the kinds of processes through which the world came to be simplified into what seemed nature on one side, and human calculation and expertise on the other.

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生活如今越来越被归入这种二元安排，呈现出一个简单的二元世界：自然对科学，物质现实对人类才智，砌石工程对蓝图，对象对观念。然而，阿斯旺的例子表明，这种二元论是特定工程与政治的人造产物。同一切二元论和一切人造物一样，它既非原初存在，也不完全稳定。人造性是一个过程的效果。[51]如果从阿斯旺工程所创造的效果——从路德维希站在已建成的大坝面前所体验到的“象征”力量——回过头去，重新找回过程本身，那么，在自然与科学、砌体与象征、待被驯服的河流与事后声称已驯服它的专门知识这些区分之中，就能发现许多事件、要素和力量，扰乱着最终人造物所产生的效果。大坝工程是一项混乱、不确定、冲突重重且充满偶然性的事业。公共财政受国际债务委员会控制，该委员会迫使工程改变坝线。原方案允许“在坝线选择上有更大的余地”，以便利用最坚固的岩石作地基。实际采用的坝基花岗岩却已风化腐坏，导致施工延误，也使大坝未来的稳定性变得不确定。延误迫使承包商放弃使用当地石灰制成砂浆进行砌筑的计划，转而进口干燥更快的成品波特兰水泥。水泥砂浆不及石灰柔韧、防水，造成坝体渗漏。流经泄水闸的水开始侵蚀坝基。工程师还忽略了热应力。大坝造价达到最初预算的两倍。[52]此后，淤泥堆积、水库渗漏与蒸发的问题如此严重，以至于水量非但没有增加，建坝后五十年中坝下年均径流量，反而比建坝前三十年减少了将近四分之一。[53]参与大坝建设的专家没有预见到这些问题中的任何一个。筑坝所需的计算形式极其复杂，后来甚至催生了一个叫作成本—收益分析的新领域。但在这个世纪里，随着世界各地兴建了数百座、最终数千座大坝，不能说计算的准确性有所提高。[54]

Life was now to be increasingly resolved into this binary arrangement, rendering up a simple, dualistic world of nature versus science, material reality versus human ingenuity, stonework versus blueprints, objects versus ideas. This dualism, however, as the Aswan example indicates, was an artifact of particular projects and politics. Like all dualisms, and all artifacts, it was neither original nor completely stable. The artifactual is the effect of a process.[51] If one turns back from the effect created by the engineering at Aswan, from the force of the “symbolic” that Ludwig was able to experience when standing before the completed dam, to recover the process itself, then the distinction between nature and science, between masonry and symbol, between the river to be tamed and the expertise that later claimed to have tamed it, one can locate any number of episodes, elements, and forces that disrupt the effect created by the final artifact. Engineering the dam was a messy, uncertain, conflict-ridden, and haphazard project. Public finances were controlled by an International Debt Commission, which forced changes in the alignment of the dam. The original plan had allowed “more play and choice of alignment” to take advantage of the soundest rock for the foundations. Instead, the granite under the foundation was rotten, causing delays to the construction and uncertainty about its future stability. The delays forced the contractors to abandon plans to build the masonry work using mortar manufactured from local lime and to import ready-made, faster-drying Portland cement instead. The cement mortar was less flexible and watertight than lime, leading to problems with leakage through the structure. The water spilling through the dam’s sluices began to erode its base. The engineers had neglected to consider thermal stresses. The dam cost twice the original budget.[52]

Subsequent problems of silt accumulation, seepage, and evaporation from the reservoir were so great that instead of increasing the water available, the mean annual discharge of water below the dam was almost one-fourth less in the fifty years following the dam's construction than in the thirty years before it was built.[53] None of these problems was foreseen by the experts at work on the dam. So complex were the forms of calculation required by dam construction, it later gave rise to the new field called cost-benefit analysis. But as hundreds and eventually thousands of large dams were built around the world in the course of the century, the accuracy of calculation cannot be said to have improved.[54]

不过，或许有人会说，科学终究逐一解决了所遇到的问题。确实，许多问题得到了克服；但这样一来，就必须承认，科学并不是作为一种预先形成的智识来指导工程师的工作。是工程本身形成了科学。[55]解决办法是在现场摸索出来的。工程学是一种在这些及类似事业中逐渐成形的专门知识。英国工程师在阿斯旺每个施工季结束后回到伦敦，在土木工程师学会会议上宣读论文。这些论文刊登在学会的《会报》上，或发表在《工程师》等专业期刊中。在工地上，工程师可以查阅 R. B. 巴克利的《灌溉袖珍手册》；其中列出的砂浆接缝黏结力、不同材料因吸湿而膨胀的数值，均基于早先工程的观察。[56]这种专门知识是混合的，不是一种从外部应用于世界的智识，而是另一种人为制成的构成体。如果再加上威尔科克斯的观点——旧有盆地灌溉系统比取代它的拦河坝与水库更为精密——那么结论就是：在某些方面，工程建设并非把知识应用于世界，而是把知识从世界中取走。英国工程师从大坝那里学到了一些东西，并把这些知识带进科学期刊和灌溉手册；但曾管理和维护旧有水利系统的农民与当地灌溉专家，却被剥夺了他们的大量知识。[57]

Still, it might be argued, science one by one solved the problems it encountered. Many of them were overcome, it is true, but then one would have to acknowledge that science did not direct the engineer's work as a preformed intelligence. The projects themselves formed the science.[55] Solutions were worked out on the ground. Engineering was an expertise given shape in these and similar undertakings. The British engineers returned to London after each season of construction in Aswan to present papers at meetings of the Institution of Civil Engineers. These were published in the *Proceedings* of the institution, or in professional journals such as *The Engineer.* On a site, engineers could refer to R. B. Buckley's *Irrigation Pocket-Book,* which quoted figures for the adhesion of mortar joints or the expansion of different materials due to the penetration of moisture, based on observations drawn from earlier projects.[56] The expertise was hybrid, not an exterior intelligence applied to the world, but another artifactual body. If one adds to this Willcock's view that the older system of basin irrigation was more sophisticated than the barrage and reservoir that replaced it, the conclusion that follows is that in some ways, rather than applying knowledge to the world, the engineering work took it away. British engineers were taught things by the dam and carried this knowledge into scientific journals and irrigation manuals, but the farmers and local irrigation experts who had managed and maintained the earlier hydraulic system had much of their knowledge taken from them.[57]

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筑坝引发的问题与争论，并不限于专业期刊上的辩论和工地工程师之间的讨论。问题向外扩散，卷入了政府官员与雇员、考古学家、埃及与欧洲的报界、企业家与投资者，以及越来越多的埃及知识分子和政治人物。坝后水库淹没了宏伟的菲莱神庙和其他古迹。考古学家在欧洲报刊和科学大会上发动反对筑坝的运动。[58]成本超支引起金融家、政府工程师、承包商和外部人士之间的冲突，持续多年，并进入了国家政治。1919 年，威尔科克斯批评公共工程部战后在苏丹再建两座大坝的计划，认为其依据的是对尼罗河流量的错误计算，并指出其中位于杰贝勒奥利亚的一座大坝将淹没宝贵农田，使大批人口流离失所、蒙受苦难。[59]科学在这些争斗中形成，这个国家新兴的民族政治也是如此。威尔科克斯提出再次加高阿斯旺大坝，以替代从苏丹控制尼罗河的方案；这一建议在 1919 年的反英起义中被民族主义运动采纳。威尔科克斯却发现自己因煽动叛乱和

The questions and disputes posed by the building of the dam were not restricted to debates within professional journals and discussions among engineers at the site. The problems spilled over and drew in government officials and employees, archaeologists, the national and European press, entrepreneurs and investors, and an increasing number of Egyptian intellectuals and political figures. The reservoir behind the dam inundated the great Temple of Philae and other ancient sites. Archaeologists campaigned against the building of the dam in the European press and at scientific congresses.[58] Cost overruns led to conflicts among financiers, government engineers, contractors, and outsiders, which continued for years and were taken up in national politics. In 1919 Willcocks criticized the postwar plans of the Ministry of Public Works to build two further dams in Sudan, claiming the plans were based on faulty calculations of the

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刑事诽谤而遭到审判。[60]争论仍在继续，尤其是调查委员会发现该部计划中还有更多、更严重的计算错误之后。

Nile flow and pointing out that one of the dams, at Jabal Aulia, would submerge valuable agricultural land and displace and cause suffering to a large population.[59] Science was formed in these wars, and so was the country's new national politics. Willcocks's proposal to increase the height of the Aswan Dam a second time as an alternative to controlling the Nile from Sudan was taken up by the nationalist movement during the 1919 uprising against the British. Willcocks found himself put on trial for sedition and criminal libel.[60] The disputes continued, especially after commissions of inquiry discovered that the ministry's program included further, more serious miscalculations.

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可以说，卷入争论的各方，旨在使自然力量在政治中“人格化”，也就是把它们的潜能转化为人的计划。正如阿布德后来试图使资本、化肥和电力的某些流转人格化一样，那些被投入运作的力量，虽被描绘成自然或物质资源，因而应当服从人的专门知识与规划，却从未完全接受这种次要角色。总有某些效果超出计算，总有某些力量超出人的意图。科学专门知识与民族政治，正是从这种张力中产生的。

The aim of those involved in the disputes, one might say, was to “personify” the forces of nature in politics, that is, to translate their potential into human projects. As with ʿAbbud's later attempt to personify certain circulations of capital, chemical fertilizer, and electricity, the forces put to work, although portrayed as nature or material resources and therefore subject to human expertise and planning, never quite accepted this secondary role. There were always certain effects that went beyond the calculations, certain forces that exceeded human intention. Scientific expertise and national politics were produced out of this tension.

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我们已经知道，在这些相互竞争的大坝计划与计算中，或在随后的技术与政治争斗中，冈比亚按蚊始终没有位置。然而，当它利用新水库和河水的流动，出人意料地抵达阿斯旺时，一场类似的斗争随即展开：各方试图把这种昆虫拉入不同的政治联盟。在两次世界大战之间的开罗，乡村的政治问题——与常年灌溉扩展、商品农业发展、大地产增长，以及法拉赫即农民日益严重的贫困、负债、失地、饥饿与寄生虫感染相联系的问题——被掌权者转译成所谓的“公共卫生”问题。它们将由政府的乡村社会改良与卫生计划加以解决。1936 年的华夫脱党政府设立了卫生部；1942 年，该党在英国帮助下重新执政后，最初采取的行动之一就是通过《改善村庄卫生法》。一位埃及政治经济学家 1940 年写道，所指的是第一次世界大战以来的时期：“战后埃及民主与民族精神的增长，使国民认识到，帮助农民不仅是一项义务，也是防范社会动荡的一种保障……1936 年卫生部的设立，以及内政部专设一个负责规划和实施乡村改革的独立部门，是公众日益关心农民的可喜迹象。”[61]掌权者把冈比亚按蚊疟疾的到来，解读为进一步证实这项社会与水利工程计划之必要性的证据。埃及的问题是自然资源有限、公共卫生不足，应由技术科学的方法加以克服。1942 年 12 月，新任卫生部长阿卜杜勒·瓦希德·瓦基勒把疟疾流行归咎于历届政府未能推行其朋友阿布德所倡导的水电计划及进一步的灌溉工程；他声称，这些工程本可提高南方的生活水平，使当地人口足够健康，能够抵御疫情。[62]

The gambiae mosquito, as we know, figured nowhere in these rival plans and calculations for the dam, or in the technical and political battles that followed. When it took advantage of the new reservoirs and river movements and arrived unexpectedly in Aswan, however, a similar struggle developed to draw the insect into a variety of political alliances. In interwar Cairo the political problems of the countryside, associated with the spread of perennial irrigation, the development of commercial agriculture, the growth of large estates, and increasing poverty, indebtedness, landlessness, hunger, and parasitic infection among the fellahs, or peasants, were translated by those in power into problems of what was called “public health.” They were to be solved by government programs of rural social improvement and hygiene. The Wafd government of 1936 created a Ministry of Health, and when the party returned to power with British help in 1942, one of its first acts was to pass the Law for the Improvement of Village Health.“The growth of the democratic and national spirit in Egypt after the war,” wrote an Egyptian political economist in 1940, referring to the period since World War I,“has made the nation aware that helping the fellah is not only a duty but also an insurance against social unrest. . . . The creation of a Ministry of Health in 1936 and of an independent Section of the Ministry of the Interior devoted to the planning and execution of rural reforms, is a welcome sign of increased public interest in the fellah.”[61] The arrival of gambiae malaria was interpreted by those in power as evidence confirming the need for this program of social and hydraulic engineering. The problems of Egypt were those of limited natural resources and a deficient public health, and were to be overcome by the methods of techno-science. In December 1942 the new minister of

health, ʿAbd al-Wahid al-Wakil, blamed the malaria epidemic on the failure of previous governments to carry through with the hydroelectric scheme promoted by his friend ʿAbbud and with further irrigation projects, arguing that these would have raised the standard of living in the south and made its population healthy enough to resist the epidemic.[62]

战前，公共卫生与公共工程计划没有给更激进地讨论乡村私有财产问题留下空间。一些政治人物借用公共卫生的语言，警告说乡村人口“就健全的民族主义生活而言，已经死去”，呼吁采取有限措施，缓解商品农业与大规模土地所有制造成的日益严重的困苦；连政府也曾试图实行农业地租管制。但财产权问题并未被提起。[63]战时肥料危机引发粮食配给后，各群体按收入获得供应，收入较高者获得的配给更多；这正体现了当时对谁有权得到什么的主流态度。[64]

Before the war, the program of public health and public works allowed no place for more radical discussions of the question of private property in the countryside. Warning that the rural population, in the language of public health, was “dead as regards healthy nationalistic life,” a number of political figures had called for limited measures to alleviate the increasing hardship caused by commercial agriculture and large-scale landownership, and even the government tried to introduce agricultural rent controls. But the issue of property rights was not raised.[63] It was symptomatic of prevailing attitudes toward entitlement that when the wartime fertilizer crisis led to food rationing, supplies were assigned to different groups according to income, with higher income groups getting bigger rations.[64]

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但蚊子到来的影响并不那么容易控制。如果阿布德的伙伴把疟疾流行转译成再次支持公共卫生与公共工程项目的论据，竞争集团也可以把蚊子引向其他方向。一群出身富裕家庭的妇女，在南方组织施粥站及其他救济项目，援助疟疾受害者；她们与反对华夫脱党政府和英国的埃及王室关系密切。通过使危机受到关注，她们为王宫提供了令政府难堪的机会。[65]但蚊子也可能反过来令她们难堪。这些富裕妇女邀请一位年轻记者前来参观并报道她们的救济工作。那位后来成为著名作家的记者，却写道，这些妇女自己就像蚊子。她们属于那个“吸人民的血，把它变成蛋糕、鱼子酱和香槟”的埃及阶级。他写道，富人才是真正的瘟疫，她们奢华的宫邸并不比蚊子孳生的死水塘更好。[66]

But the impact of the mosquito’s arrival was not so easily controlled. If ʿAbbud’s associates translated the malaria epidemic into renewed arguments for projects of public health and public works, the mosquito could be taken up by rival groups in other directions. A group of women from wealthy families closely associated with the Egyptian royal family, which opposed the Wafd government and the British, organized soup kitchens and other relief projects in the south to aid the malaria victims. By drawing attention to the crisis they provided the royal palace with an opportunity to embarrass the government.[65] But they in turn could be embarrassed by the mosquito. The wealthy women invited a young journalist to visit and describe their relief efforts. The journalist, later to become a leading writer, reported instead that the women themselves were like mosquitoes. They belonged to the class of Egyptians who “suck the people’s blood and turn it into cakes, caviar and champagne.” The rich were the real epidemic, he wrote, and their opulent palaces were no better than the stagnant pools in which the mosquitoes bred.[66]

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统治秩序的批评者利用蚊子，改变了全国讨论的议题与方式。当南方的疟疾疫情在开罗成为公开的知识时，一些改革者把危机转化为主张更激进变革的论据。他们拒绝政府的解释——高死亡率反映了乡村卫生条件恶劣，以及进一步兴办公共工程的必要——转而把危机同土地分配不均联系起来。一位议员声称，在土地公有的苏联，生活水平高得多；另一些人则提请注意东欧成功的土地改革计划。1944 至 1947 年，以及 1951 年，仿照这些改革的法案被提交议会，提议禁止拥有五十费丹以上土地者继续购入土地。[67]这些措施在议会遭到阻挠，没有一个政党把土地改革作为政治议题。[68]政府转而在 1948 年 3 月启动了一项计划：把从沙漠开垦出的土地以五费丹为一块分给小农；他们还获得了“卫生住房”，集中在四个村庄中，每村都配有学校、清真寺、卫生站和公共浴室。[69]借助这些因阿斯旺大坝加高而

The mosquito was put to work by critics of the ruling order to alter the terms of national debate. As the malaria epidemic in the south became public knowledge in Cairo, a number of individual reformers turned the crisis into an argument for more radical change. Rejecting the government view that the high death toll reflected poor sanitary conditions in the countryside and the need for further public works, they linked the crisis to the unequal distribution of land. One deputy in parliament claimed that the standard of living in the Soviet Union, where land was held in common, was much higher, and others drew attention to the successful land reform programs of Eastern Europe. From 1944 to 1947 and again in 1951, bills were introduced into parliament modeled on these reforms, proposing to bar owners of more than fifty acres from acquiring additional land.[67] The mea-

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成为可能的措施，艾哈迈德·阿布德这样的人似乎没有面临直接的权力威胁，反而能够巩固自己的政治经济地位。然而，得益于疟疾疫情，以及它使之显露的粮食短缺与贫困，土地改革的问题如今已开始流传。

sures were blocked in parliament and no party made land reform an issue.[68] Instead, in March 1948 the government launched a program to distribute land reclaimed from the desert in five-acre plots to small farmers, who also received "hygienic houses" grouped in four villages, each equipped with a school, mosque, health unit, and public bath.[69] With such measures, made possible by the raising of the dam at Aswan, there seemed no immediate threat to the power of men like Ahmad ʿAbbud, who were able to consolidate their political and economic position. Yet thanks to the malaria epidemic and to the food shortages and poverty it had made visible, the question of land reform was now in circulation.

1944 年 10 月，阿布德的政敌成功推翻华夫脱党政府，他的电力与肥料计划随之中断。然而，五年后，他同竞争对手米斯尔集团结盟；1950 年，双方共同帮助华夫脱党重新执政，并保障了自身的经济垄断地位。阿布德的帝国通过收购旅游业和纺织制造业的企业而扩张，又创办了造纸和香水企业，两者都利用制糖业的副产品。[70]他放弃了在阿斯旺兴建水电驱动硝酸盐工厂的提议，该计划随后将成为政府项目。他转而利用来自美国的资本贷款和新型固氮技术，在苏伊士建立了一座肥料厂；其动力不再来自尼罗河水，而来自附近壳牌油田的废气。

ʿAbbud's electricity and fertilizer schemes were interrupted in October 1944, when his political rivals managed to bring down the Wafd government. Five years later, however, he formed an alliance with the rival Misr group, and together in 1950 they helped put the Wafd back in power and secure their monopolistic economic positions. ʿAbbud's empire expanded with acquisitions in tourism and textile manufacturing, and new ventures to manufacture paper and perfumes, both using byproducts of the sugar industry.[70] He abandoned the proposal to build a nitrate plant at Aswan powered by hydroelectricity, which was to become a government project. In its place he took advantage of capital loans and a new nitrogen-fixing technology available from the United States to build a fertilizer factory at Suez, powered not with Nile waters but waste gases from the nearby Shell oil field.

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这座肥料厂由美国出资，象征着美国战后在埃及的角色。正如美国使馆提醒华盛顿的那样，它是"美国对埃及经济援助最实质、最有形的例子"。[71]美国人计划通过"技术援助"建立政治影响，同时补贴本国工业技术；这一计划将围绕技术—经济发展的政治，组织战后国际关系。除了肥料厂，援助计划在最初几年还资助了引入杂交玉米的试点；从加利福尼亚州帕洛阿尔托的联合直升机公司供应六架直升机，用于向农作物喷洒新型化学杀虫剂——稍后还会谈到；开展井式排水示范项目，修复那些拜大坝所赐、正"因过度灌溉与盐分而退化"的土地；以及一项用泥砖建房的新技术。[72]当然，尼罗河谷居民已经用泥砖建房数千年了，因此最后这一项需要解释。为美国援助团提供咨询的波士顿公司亚瑟·D. 利特尔认定，改良泥砖是埃及技术—经济发展"必不可少的一部分"。使馆向华盛顿报告，农民的房子"从来不干净"。"泥砖的性质本身就助长灰尘，而不是洁净。它的表面多孔，不容易刷上石灰浆或涂料。"[73]顾问们获得合同，要用新工艺建造二十座泥砖房，以"一种专门的制泥砖机器"取代传统木模，并在通常的泥土与秸秆混合物中加入油，以改良砖料。人们希望通过这二十座示范房，"建造这种房屋的知识将传遍埃及"。[74]

The fertilizer factory was funded by the United States as an emblem of its postwar role in the country, "the most substantial, tangible example of American economic assistance to Egypt," as the U.S. Embassy reminded Washington.[71] The Americans planned to build political influence and at the same time subsidize their own industrial technology through a program of "technical assistance," which would organize postwar international relations around a politics of techno-economic development. Besides the fertilizer factory, in its first years the assistance program also funded a pilot scheme for the introduction of hybrid corn, the supply of six helicopters from United Helicopters of Palo Alto, California, for spraying crops with new chemical pesticides (more on that in a moment), a demonstration project for well drainage to restore land that, thanks to the dam, was "deteriorating from excess irrigation and salinity," and a new technology for building houses of mud brick.[72] People in the Nile valley had been building their houses with mud brick for several millennia, of course, so this last item needs explanation. Arthur D. Little, the Boston-based consulting firm advising the U.S. mission, had determined that an improvement to mud bricks was "an essential part" of Egypt's techno-economic development. The peasant's house "is never clean," the embassy reported to Washington. "The very nature of the mud brick promotes dust rather than cleanliness. Its surface is porous and will not readily take whitewash or paint."[73] The consultants received a contract to build twenty mud

brick houses by a new process, using "a special mud brick making machine" instead of the traditional wooden mold, and improving the normal mixture of mud and straw by the addition of oil. From the twenty demonstration houses, it was hoped, "the knowledge of how to build such a house will be spread throughout Egypt."[74]

这种以技术专长为基础的新政治，有三个重要特征。首先，同阿斯旺大坝一样，它体现的是知识的集中与重组，而不是把专门知识带入此前从未运用过它的地方。技术知识将集中于试点项目和示范地点，再从那里传播到全国。埃及村庄本来就有一种直接有效的泥砖抹灰方法：将某些当地黏土与秸秆混合，在房屋需要更光滑或更气派的墙面时使用。但既有实践同旧有灌溉知识一样，所包含的专门知识分布得过于广泛，无法成为建立帝国权力——或为一家波士顿咨询公司赚取利润——的手段。

There were three significant features of this new politics based on technical expertise. First, as with the dam at Aswan, it represented a concentration and reorganization of knowledge rather than an introduction of expertise where none had been in use before. Technical knowledge was to be focused into pilot projects and demonstration sites, from where it would spread throughout the land. Villages in Egypt already had a straightforward method of plastering over mud brick, using particular local clays mixed with straw, employed whenever a house needed smoother or more impressive walls. But existing practice, like the old knowledge of irrigation, involved an expertise that was too widely dispersed to provide a means for building imperial power—or the profits of a Boston consulting firm.

其次，与阿斯旺工程一样，这些项目不断遇到实际困难。事实上，每一个都失败了。杂交玉米幼苗"枯萎"了，加油稳定的泥砖失败了，直升机的使用"遇到了各种复杂情况"——也就是机器坏了——而用于制造肥料的新型固氮技术，也没有按计划运转。1951 年 11 月，使馆向华盛顿报告："肥料厂要达到令人满意的生产状态，还需要很长时间。""基本工艺从设计角度看就有缺陷。一部分工程在纽约完成，另一部分在伦敦完成，结果证明做得很差。"[75]如同在阿斯旺一样，技术专家试图从这些失败中学习。他们临时修补，因势引入替代方案，重新制定目标。但这意味着，技术专长的运作，并不是把科学与技术带来开发自然资源。事情同样会反过来发生，而且往往以不完整或无法实现的方式发生。所谓的自然塑造了专门知识，而后者从未完全摆脱其令人尴尬的起源。

Second, as with the engineering at Aswan, the projects encountered continuous practical difficulties. In fact, every one of them failed. The seedlings of hybrid corn "withered," the oil-stabilized mud brick was a failure, the use of helicopters had "run into various complications" (they broke down), and the new nitrogen-fixing technology for the manufacture of fertilizer did not work as planned. "It will be a long time before the fertilizer plant will produce satisfactorily," the embassy reported to Washington in November 1951. "The basic process is faulty from the design standpoint. Some of the engineering was done in New York and parts in London and it turns out to have been a weak job."[75] As at Aswan, the technical experts tried to learn from these failures. Repairs were improvised, opportunistic alternatives were introduced, and goals were reformulated. But what this means is that technical expertise did not work by bringing science and technology to develop natural resources. It happened just as much the other way around, and in ways that tended to be incomplete or unrealizable. So-called nature formed the expertise, which never completely escaped its compromising origins.

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然而，第三点是，忽略这些失败与调整——实际上是主动加以掩盖——构成了技术专长政治的一个重要方面。技术科学必须遮蔽其科学之外的起源。首先，没有任何地方提到，这些技术中的每一种——作物喷药、高产玉米、排水机制、肥料厂，或更有利于防病的泥砖——本身都是对早先技术科学项目、尤其是阿斯旺大坝所造成问题的回应，而且是不成功的回应。不仅如此，根本性的困难被说成是些小问题：计划执行不当、意外的复杂情况、官僚拖延，或需要后续跟进。人们决定，杂交玉米需要用更多杀虫剂重新培育；直升机需要从加利福尼亚获得数量更大、供应更持续的备件；新型泥砖技术需要克服社会事务部的政治反对，该部认为现代住房应由混凝土建造。这种模式从阿布德在苏伊士的硝酸盐工厂一开始就已确立。既然工厂不仅要制造肥料，也要制造帝国权力的政治效力，那么设计和工程错

Third, however, it was an important aspect of the politics of technical expertise that these failures and adjustments were overlooked, in fact actively covered up. Techno-science had to conceal its extrascientific origins. Nowhere, first of all, was it mentioned that every one of these technologies—crop spraying, high-yield corn, drainage mechanisms, fertilizer plants, or a mud brick more resistant to disease—were themselves responses (and unsuccessful responses) to problems caused by earlier techno-scientific projects, in particular the Aswan Dam. Beyond this, the fundamental difficulties were presented as minor issues of the improper implementation of the plans, unexpected complications, bureaucratic delays, or the need to follow up. The hybrid corn, it was decided, needed to be recultivated with greater quantities of pesticide. The helicopters needed a larger

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误就不能公之于众。不能让任何人知道工程做得差，也不能让人知道整座工厂用行话说“从设计角度看”就是错的。使馆向阿布德及其美国伙伴保证：“我们将全力配合，使此事不声张。有人询问时，我们将继续表示，任何采用新化学工艺的工厂，在运行初期出现困难都是可以预料的。”[76]

and more continuous supply of spare parts from California. The new mud brick technology had to overcome political objections from the Ministry of Social Affairs, which believed that modern housing should be built of concrete. The pattern was set from the very start with ʿAbbud's nitrate factory at Suez. Since the factory was built to manufacture not just fertilizer but the political effectiveness of an imperial power, the mistakes in design and engineering could not be made public. No one was to be told that the engineering was a weak job, or that the entire plant was wrong, as the jargon put it,“from the design standpoint.”The embassy promised ʿAbbud and his U.S. partners that “we would cooperate fully in keeping the situation quiet. In response to inquiries we would continue to say that operating difficulties in the early phases are to be expected from any new chemical process factory.”[76]

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技术政治始终是一种技术性构成体，一种必须从制造过程中生成的合金；其成分既有人也有非人，既有意图也有非意图，而有意图的或人的部分，总会在某种程度上被非预期之物冲破。但它是一种特殊的制造形式，是以某种方式组织人和非人、事物与观念的混合物，使人的、智识的、意图和观念的领域看起来先行存在，并控制和组织非人之物。[77]

Techno-politics is always a technical body, an alloy that must emerge from a process of manufacture whose ingredients are both human and nonhuman, both intentional and not, and in which the intentional or the human is always somewhat overrun by the unintended. But it is a particular form of manufacturing, a certain way of organizing the amalgam of human and nonhuman, things and ideas, so that the human, the intellectual, the realm of intentions and ideas seems to come first and to control and organize the nonhuman.[77]

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阿布德的新一轮好运并不长久。战后，抗议乡村状况的活动不断加剧，反对英国在埃及所扮演角色的民众运动，在 1952 年 1 月 26 日的开罗大火中达到高潮：象征英国存在的谢泼德饭店被烧毁。一个新任政府开始打击阿布德的商业垄断。财政部未能让他缴清五百万埃及镑的欠税，遂决定将其糖业公司国有化。据报道，阿布德以贿赂国王作为回应，国王在政府执政四个月后将其解散。[78]政治危机不断发展，1952 年 7 月 23 日，由贾迈勒 • 阿卜杜勒 • 纳赛尔领导的埃及陆军中下级军官发动政变。六周之内，他们通过了土地改革法，并宣布将推进战后提出的方案，在阿斯旺兴建第二座、规模大得多的大坝，作为独立后国家建设的核心。

ʿAbbud's new fortunes were short-lived. Postwar protests against conditions in the countryside intensified, and a popular campaign against the British role in Egypt culminated in the Cairo fire of January 26, 1952, when Shepeard's Hotel, symbol of the British presence, was burned down. A newly appointed government began to move against ʿAbbud's business monopoly. After failing to get him to pay E£5 million in tax arrears, the Ministry of Finance decided to nationalize his sugar company. ʿAbbud in response was reported to have bribed the king, who dismissed the government after four months in office.[78] As the political crisis developed, on July 23, 1952, junior officers in the Egyptian army led by Jamal Abdul Nasser carried out a coup d'état. Within six weeks they passed a land reform law and announced that postwar proposals to build a second and far larger dam at Aswan would go ahead, as the centerpiece of postindependence state building.

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这些行动，以及随后的 1956 年苏伊士运河国有化和苏伊士危机——后两者由美国突然撤回对阿斯旺高坝的支持所触发——如今被铭记为埃及政治的转折点。但纳赛尔及其同僚夺取权力时，目标并不是实行土地改革，或围绕阿斯旺工程建设后殖民国家。他们主要关心军队高级指挥层的无能与腐败，在突然担忧自己将被捕时才夺取了控制权。[79]他们迫使国王退位、让位给尚在襁褓中的儿子，并任命一位改革派首相，意在恢复一个较少腐败、较少寡头色彩的议会秩序。不过，土地改革提案已经在流传，甚至得到美国使馆的倡导，被当作防范想象中的共产主义威胁的办法；因此，它们为立足未稳的新政权提供了赢得民众赞同、削弱阿布德等几十名阻挠政治改革的寡头的手段。军人政权随后没收了王室的

These actions, followed by the 1956 nationalization of the Suez Canal and the Suez crisis, brought on by the abrupt U.S. withdrawal of support for the Aswan High Dam, are now remembered as a turning point in Egypt's politics. But Nasser and his fellow officers had not seized power with the aim of carrying out land reform or building a postcolonial state around the Aswan project. Concerned principally with the incompetence and corruption of the army high command, they took control when they suddenly feared their own arrest.[79] They forced the king to abdicate in favor of his infant son and appointed a reformist prime minister with the aim of restoring a less corrupt and oligarchic parliamentary order. Since proposals for land reform were circulating,

全部地产，但对其他人设定的土地持有上限相对较高，为三百费丹。[80]较高的上限，加上规避起来十分容易，使得实际重新分配的土地相对很少；但改革确立了地租管制和佃农权利，改善了每个村庄的状况，直至这些措施于 1997 年 10 月被废除。阿布德失去了六千费丹甘蔗种植园中的大部分土地；后来，随着军政府打击“垄断集团”和其他竞争性权力来源，他又失去了商业帝国。[81]水政治使阿布德致富，却也启动了其他力量，尤其是蚊子；这些力量合在一起，促成了他的垮台。

however, and were even advocated by the U.S. Embassy (as a defense against an imagined communist threat), they offered the means for an insecure new regime to win popular approval and weaken the few dozen oligarchs like ʿAbbud obstructing political reform. The army regime went on to expropriate all the estates of the royal family, but for others set the maximum holding relatively high, at three hundred acres.[80] The high limit and the ease with which it was evaded ensured that relatively little land was redistributed, but the reform did establish rent controls and tenant rights that improved conditions in every village, until their abrogation in October 1997. ʿAbbud lost most of his six-thousand-acre sugar plantation, and subsequently, as the military government moved against “the monopolies” and other rival sources of power, his business empire as well.[81] Hydropolitics had made ʿAbbud rich, but had also set in motion other forces, not least the mosquito, that combined to bring him down.

类似的组合构成了高坝的谱系，其中涉及水电、肥料、经济崩溃与战争之间的交换。第一次世界大战期间，美国既畏惧德国的新型硝酸盐技术，又依赖智利这个唯一的天然硝酸盐来源，因此国会被说服，在 1916 年《国防法》中列入资金，在阿拉巴马州北部马斯尔肖尔斯兴建一座巨型固氮厂。项目还包括在附近修建一座大坝，并配设水电站，供应固氮所消耗的大量电力。战争结束时，联邦已经投入一亿美元，而工厂及尚未完工的大坝和电站却既不经济，也无用处。[82]然而，这次技术失败使一个更大的结果成为可能。1921 年 7 月，亨利 • 福特提出一个方案，不仅针对阿拉巴马州北部这一角，而是面向它所属的整个流域，把工业、水电、输电网、河运、利用人工肥料改良土壤，以及科学农业联系在一起。该提案设想把福特首创的工业协调方案，从汽车工厂和郊区生活方式的尺度，扩展到整个地理区域生态的转变；这一转变围绕技术化的空间展开，即给一套河流系统筑坝，并把它的能量转化为无限的水电。福特要求联邦提供巨额支持，必然招致其商业对手反对，提案因此遭到否决。但随着大萧条带来的经济崩溃，联邦政府重新启动了这个项目。1933 年，国会通过法案，把福特的方案付诸实施，成为全国最大的公共工程项目——田纳西河流域管理局。[83]

Similar combinations formed the genealogy of the High Dam, involving exchanges among hydropower, fertilizers, economic collapse, and war. During World War I American fear of Germany’s new nitrate technology, and its own dependence on the single natural source of nitrates in Chile, persuaded Congress to include in the National Defense Act of 1916 the funds to build a mammoth nitrogen-fixing plant at Muscle Shoals in northern Alabama. The project included the construction of a large dam nearby, with a hydroelectric power station to supply the great quantities of electricity consumed in nitrogen fixing. At the war’s end, after a $100 million federal investment, both the factory and the partially completed dam and power station were uneconomic and useless.[82] This technical failure, however, enabled something much larger to result. In July 1921, Henry Ford proposed a scheme not just for this corner of northern Alabama, but for the entire river basin to which it belonged, linking together industry, hydroelectric power, transmission grids, river navigation, soil improvement through artificial fertilizer, and scientific agriculture. The proposal envisioned an expansion of the industrial coordination schemes that Ford had pioneered, from the scale of car factories and suburban lifestyles, to the transformation of the entire ecology of a geographic region, organized around the technicized space of the damming of a river system and the transforming of its energy into unlimited hydropower. The scale of federal support that Ford demanded for the project ensured the opposition of his business rivals, and the proposal was rejected. But with the economic collapse of the Great Depression the project was revived by the federal government. In 1933 an act of congress put Ford’s scheme into effect as the country’s largest public works project, the Tennessee Valley Authority.[83]

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田纳西河流域管理局是早先技术与政治失败的产物，却成为发展与规划新可能性的典范，在中东等干旱地区尤其如此。大型水坝提供了一种途径，不仅建设灌溉和电力系统，也建设民族国家本身。1949 年，联合国向中东派出经济调查团，团长是田纳西

The TVA, the child of earlier technical and political failures, came to epitomize the new possibilities of development and planning, especially in arid regions such as the Middle East. Large dams offered a way to build not just irrigation and power systems, but nation-states

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河流域管理局董事会主席戈登·克拉普。次年，两位世界银行专家在论述中东发展时说明，自 1930 年以来，“开发整个河流系统的想法已经俘获了公众的想象力”。[84]随后几年，阿布德及其对手早年在阿斯旺兴建水电站和肥料厂的方案，被埃及新的军政府接纳。但它们如今成为受田纳西河流域管理局启发的一项计划的组成部分：在阿斯旺现有大坝稍上游，兴建规模庞大的第二座坝。工程于 1964 年开始，1971 年完成。阿斯旺高坝忽略了盐碱化、渍涝、土壤肥力下降、努比亚人口被迁移、考古遗产丧失、疾病增加、海岸侵蚀、大型渔业遭到破坏，以及蒸发和渗漏造成的水量损失等代价，也忽略了第一座大坝已显现的其他问题，甚至没有尝试开展成本与收益研究，却成为战后埃及民族国家建构的核心。[85]

〔校注：底本写 1964 年开工、1971 年完成，照录。埃及水资源与灌溉部的工程年表列 1960 年 1 月 9 日为第一阶段开工、1964 年 5 月为截流、1970 年 7 月为主体完成、1971 年 1 月 15 日为正式落成典礼；须区别这些节点。〕

themselves. In 1949 the United Nations sent an economic survey mission to the Middle East. Its head was Gordon Clapp, the chairman of the TVA board. The following year, two World Bank experts writing on development in the Middle East explained how since 1930, “the popular imagination has been captured by the idea of the development of entire river systems.”[84] In the following years the old schemes of ʿAbbud and his rivals for a hydroelectric plant and fertilizer factory at Aswan were taken up by the new military government in Egypt. But they now formed part of a TVA-inspired scheme to build the second dam, on a mammoth scale, just above the existing dam at Aswan. Work began in 1964 and was completed in 1971. Ignoring the costs of salinization, waterlogging, declining soil fertility, the displacement of the people of Nubia, the loss of an archaeological heritage, increased disease, coastal erosion, the destruction of a large fishing industry, the loss of water due to evaporation and seepage, and other problems already evident from the first dam, and without even attempting studies of costs and benefits, the Aswan High Dam became the centerpiece of postwar nation making in Egypt.[85]

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马克思写过几句关于昆虫的名言——不是蚊子，而是蜜蜂。他写道，尽管蜜蜂能为自己建造精巧的蜂巢，它却不是建筑师，因为建筑师“在现实中建起结构之前，已先在想象中把它建起来”。[86]自马克思写下这些话以来，我们愈发相信，这种把心灵当作建筑师事务所的笛卡尔式观念，抓住了我们与自然之间的差别。想象工作先组织起计划、图像、理想结构——实际上是整套文化与意义系统——然后再把它们带到外面，在现实中建造出来。我们依赖这种简单得具有误导性的世界观已经太久了，而马克思本人也曾对它提出质疑。我已通过描述阿斯旺工程师的工作，说明它何以具有误导性，并在别处给出过其他例子。[87]只要回想一下建筑师实际所做的工作，我就能再次说明这一点：动笔绘图前的现场考察和客户商谈；长时间盯着计算机辅助设计程序、令人眼睛疲劳的工作；图纸的印制与分发；围坐桌旁、摊开图纸，就建筑规范与规划条例进行协商的会议；对建筑承包商的日常监督；在电气、给排水与通风承包商之间进行调解，因为他们安装的电缆、管线与风道网络彼此争夺空间；对不上的尺寸；被忽视的细节；以及拆掉重做时的改变主意。无可争辩，这一切都包含持续的想象工作，但没有哪一项先于现实中的做事，或独立于现实中的做事。不存在另一个更真实的世界。没有哪一个时刻，人会突然从想象迈入现实，从计划走向实物；阿斯旺的工程师如此，建筑师也一样。

Marx published some famous lines about an insect—not the mosquito, but the bee. Although it builds itself an elaborate hive, he wrote, the bee is no architect, for the architect “raises his structure in imagination before he erects it in reality.”[86] Since Marx wrote those words we have come to believe more and more that this Cartesian notion of the mind-as-architect’s-office is what captures the difference between ourselves and nature. The work of imagination puts together plans, images, ideal structures—in fact entire systems of culture and meaning—before they are taken outside and erected in reality. We have made do for too long with this misleadingly simple view of the world that Marx himself placed in question. I have already suggested by describing the work of the engineers at Aswan why this is misleading, and have offered other examples elsewhere.[87] I could make the point again simply by recalling the work an architect actually does: the visits to the site and consultations with the client that precede any attempt at drawing; the long, eye-straining hours with the CAD program; the printing and distribution of drawings; the meetings around a table with the plans spread out to negotiate the rules of building codes and planning regulations; the day-to-day supervision of the building contractor; the arbitrating among electrical, plumbing, and ventilation contractors installing rival networks of cables, pipes, and ducts; the measurements that do not fit; the overlooked details; and the changes of mind as things are taken apart and redone. There is no disputing that all this involves a constant work of imagination, but none of it precedes or stands apart from doing things in reality. There is no other, more real world. Nowhere does one suddenly step from imagination to reality, from plan to real thing, any more than did the engineers at Aswan.

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不过，如果回到埃及的抗疟运动，或许有人会说，蚊子与人类专家之间的差别显然很清楚。冈比亚按蚊能够穿越非洲大陆，或许颇为聪明，但最终仍敌

Yet if we return to the case of the anti-malaria campaign in Egypt, it might be said, surely the difference between the mosquito and the human expert is clear.

不过化学科学、人类才智与规划的力量。也许确实如此。然而，在这里，把故事讲成专门知识对抗自然，仍然过于简单。毕竟，灭蚊队并不是徒手杀死蚊子的。他们需要许多非人助手。第一次世界大战结束后，瑞士的 J. R. 嘉基公司一直试图寻找一种安全、有效、持久的物质，用作纺织品防蛀剂。这家公司为纺织业生产染料，也是法本化学卡特尔的成员。[88]1941 年，隆美尔正在北非推进，冈比亚按蚊正从苏丹北上，公司一位名叫保罗·穆勒的化学家发现了二氯二苯三氯乙烷的毒性。人们发现，把这种化学品以 5% 的比例混入惰性粉末，能够杀死衣蛾及许多其他昆虫，却似乎不影响温血动物。由于它通过接触而非经口摄入发挥毒性，即使对不吞食毒物的吸血昆虫也能有效杀灭，其中包括传播斑疹伤寒的寄生虫——人虱。[89]正是斑疹伤寒在次年把洛克菲勒基金会的弗雷德·索珀带到了开罗；它对战场上的士兵和饱受战乱城市中的平民构成重大威胁。英国供应部用这种新型瑞士化学品名称的首字母为其命名，于 1943 年 4 月开始生产 DDT，并将其生产同雷达和青霉素一道，列为战时最高优先事项。[90]次年，索珀说服埃及政府，在南部根除冈比亚按蚊的运动中，用 DDT 取代除虫菊——菊花制成的粉末。房屋用这种化学品处理，火车车厢顶棚则喷涂 DDT 与煤油的混合物；这一创新后来被世界各地仿效。[91]

〔校注：底本将穆勒发现 DDT 杀虫毒性的年份写为 1941，照录。法兰西公学院 2022 年相关课程资料将这一发现列于 1939 年；此处年份存在差异，不能与 DDT 首次合成年份混同。〕

Anopheles gambiae may have been clever to make its way across the African continent, but it was no match in the end for the powers of chemical science, human ingenuity, and planning. Perhaps not. Yet here, too, the story of expertise versus nature is too simple. After all, the eradication teams did not kill the mosquitoes barehanded. They needed a lot of nonhuman assistants. Since the end of World War I, the J. R. Geigy company in Switzerland, a manufacturer of dyestuffs for the textile industry (and member of the I. G. Farben chemicals cartel), had been trying to find a safe, effective, and long-lasting substance to use as a mothproofing agent for textiles.[88] In 1941, with Rommel advancing in North Africa and the gambiae mosquito making its way northwards from Sudan, a company chemist named Paul Müller discovered the toxicity of dichlorodiphenyl-trichloroethane. Mixed 5 percent in an inert powder, the chemical was found to kill clothes moths and many other insects yet seemed to have no effect on warm-blooded animals. Since it was a contact rather than an oral poison, it proved to be a potent killer even of bloodsucking insects, which do not ingest poisons, including human lice, the parasite responsible for transmitting typhus.[89] The disease that brought Fred Soper of the Rockefeller Foundation to Cairo the following year, typhus was a major threat to soldiers at war and civilians in war-torn cities. The British Ministry of Supply named the new Swiss chemical after its initials and began manufacturing DDT in April 1943, giving its production the highest wartime priority alongside radar and penicillin.[90] The following year, Soper persuaded the Egyptian government to replace pyrethrum—the powder of chrysanthemum flowers—with DDT in the gambiae eradication campaign in southern Egypt. Houses were treated with the chemical, and the ceilings of trains were spray-painted with a mixture of DDT and kerosene, an innovation later copied around the world.[91]

无论制造 DDT 的公司，还是在埃及使用它的灭蚊队，都不知道这种化学品如何起作用。他们只是赞叹它的威力。事实上，没有人知道它如何起作用，连把它的力量引入应用的人也不知道。1948 年，穆勒获得诺贝尔生理学或医学奖，获奖原因是他证明了 DDT 能够杀死节肢动物，而不是他知道其中的原因。[92]要让它的力量发挥作用，需要数年有条不紊的工作：在被关在皮特—格雷迪试验室中的苍蝇身上，测试数百种合成有机物。这是一种封闭空间或房间，模拟昆虫的环境，例如沙漠、雨林或公寓厨房。昆虫和化学药剂被放入其中，观察者则透过单向玻璃观看。[93]DDT 能杀死苍蝇这一发现，是在皮特—格雷迪试验室里作出的，而不是在穆勒脑中作出的。技术科学所涉及的，与其说是预先规划，或在想象中搭建结构，不如说是在研究实验室内建起一个房间。这个房间重新安排了所谓的自然，很像阿斯旺大坝：把各种要素集中在一处，把雨林搬进一家化学公司的场所，并提供一个可以持续观察它的地点。从皮特—格雷迪试验室到田野，DDT 经历的一连串事件，是借用、转译，以及为一种目的而发明的事物被其他力量接管的过程；这一切又受制于英美争夺埃及的政治、战争需要、洛克菲勒体系

Neither the companies manufacturing DDT nor the eradication teams using it in Egypt had any idea how the chemical worked. They just admired its potency. In fact, no one knew how it worked, not even the man who introduced its powers. When Müller received the Nobel Prize in Medicine or Physiology in 1948, he won the award for having demonstrated that DDT killed arthropods, not for knowing why.[92] Bringing its powers into action had required several years of methodical work, testing hundreds of synthesized organic substances on flies trapped inside a Peet-Grady chamber. This was an enclosure or room that simulated an insect's environment, such as a desert, a rainforest, or the kitchen of an apartment. The insects and the chemical agent were introduced into the chamber while an observer watched through one-way glass.[93] The discovery that DDT killed the flies was made in the Peet-Grady chamber, not in Müller's head. Techno-science involved not so much planning in advance or raising structures in the imagination; it involved erecting a room inside the research lab, which rearranged so-called nature, much like the dam at Aswan, concentrating its elements in one place, trans-

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中一段职业生涯的偶然与雄心，以及甘蔗生产和灌溉工程的影响。[94]

porting the rainforest onto the premises of a chemical company, and providing a place where it could be continuously observed. The chain of events that took DDT from the Peet-Grady chamber to the field was a process of borrowing, translations, and things invented for one purpose taken over by other forces, all modulated by the politics of U.S.-British rivalry over Egypt, the needs of war, the accidents and ambitions of a Rockefeller career, and the impact of sugarcane production and irrigation works.[94]

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更重要的是，如今我们知道，抗疟运动虽然运用了 DDT 的新力量，但这种杀虫剂也有它自己的目的，远远超出研究化学家和灭蚊队的意图。1944 年，美国陆军公共卫生机构和美国昆虫学会已经开始发出警告：DDT 不仅杀死有害昆虫，也会杀死益虫；它对鱼类有毒，并可能伤害一切形式的植物和动物生命。警告遭到忽视。在这种新化学品于埃及取得“成功”之后，以及在索珀监督下于那不勒斯消灭头虱、取得一次更著名的成功之后——事实上，两场运动主要使用的都是除虫菊，DDT 只是在疫情大体已经过去之后才于末期使用——索珀赞同 DDT 倡导者的说法，称它是“一种几乎完美的杀虫剂”。[95]

What is more important, as we now know, while the malaria campaign used the new power of DDT, the pesticide had purposes of its own, well beyond the intentions of the research chemists and the eradication teams. In 1944 the U.S. Army Public Health Service and the American Entomological Society had already begun to issue warnings: that DDT would kill beneficial as well as harmful insects, was poisonous to fish, and was potentially harmful to all forms of plant and animal life. The warnings were ignored. After the “success” of the new chemical in Egypt, and a more famous success in eliminating head lice under Soper’s supervision in Naples (in fact both campaigns mostly employed pyrethrum, with DDT used only at the end, after the epidemics had largely passed), Soper agreed with the advocates of DDT that it was “an almost perfect insecticide.”[95]

事实上，DDT 的杀虫毒力并不高于除虫菊或它所取代的其他化学品。同除虫菊一样，它不直接攻击疟原虫。疟原虫的孢子太小、太多，毒物难以触及。这种新化学品只是在其繁殖周期最脆弱的一环介入，从而将其打断：在这个环节，数以百万计的孢子集中于数量相对较少、体型大得多的宿主——雌蚊——体内。DDT 对蚊子更有效，是由于其化学结构非常稳定。它几乎不溶于水，也能抵抗阳光或土壤中的降解。因此，它在环境中停留的不是几天或几周，而是几年、几十年。后来人们发现，即使它分解了，产物中也包括 DDD；后者同样有毒，并可抵抗分解长达 190 年。DDT 喷进房屋后会持续残留，因此它为房屋“接种”的保护期足以打断蚊子的繁殖周期，而且不需要像除虫菊喷雾那样，把房屋封闭起来——那在实践中很难做到。

DDT was not in fact a more lethal insecticide than pyrethrum or other chemicals it replaced. Like pyrethrum, it did not attack the malaria parasite directly. The plasmodium spores were too small and too numerous to reach with poison. The new chemical simply interrupted their breeding cycle by intervening at its most vulnerable point, when millions of spores were concentrated in the bodies of a relatively small number of much larger hosts, the female mosquitoes. DDT’s greater effectiveness against mosquitoes was due to its very stable chemical structure. It was practically insoluble in water and resisted degradation by sunlight or soils. So it remained in the environment not for days or weeks, but years and decades. (When it does break down, it was later discovered, the resulting products include DDD, which is also toxic and resists decomposition for up to 190 years.) When sprayed in a house DDT lingered, so it “vaccinated” the place long enough to interrupt the mosquito’s breeding cycle, and without the impractical need to seal up the house, as the pyrethrum spray required.

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然而，在埃及，DDT 及除虫菊的效力，也来自冈比亚按蚊的特殊特征——更确切地说，来自蚊子与人类宿主之间社会关系的特殊性。同巴西一样，这种昆虫是新移民，尚未在当地群落中稳固立足，因此相对容易隔离。同时，冈比亚按蚊又是最具社会性的传疟蚊。它尤其依赖人类宿主，比起动物血液更偏好人血。[96]因此，它通常只出现在人类居住地周围。这使它无法远距离迁移，也就凸显了船、火车和漂浮水草对于它移动的重要性。但对人的依赖也使它更容易被根除，因为只对房屋和车辆喷药就相对有效。由于这些原因，一场有条不紊且毫不松懈的媒介根除运动，逐个水池、逐座房屋、逐个村庄

In Egypt, however, DDT (and pyrethrum) also gained their effectiveness from special features of the gambiae mosquito—or rather, of the social relations between the mosquito and its human hosts. As in Brazil, the insect was a new immigrant, so it was not well established in the local community and was comparatively easy to isolate. At the same time, the *Anopheles gambiae* is the most social form of malarial mosquito. It is especially dependent on its human hosts, preferring human blood to animal.[96] Thus it is generally found only around human habitation. This makes it unable to travel long distances, hence the importance of boats, trains, and float-

地对付蚊子，取得了成功。

ing weeds for its travel. But the dependence on humans also made it easier to eradicate, because the spraying of houses and vehicles only was relatively effective. For all these reasons, a methodical and relentless vector eradication campaign, taking on the mosquito pool by pool, house by house, and village by village, was successful.

结果，疟疾专家从埃及得出了错误的教训。那里的成功使他们误以为，能够通过搜捕歼灭运动和杀虫剂的杀伤力，在全球根除一个物种。1946 年，索珀与洛克菲勒基金会在撒丁岛启动抗疟运动，旨在表明 DDT 不仅可用于控制疟疾，还可以消除疟疾。他们用飞机和直升机喷洒药剂，地面队伍总共雇用了两万四千人，装备甚至包括火焰喷射器。然而，尽管运动持续了五年，他们还是未能根除蚊子。它已经扎根得太深。数十万磅 DDT 被撒向撒丁岛各地，但试验表明，蚊幼虫能在根除运动所用浓度二十五倍的药剂中存活，其原因无人能够完全解释。[97]

As a result, the malaria experts drew the wrong lessons from Egypt. Success there suggested, mistakenly, the possibility of a worldwide species eradication using hunt-and-destroy campaigns and the killing power of the pesticide. In 1946 Soper and the Rockefeller Foundation embarked on a malaria campaign in Sardinia, designed to show that DDT could be used not just to control malaria, but to eliminate it. They sprayed the chemical from airplanes and helicopters and employed a total of twenty-four thousand men in ground teams, whose equipment included flame-throwers. Yet despite extending the campaign for five years, they failed to eradicate the mosquito. It was too well entrenched. Hundreds of thousands of pounds of DDT were spread over the Sardinian landscape, but tests showed that the mosquito larvae survived concentrations of the chemical twenty-five times greater than those used in the eradication campaign —for reasons no one could quite explain.[97]

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这次失败没有使他们退缩。四年后，即 1955 年，已经从洛克菲勒基金会接手全球抗疟运动管理责任的世界卫生组织（WHO），采纳了使用 DDT 在全球根除疟疾的计划。在按蚊分布相对稀疏的国家，疾病媒介数量减少，甚至被消除，欧洲尤其如此；但在更多地方，根除并未奏效。尽管被称为“全球”计划，这项根除计划除了一两个试点外，却忽略了世界上最主要的疟疾地区——非洲。在其他地方，寄生虫逐渐对奎宁和其他药物产生抗药性，并大规模卷土重来。[98]与此同时，DDT 产生了其他更具破坏性的影响。直到 1969 年，世卫组织才同意转向媒介管理计划，而非根除计划，并开始警告 DDT 的风险，最终促成了至少在农业上的禁用，但并未使其绝迹。20 世纪 90 年代末，联合国环境规划署推动谈判，拟在 2007 年前彻底停止使用 DDT。[99]

〔校注：这里的 2007 年是底本所述谈判目标，不表示该年已实现全球全面禁用。斯德哥尔摩公约秘书处将 DDT 列于附件 B，在规定条件下仍保留病媒控制的可接受用途。〕

Undeterred by this failure, four years later, in 1955, the World Health Organization (WHO), which had taken over responsibility for the worldwide administration of antimalarial campaigns from the Rockefeller Foundation, adopted a plan for the global eradication of malaria using DDT. Countries where the anopheles mosquito was relatively thinly established reduced or even eliminated the vector, especially in Europe, but in many more places eradication was not effective. Although described as “global,” the eradication program ignored Africa, the world’s major malarial region, aside from one or two pilot schemes. Elsewhere the parasite gradually developed resistance to quinine and other drugs and returned in large numbers.[98] Meanwhile, DDT produced other, more destructive effects. Only in 1969 did the WHO agree to move toward programs of vector management rather than eradication and begin to warn of the risks of DDT, leading to its banning (but not elimination), at least in agriculture. At the end of the 1990s the United Nations Environment Program sponsored negotiations to end the use of DDT altogether by 2007.[99]

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到这时，人们对 DDT 的长期影响已有了更多理解。它虽然几乎不溶于水，却很容易溶于脂肪，因此会在动物脂肪组织中累积，并经食物网逐级放大。尽管仍没有人完全知道它如何起作用，但人们认为，它会像激素一样，模仿或扰乱体内的化学信使，影响有机体的发育与功能。它削弱免疫系统、减少泌乳，使雄性动物发育出雌性生殖器官，并造成其他性发育紊乱。[100]

By this time there was a better understanding of DDT’s long-term effects. While almost insoluble in water, it dissolves easily in fat, so it accumulates in the fatty tissue of animals, an accumulation that is magnified through the food web. Although no one still quite knew how it worked, it was believed that it acts like a hormone, mimicking or disrupting chemical messengers in the body, affecting the development and functioning of the organism. It weakens the immune system, decreases lactation, causes male animals to develop female reproductive organs, and leads to other disruptions of sexual development.[100]

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由于这些力量并不限于杀死当初开发 DDT 所针

Since these powers were not limited to killing the lice,

对的虱子、衣蛾和蚊子，它的用途迅速扩展到其他领域，尤其是农业。世界范围内，为配合合成肥料使用的增加，农业使用这种杀虫剂的数量，远远超过公共卫生计划中的用量。其中最普遍的应用之一，就是保护棉花作物。

clothes moths, and mosquitoes for which DDT had been developed, its use quickly spread to other areas, especially agriculture. Far more of the pesticide was used worldwide in farming, to support the increased use of synthetic fertilizers, than in public health programs. One of the most popular worldwide applications was in the protection of cotton crops.

在埃及，到 1950 年，化肥使用量已经恢复到战前水平，仍为世界最高。据报告，肥料造成“植株繁茂、开花旺盛”，因而助长了害虫，尤其是棉叶虫。[101]两家本地公司开始进口 DDT，以对付这种棉花害虫。在美国帮助下，政府从联合直升机公司购入了那六架直升机，用于从空中喷洒药剂。[102]

In Egypt, by 1950 the use of chemical fertilizers had returned to their prewar levels, the highest in the world. The fertilizers were producing “lush vegetation and flowering,” it was reported, which encouraged insect pests, in particular the cotton leaf worm.[101] Two local companies began importing DDT to combat the cotton pest. With U.S. help the government acquired the six helicopters from United Helicopters to use for spraying the chemical from the air.[102]

与此同时，前文提到权力不断增长的两大垄断企业集团——阿布德集团和米斯尔集团——通过结成政治经济联盟巩固了控制，并于 1950 年帮助华夫脱党重新执政。一年前，两集团同意共同投资，与美国化学公司孟山都建立一家新的合资企业，在当地建厂生产 DDT，为这次合作铺平了道路。[103]DDT 的一种主要原料是乙醇，将由埃及唯一的商业酿馏厂——埃及蒸馏公司（Société Egyptienne de Distillerie）——制造；该厂属于糖业大亨艾哈迈德 • 阿布德，使用其糖业垄断企业提供的糖蜜。[104]阿布德靠参与兴建阿斯旺大坝发了财；大坝使甘蔗种植园得以扩张，却也要求引入肥料。肥料使用带来了害虫，需要用 DDT 处理。而如今，DDT 又将由阿布德的糖来制造。

Meanwhile, the country’s two dominant business monopolies, the ʿAbbud and Misr groups, whose increasing power I mentioned earlier, consolidated their control by forming a political and economic alliance, which in 1950 helped bring the Wafd party back to power. The two groups prepared the way for this collaboration a year earlier by agreeing to coinvest in a new joint venture with the U.S. chemical company Monsanto—to build a local plant to manufacture DDT.[103] A major ingredient of DDT is ethyl alcohol, which was to be made by Egypt’s only commercial distillery, the Société Egyptienne de Distillerie, owned by the sugar baron Ahmad ʿAbbud, using molasses supplied by his sugar monopoly.[104] ʿAbbud had got rich helping to build the Aswan Dam, which enabled the spread of sugar plantations but also required the introduction of fertilizer. The use of fertilizers brought insect pests, which needed treatment with DDT. Now the DDT would be made from ʿAbbud’s sugar.

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两年后，纳赛尔发动军事政变时，政府已经决定借助国际卫生机构的援助，自行兴建杀虫剂工厂。政变一周后，即 1952 年 8 月 2 日，新政府在可能是其第一次国际行动中，与世卫组织及联合国儿童基金会签署协议，在卡夫尔扎亚特建厂，每年生产二百吨 DDT 成品。[105]有卫生机构出资大规模生产这种新杀虫剂，美国使馆便能乐观地报告：尽管“埃及目前的杀虫剂消费量，低于普通农民受到更好教育时本可达到的水平”，但随着教育程度提高，“对这些产品的需求应当增长”。[106]需求果然增长了。人们很快发现，标准剂量的杀虫剂已不起作用。DDT 一直在杀死自然天敌，于是在药剂作用下存活的害虫得以爆发式繁殖。用量不得不先翻倍，再增至三倍。政府无需为农民的进一步教育花一分钱，杀虫剂的使用便通过这种化学品自身尚未被充分理解的力量不断推进。

By the time of Nasser’s military coup two years later, the government had decided to build the pesticide factory itself using the assistance of international healthcare agencies. One week after the coup, on August 2, 1952, in perhaps its first international act, the new government signed an agreement with WHO and UNICEF to build a factory at Kafr Zayat that would produce two hundred tons a year of finished DDT.[105] With healthcare agencies financing this mass production of the new pesticide, the U.S. Embassy was able to report with optimism that while “Egypt consumes at present less pesticides than would be the case if the average farmer were better educated,” as his education progressed, “increased demand for such products should develop.”[106] Demand did indeed develop. Standard doses of pesticide were soon found to be ineffective. The DDT had been killing off natural predators, so the pests that survived the chemical were able to reproduce explosively. The quantities used had to be doubled and then trebled. Without the government having to spend a penny on further education for farmers, but through the working of the chemical’s own poorly understood powers, the use of pesticides progressed.

如今，冈比亚按蚊已经从埃及政治的故事中消失。甚至我们拥有的那部有关疟疾疫情的出色叙述——南希·加拉格尔的著作——也没有给蚊子或它的寄生虫多少功劳。和对这类政治的其他一切解释一样，历史中的行动者数量有限，而从南方到来的昆虫并不在其中。有英国人，一面操纵埃及政治，一面抵抗美国人在战后开始取代其角色；有民族精英——君主制与人数不多的土地贵族——其权力逐渐落入更具活力的商业地主、企业家和军官阶层手中；偶尔，还有底层社群——乡村人口、城市工人阶级、妇女——构成社会秩序的其余部分。蚊子却被说成属于自然。它不能说话。

Today the *Anopheles gambiae* has disappeared from the story of Egyptian politics. Even the one good account we have of the malaria epidemic, by Nancy Gallagher, does not give the mosquito or its parasite much credit. As in every other explanation of this kind of politics, history has a limited number of actors, and the insect arriving from the south is not one of them. There are the British, manipulating Egyptian politics while resisting the incipient postwar usurpation of their role by Americans; there are the national elites—the monarchy and the small landed aristocracy—losing their power to a more dynamic class of commercial landowners, entrepreneurs, and military officers; and, now and again, there are the subaltern communities—the rural population, the urban working classes, women—making up the rest of the social order. The mosquito, on the other hand, is said to belong to nature. It cannot speak. p. 50

作为自然的一部分，冈比亚按蚊成了公共卫生问题。在蚊子的帮助下，卫生、疾病、住房和无知这些问题，成为处理埃及乡村处境的主要方式。国家政治围绕改善健康、乡村重建、技术发展的计划展开，尤其围绕尼罗河的工程改造，以及把它的力量转化为电力、肥料、灌溉和农业与制造业增长的计划展开。自然的资源与限度，进而乡村社会的资源与限度，都将通过技术发展的能动活动而改变；而这要求运用科学和社会科学的专门知识。

As part of nature, the gambiae mosquito became a problem of public health. With the mosquito's help, questions of hygiene, disease, housing, and ignorance emerged as the principal way of addressing the situation of rural Egypt. National politics was organized around programs of health improvement, rural reconstruction, technical development, and above all the engineering of the river Nile and the transformation of its power into electricity, fertilizers, irrigation, and the growth of agriculture and manufacturing. The resources and limits of nature and, by extension, of rural society were to be transformed by the dynamic activity of technical development, which required the application of scientific and social scientific expertise.

这些项目开始把世界安排成这样一个世界：科学与自然相对立，技术专长声称能够克服社会改良的障碍。根除疟疾的运动提供了一个机会，让医学科学的智慧，连同其化学、卫生、过往经验和全球信息资源，作用于昆虫媒介、原生动物寄生虫、发热、贫困和营养不良；这些构成了物质世界的缺陷，必须被击败。在灌溉项目中，技术援助与工程的力量将克服自然资源的限制。在阿拉曼——技术化战争的第一场大战——据说，两名敌对的将军结合机械化武器的机动力与大规模布雷的新部署，决定了历史进程。这些计划与运动制造出一个世界，呈现为自然资源对技术、身体对卫生、人对机器、河流对人类才智。

These projects began to arrange the world as one in which science was opposed to nature and technical expertise claimed to overcome the obstacles to social improvement. The malaria eradication campaign presented an opportunity to bring the intelligence of medical science, with its resources of chemistry, hygiene, past experience, and worldwide information, to work upon the insect vectors, protozoan parasites, fevers, poverty, and malnutrition that made up the defects of the material world and had to be defeated. In irrigation projects, the power of technical assistance and engineering was to overcome the limits of natural resources. At al-Alamein, the first great battle of technicized warfare, two opposing generals, it is said, combined the mobile powers of mechanized weapons and the new, large-scale deployment of mines to determine the course of history. Such programs and campaigns manufactured a world that appeared as natural resources versus technology, bodies versus hygiene, men versus machines, the river versus human ingenuity. p. 51

然而，产生这个二元世界的项目，只有调用一系列其他逻辑、力量和化学作用才能出现：尼罗河的水力能量、氨的化学性质、按蚊的摄食模式、洛克菲勒体系中一位流行病学家的职业经营、维系战时军队的补给线、疟原虫的繁殖周期、埃及民族主义的反殖民斗争、世界对糖日益增长的化学性依赖，以及 DDT 对脂肪组织的偏好——这里只列举了其中一些。尽管技术发展把世界描绘成被动的，描绘成

Yet the projects that produced this binary world could emerge only by engaging a series of other logics, forces, and chemistries: the hydraulic energy of the Nile River, the chemical properties of ammonia, the feeding patterns of the anopheles mosquito, the career making of a Rockefeller epidemiologist, the supply lines sustaining an army at war, the reproductive cycle of the plasmodium parasite, the anticolonial struggle of Egyptian

有待克服的自然或有待开发的物质资源，但科学与发展的关系，只有与这些力量一道工作才能形成。

nationalism, the world's increasing chemical addiction to sugar, and DDT's preference for fatty tissue, to name a few. Although technical development portrayed the world as passive, as nature to be overcome or material resources to be developed, the relations of science and development came into being only by working with such forces.

所谓资本主义的发展，也是如此。阿布德试图控制并转化为利润来源的回路，涉及家族网络、糖和硝酸盐的性质、收割甘蔗者的劳动、帝国关系，以及战争带来的短缺。只有在这些其他力量与储备之中运作，并使其转变，利润或剩余价值的生产才会发生。因此，“资本主义发展”这样的用语，涵盖了一系列能动作用、逻辑、连锁反应与偶然互动；资本自身特定的回路与关系，只构成其中一部分。

The same was true of what was called the development of capitalism. The circuits that ʿAbbud tried to control and transform into sources of profit involved family networks, the properties of sugar and nitrates, the labor of those harvesting cane, imperial connections, and the shortages brought by war. The production of profit, or surplus value, came about only by working within and transforming such other forces and reserves. Thus a term like "capitalist development" covers a series of agencies, logics, chain reactions, and contingent interactions, among which the specific circuits and relations of capital formed only a part.

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引入这些其他力量，并不是要描述自然或物质条件的抵抗，也不是要承认那些对抗人类专门知识、或阻碍技术进步与资本主义发展的非人力量。描述阿斯旺大坝的问题、根除疟疾的挫折，或技术援助计划失败的报告，常常用这类说法来表达人的意图与经验世界之间的艰难关系。然而，专门知识并不是先已完整形成，然后才在外部遭遇这种抵抗；资本的力量也不是。计划、意图、科学专门知识、技术权力和剩余价值，都是与这些其他力量或要素结合而创造出来的。筑坝技术形成于阿斯旺工地，也形成于此前和此后的工程。巴西与埃及发展出的灭蚊方法，是在特定地点、在一群新的人类宿主之中同冈比亚按蚊打交道的结果。所谓自然或物质世界，像疟原虫一样进出于人的形态之中，或者像尼罗河一样，作为同时具有社会性与自然性、技术性与物质性的安排而存在。技术政治从中涌现的世界，是理性、力量、想象和资源一种先在而未获化解的组合。观念与技术并非先于这种混合物而存在，再以纯粹思想的形式作用于混乱的现实世界。它们从混合物中涌现，并在过程本身之中被制造出来。

Introducing these other forces is not a question of describing the resistance of nature or material conditions. It is not a matter of acknowledging nonhuman forces that worked against human expertise or created obstacles to technical progress and capitalist development. The reports describing the problems of the Aswan Dam, the setbacks in malaria eradication, or the failure of technical assistance programs often used such formulations to express the difficult relationship between human intention and the world of experience. Expertise, however, did not confront such resistance externally, after it was already complete, nor did the power of capital. Plans, intentions, scientific expertise, techno-power, and surplus value were created in combination with these other forces or elements. The technology of dam construction was formed at the construction site in Aswan, and in earlier and subsequent projects. The methods of mosquito eradication developed in Brazil and Egypt were the outcome of working with *Anopheles gambiae* in particular locations, among a new population of human hosts. What is called nature or the material world moves, like the plasmodium, in and out of human forms, or occurs as arrangements, like the river Nile, that are social as well as natural, technical as well as material. The world out of which techno-politics emerged was an unresolved and prior combination of reason, force, imagination, and resources. Ideas and technology did not precede this mixture as pure forms of thought brought to bear upon the messy world of reality. They emerged from the mixture and were manufactured in the processes themselves.

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把这些过程归结为理性对力量、智识对自然，或想象对实在，会误解其中的复杂性。但这种误解又是必要的，因为技术权力的生产恰恰如此进行。忽视事情以混合方式发生，甚至制造出理性与现实世界、观念与其对象、人和非人分属整齐分隔领域的效果，正是权力在埃及、乃至整个 20 世纪开始发挥作用的方式。

Resolving these processes into reason versus force, intelligence versus nature, or the imagined versus the real misapprehends the complexity. But this misapprehension was necessary, for it was exactly how the production of techno-power proceeded. Overlooking the mixed way things happen, indeed producing the effect of neatly separate realms of reason and the real world, ideas and their objects, the human and the nonhuman, was how power was coming to work in Egypt, and in the twentieth century in general.

社会科学通过把特定事件同普遍理性联系起来，并把人类能动性视为既定事实，模仿了这种权力形式。常规分析方法最终受它所制造的效果迷惑，再生产了这种权力。事实上，社会科学帮助把世界格式化为这种二元秩序，从而构成并巩固人们对能动性与专门知识的体验。在社会科学的许多领域，这完全是有意为之。它试图在智识上掌握社会过程，就像大坝似乎能够掌握河流、人工硝酸盐似乎能够掌握甘蔗生产、DDT 似乎能够掌握节肢动物一样。是否理解事物如何运作，变得不那么重要；眼前结果是否有效，反而更重要。但更为审慎的历史或文化分析，也可能以不那么明显的方式做同样的事：不去审视技术，或把那些显然不止是社会性的事物，说成是“社会建构”的。

Social science, by relating particular events to a universal reason and by treating human agency as given, mimics this form of power. The normal methods of analysis end up reproducing this kind of power, taken in by the effects it generates. In fact, social science helps to format a world resolved into this binary order, and thus to constitute and solidify the experience of agency and expertise. In much of social science this is quite deliberate. It tries to acquire the kind of intellectual mastery of social processes that dams seem to offer over rivers, artificial nitrates over sugarcane production, or DDT over arthropods. It is less important whether one understands how things work, more important how effective are the immediate results. But more careful forms of historical or cultural analysis can do the same thing in less obvious ways, by leaving technics unexamined, or talking about the “social construction” of things that are clearly more than social.

质疑这些区分，以及它们使之成为可能的关于能动性与历史的预设，并不意味着要引入无限多的行动者和网络，并把它们都视为具有同等意义和力量。相反，这意味着让权力和能动性本身成为问题，而不是预先已知的答案。这意味着承认某种无法化解的张力、不可分割的混合、难以成立却又存在的多重性，意图和专门知识必须从中涌现。它要求承认，人类能动性同资本一样，是一种技术性构成体，是被制造出来的东西。我们不必援引理性、自利、科学或资本的力量和逻辑，把世界上发生的事情归因于这些被赋予魔力的力量与过程的运作；而可以像我在此尝试的那样，打开这样一个问题：究竟是哪些混合的能动作用、联结、互动与暴力形式，能够把自己的行动描绘成历史，描绘成人类专门知识克服自然、理性与现代性的进步，或资本主义的扩张与发展？

To put in question these distinctions, and the assumptions about agency and history that they make possible, does not mean introducing a limitless number of actors and networks, all of which are somehow of equal significance and power. Rather, it means making this issue of power and agency a question, instead of an answer known in advance. It means acknowledging something of the unresolvable tension, the inseparable mixture, the impossible multiplicity, out of which intention and expertise must emerge. It requires acknowledging that human agency, like capital, is a technical body, is something made. Instead of invoking the force and logic of reason, self-interest, science, or capital and attributing what happens in the world to the working of these enchanted powers and processes, we can open up the question, as I have attempted here, of what kinds of hybrid agencies, connections, interactions, and forms of violence are able to portray their actions as history, as human expertise overcoming nature, as the progress of reason and modernity, or as the expansion and development of capitalism.

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章末原注 / Endnotes

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↑1. Jill Edwards 编，《重访阿拉曼：阿拉曼战役及其历史意义》（Al-Alamein Revisited: The Battle of Al-Alamein and Its Historical Implications；Cairo: American University in Cairo Press, 2000），第 41 页。阿拉曼与其说是决定性的转折点，不如说显示了双方相对实力的变化，而这些变化由直接战场之外的因素决定。同书，第 55—98 页。

1. Jill Edwards, ed., *Al-Alamein Revisited: The Battle of Al-Alamein and Its Historical Implications* (Cairo: American University in Cairo Press, 2000), 41. Al-Alamein may have been less a decisive turning point than an indicator of changes in the relative strengths of the two sides determined by factors outside the immediate battlefield. Ibid., 55–98

↑2. 阿拉曼是大规模地雷战新方法的试验场，其结果往往无法预测，尤其因为地雷并不始终受布雷者控制。双方都曾在不同时期把对方的雷区纳入自己的防御体系。美国国务院政治军事事务局，《隐蔽的杀手：全球地雷危机》（“Hidden Killers: The Global Landmine Crisis”），1995 年 1 月 27 日。地雷数量见 http://www.icbl.org/resources/mideast4 及 Middle East Times，1999 年 8 月 20 日；其对当地社区的影响见 Lila Abu-Lughod，《书写妇女的世界：贝都因人的故事》（Writing Women’s Worlds: Bedouin Stories；Berkeley: University of California Press, 1993），第 56—65 页。

2. Al-Alamein was the testing ground for the new methods of large-scale mine warfare, which were often unpredictable, especially since the mines did not remain under the control of those who laid them. Each side at various times incorporated the other’s minefields in its own defenses. U.S. Department of State, Bureau of Political-Military Affairs, “Hidden Killers: The Global Landmine Crisis,” Jan. 27, 1995. On the number of mines see http://www.icbl.org/resources/mideast4 and *Middle East Times,* Aug. 20, 1999; on their impact on the local community see Lila Abu-Lughod, *Writing Women’s Worlds: Bedouin Stories* (Berkeley: University of California Press, 1993), 56–65.

↑3. 疟疾专家区分四种疟疾，由疟原虫属（Plasmodium）中的四种原生动物引起：常常致命的 Plasmodium gambiae，以及危险性较低的间日疟原虫（P. vivax）、三日疟原虫（P. malariae）和卵形疟原虫（P. ovale）。在已确认的 380 种按蚊中，约 60 种能把疟原虫传播给人类。

〔校注：底本清楚印为 Plasmodium gambiae，疑将冈比亚按蚊（Anopheles gambiae）与恶性疟原虫（Plasmodium falciparum）混淆；本章正文原 19 页使用的是后者。CDC DPDx 的疟疾病原说明支持这一辨别。此处保留错误名称以便对勘；物种数量亦保留底本年代的说法。〕

3. Malaria specialists recognize four forms of malaria, caused by four species of protozoans in the genus *Plasmodium:* the often lethal *Plasmodium gambiae* and the less dangerous *P. vivax, P. malariae,* and *P. ovale.* The plasmodium parasite is transmitted to humans by about 60 out of the 380 recognized species of anopheline mosquito.

↑4. 我对疫情历史的叙述，主要取自南希 • E. 加拉格尔内容丰富的著作《埃及的其他战争：流行病与公共卫生政治》（Nancy E. Gallagher, Egypt’s Other Wars: Epidemics and the Politics of Public Health；Syracuse, N.Y.: Syracuse University Press, 1990），第 20—95 页。在阅读了桑德拉 • 苏菲安关于巴勒斯坦疟疾与犹太复国主义政治的研究后，我对疟疾问题的兴趣进一步扩展。见 Sandra Sufian，《医治土地与民族：托管时期巴勒斯坦的疟疾与犹太复国主义工程，1920—1947》（“Healing the Land and the Nation: Malaria and the Zionist Project in Mandatory Palestine, 1920–1947”），纽约大学博士论文，1999 年。

4. I have taken the history of the epidemic mostly from the informative account in Nancy E. Gallagher, *Egypt’s Other Wars: Epidemics and the Politics of Public Health* (Syracuse, N.Y.: Syracuse University Press, 1990), 20 – 95. My interest in the question of malaria was widened after reading the study of malaria and Zionist politics in Palestine by Sandra Sufian, “Healing the Land and the Nation: Malaria and the Zionist Project in Mandatory Palestine, 1920 – 1947,” Ph. D. diss., New York University, 1999.

↑5. 我更改了村中人物的姓名。

5. I have changed the names of people from the village.

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↑6. 第一座阿斯旺大坝建于 1898—1902 年，1907—1912 年加固并加高，1929—1933 年再次加高。Willcocks 与 Craig，《埃及灌溉》（Egyptian Irrigation），两卷本，第 3 版（London: E. & F. N. Spon, 1913），第 2 卷，第 718—758 页。

6. The first Aswan dam was built in 1898 – 1902, strengthened and raised in height in 1907–12, and raised again in 1929–33. Willcocks and Craig, *Egyptian Irrigation,* 2 vols., 3rd ed. (London: E. & F. N. Spon, 1913), 2:718–58.

↑7. Doreen Warriner，《中东的土地与贫困》（Land and Poverty in the Middle East；Westport, Conn.: Hyperion Press, 1948），第 30—31 页。

7. Doreen Warriner, *Land and Poverty in the Middle East* (Westport, Conn.: Hyperion Press, 1948), 30–31.

↑8. Charles Issawi，《革命中的埃及：经济分析》（Egypt in Revolution: An Economic Analysis；Ox-

8. Charles Issawi, *Egypt in Revolution:An Economic Analysis* (Oxford: Oxford University Press, 1963), 35.

ford: Oxford University Press, 1963），第 35 页。

↑9. 埃及是德国合成硝酸盐最大的出口市场。Robert Vitalis，《当资本家发生冲突：埃及的商业冲突与帝国终结》（When Capitalists Collide: Business Conflict and the End of Empire in Egypt；Berkeley: University of California Press, 1995），第 88—89 页。

9. Egypt was Germany's largest export market for synthetic nitrates. Robert Vitalis, *When Capitalists Collide: Business Conflict and the End of Empire in Egypt* (Berkeley: University of California Press, 1995), 88–89.

↑10. 战前实施的措施规定，地主种植棉花的面积最多只能占其土地的 50%。战争期间，政府根据地区不同，把上限降至 16% 至 22%，并在上埃及部分地区完全禁止种棉。作物减产也被部分归因于轮作周期被打乱。Jean Anhouri，《战争对埃及农业的影响》（"Les Répercussions de la guerre sur l'agriculture égyptienne"），L'Egypte Contemporaine 38，第 238—239 期（1947 年 3—4 月）：第 233—251 页，所引见第 241、251 页。

10. Measures introduced before the war had limited landowners to growing cotton on a maximum of 50 percent of their land. During the war the government lowered the limit to between 16 and 22 percent, according to the region, and banned cotton cultivation altogether in parts of Upper Egypt. The decline in crop yields was also attributed in part to the disruption of rotation cycles. Jean Anhouri, "Les Répercussions de la guerre sur l'agriculture égyptienne," *L'Egypte Contemporaine* 38, nos. 238–39 (Mar.–Apr. 1947): 233–51, at 241, 251.

↑11. 从 1935—1939 年到 1942—1945 年，甘蔗的平均每英亩产量下降了 30%；但尽管单产下降，战争结束时的总产量仍高于前一时期，反映出用于种糖作物的面积增加了（Anhouri，《战争的影响》，第 241 页）。棉花是夏季作物，同一块土地冬季还可种植谷物、豆类或饲料；甘蔗则全年占地，无法与谷物生产相结合。

11. Average sugarcane yields per acre dropped by 30 percent between 1935–39 and 1942–45, yet despite the drop in yields total production at the end of the war was higher than in the preceding period, reflecting the increased acreage devoted to sugar (Anhouri, "Les Répercussions de la guerre," 241). Cotton is a summer crop and allows the cultivation of grain, pulses, or fodder on the same land in the winter, whereas cane occupies the ground year-round and cannot be combined with grain production.

↑12. Gallagher，《埃及的其他战争》，第 83 页。

12. Gallagher, *Egypt's Other Wars,* 83.

↑13. 同书，第 33 页。

13. Ibid., 33.

↑14. 关于大型水坝的影响，见世界水坝委员会，《水坝与发展：决策的新框架》（World Commission on Dams, Dams and Development: A New Framework for Decision Making，2000），可见 http://www.damsreport.org 。

14. On the impact of large dams see World Commission on Dams, *Dams and Development:A New Framework for Decision Making* (2000), available at http://www.damsreport.org.

↑15. Mike Samuels 与 Hal Zina Bennett，《健康的身体，健康的地球》（Well Body, Well Earth；San Francisco: Sierra Club, 1985）。

15. Mike Samuels and Hal Zina Bennett, *Well Body, Well Earth* (San Francisco: Sierra Club, 1985).

↑16. 据估计，它每年导致一百万至二百万人死亡，并可能是另外约一百万人死亡的促成因素。见 http://www.who.int/ctd/html/malaria.html 。

16. It was estimated to cause one to two million deaths a year, and to be a contributing factor in perhaps another million. See http://www.who.int/ctd/html/malaria.html.

↑17. William H. McNeill，《瘟疫与人》（Plagues and Peoples；Garden City, N.Y.: Anchor Press / Doubleday, 1976），第 279—280 页。失去爪哇促使人们开发奎宁的合成替代品。

17. William H. McNeill, *Plagues and Peoples* (Garden City, N.Y.: Anchor Press / Doubleday, 1976), 279–80. The loss of Java spurred the development of synthetic alternatives to quinine.

↑18. 除虫菊喷雾由德国科学家 G. Giemsa 于 1910—1913 年发明并作为杀虫剂试验，但直到 1929—1935 年南非疫情期间，它才被用于应对疟疾流行。Gordon Harrison，《蚊子、疟疾与人：1880 年以来的交战史》（Mosquitoes, Malaria and Man: A History of the Hostilities Since 1880；New York: E. P. Dutton, 1978），第 209—211 页。

18.The pyrethrum spray was invented and tested as an insecticide by a German scientist, G. Giemsa, in 1910–13, but was ignored in the treatment of malaria epidemics until the South African epidemic of 1929–35. Gordon Harrison, *Mosquitoes, Malaria and Man: A History of the Hostilities Since 1880* (New York: E. P. Dutton, 1978), 209–11.

↑19. Emil Ludwig，《尼罗河：埃及传奇之河的宏大故事——六千年的惊心历史》（The Nile: The Mighty Story of Egypt's Fabulous River—6,000 Years of Thrilling History），Mary H. Lindsay 译（New York: Viking Press, 1937），第 265 页。血吸虫病此前已在下

19. Emil Ludwig, *The Nile: The Mighty Story of Egypt's Fabulous River—6,000 Years of Thrilling History,* trans. Mary H. Lindsay (New York: Viking Press, 1937), 265. Schistosomiasis was already prevalent in Lower Egypt, brought there by the earlier development

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埃及流行，是更早发展的常年灌溉将其带到那里。阿斯旺大坝把这种疾病带到上埃及，感染率达到 80% 甚至更高。自 20 世纪 20 年代起，卫生当局开始大规模注射治疗血吸虫病。所用针头虽然按照当时认为足够的程序消毒，却传播了丙型肝炎，一种可能导致肝衰竭的病毒性感染。到 20 世纪 90 年代，多达 20% 的人口检测出这种慢性病阳性。The Lancet，2000 年 3 月 11 日。

of perennial irrigation. The Aswan Dam brought the disease to Upper Egypt, where infection rates reached 80 percent or more. From the 1920s, health authorities began mass injection campaigns against schistosomiasis. The needles they used, although disinfected with procedures then considered sufficient, spread hepatitis C, a viral infection that can lead to liver failure. By the 1990s up to 20 percent of population tested positive for the chronic illness. The *Lancet,* Mar. 11, 2000.

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†20. Gallagher，《埃及的其他战争》，第 18 页。这一承认并非认真提出，因为报告又补充说，尽管幸存者会更健康，但一半人口将死于饥饿。然而，大坝的主要目的，是支持工业作物——棉花和甘蔗——而非粮食生产；因此，恢复盆地灌溉、种植更多小麦、蚕豆和鹰嘴豆，受到损害的并不会是乡村人口。

20. Gallagher, *Egypt's Other Wars,* 18. The acknowledgment was not made seriously, for the report added that although the survivors would be healthier, half the population would die of hunger. The main purpose of the dams, however, was to support industrial crops (cotton and sugarcane) rather than food production, so it was not the rural population who would have suffered from a return to basin irrigation and the growing of more wheat, beans, and chickpeas.

†21. 同书，第 24 页。

21. Ibid., 24.

†22. 同书，第 33 页。

22. Ibid., 33.

†23. Luis Camargo，《回复：疟疾治疗期间的甘蔗汁》（“Re: Sugar cane juice during malaria treatment”），1997 年 5 月 20 日发表于疟疾讨论邮件列表，http://www.wehi.edu.au/MalDB-www/discuss/listserv.html。古巴研究过甘蔗与血流相关的药理性质，当地制药业从甘蔗中开发出一种药物 policosanal（PPG），用来对抗动脉硬化。

23. Luis Camargo, “Re: Sugar cane juice during malaria treatment,” message posted May 20, 1997, on the Malaria Discussion List, http://www.wehi.edu.au/MalDB-www/discuss/listserv.html. The pharmacological properties of sugarcane in relation to blood flow were investigated in Cuba, where the pharmaceutical industry developed a drug from sugarcane, policosanal (PPG), to combat arteriosclerosis.

†24. Gallagher，《埃及的其他战争》，第 25、38 页。

24. Gallagher, *Egypt's Other Wars,* 25, 38.

†25. McNeill，《瘟疫与人》，第 280—282 页。

25. McNeill, *Plagues and Peoples,* 280–82.

†26. 关于索珀在巴西的运动，见 R. M. Packard 与 P. Gadelha，《蚊子遍地的土地》（“A Land Filled with Mosquitoes”），Parassitologia 36（1994）：第 197—213 页。

26. On Soper's Brazil campaign see R.M. Packard and P. Gadelha, “A Land Filled with Mosquitoes.” *Parassitologia* 36 (1994), 197–213.

†27. Gallagher，《埃及的其他战争》，第 27 页。

27. Gallagher, *Egypt's Other Wars,* 27.

†28. 基勒恩勋爵（迈尔斯 • 兰普森爵士），《日记》，私人文书收藏，牛津大学圣安东尼学院中东中心，1944 年 1 月 14 日条目。

28. Lord Killearn (Sir Miles Lampson), *Diaries,* Private Papers Collection, Middle East Centre, St. Antony's College, Oxford, entry dated Jan. 14, 1944.

†29. Harrison，《蚊子、疟疾与人》，第 220—222 页；Gallagher，《埃及的其他战争》，第 28—31、77—95 页。

29. Harrison, *Mosquitoes, Malaria and Man,* 220–22; Gallagher, *Egypt's Other Wars,* 28–31, 77–95.

†30. Bruno Latour，《我们从未现代过》（We Have Never Been Modern；Cambridge, Mass.: Harvard University Press, 1993），第 1 页。

30. Bruno Latour, *We Have Never Been Modern* (Cambridge, Mass.: Harvard University Press, 1993), 1.

†31. 正如迪佩什 • 查克拉巴蒂所说，欧洲是“一切历史中居于主权地位的理论主体”。见《后殖民性与历史的人为构造：谁为“印度”的过去发言？》（“Postcoloniality and the Artifice of History: Who Speaks for ‘Indian’ Pasts?”），Representations 37，第 1 期（1992）：第 1—26 页，所引见第 1—3 页。另见 Dipesh Chakrabarty，《资本的两种历史》（“The Two Histories of Capital”），载《将欧洲地方化：后殖民思想与历史差异》（Provincializing Europe: Postcolonial Thought and Historical Difference；Princeton, N.J.: Princeton University Press, 2000），第 47—71 页；

31. Europe, as Dipesh Chakrabarty says, is “the sovereign, theoretical subject of all histories.” “Postcoloniality and the Artifice of History: Who Speaks for ‘Indian’ Pasts?” *Representations* 37, no. 1 (1992): 1–26, at 1–3. See also Dipesh Chakrabarty, “The Two Histories of Capital,” in *Provincializing Europe: Postcolonial Thought and Historical Difference* (Princeton, N.J.: Princeton University Press, 2000), 47–71; Gyan Prakash, “Can the ‘Subaltern’ Ride? A Reply to O’Hanlon and Washbrook,” *Comparative Studies in Society and History* 34, no. 1 (1992): 168–84; and Gaya-

Gyan Prakash，《“底层”能骑乘吗？答奥汉隆与沃什布鲁克》（“Can the ‘Subaltern’ Ride? A Reply to O’Hanlon and Washbrook”），Comparative Studies in Society and History 34，第 1 期（1992）：第 168—184 页；Gayatri Chakravorty Spivak，《底层能说话吗?》（“Can the Subaltern Speak?”），载 C. Nelson 与 L. Grossberg 编，《马克思主义与文化阐释》（Marxism and the Interpretation of Culture；Basingstoke: Macmillan Education, 1988），第 271—313 页，后重印于 Patrick Williams 与 Laura Chrisman 编，《殖民话语与后殖民理论》（Colonial Discourse and Post-Colonial Theory；New York: Columbia University Press, 1994），第 66—111 页。

tri Chakravorty Spivak, “Can the Subaltern Speak?” in *Marxism and the Interpretation of Culture,* ed. C. Nelson and L. Grossberg (Basingstoke: Macmillan Education, 1988), 271–313, reprinted in *Colonial Discourse and Post-Colonial Theory,* ed. Patrick Williams and Laura Chrisman (New York: Columbia University Press, 1994), 66–111.

↑32. Emile Durkheim，《社会学方法的准则》（The Rules of Sociological Method；New York: Free Press, 1938），第 xlvi 页。关于非人能动性以及预先知道主角是谁的问题，见 Bruno Latour，《法国的巴斯德化》（The Pasteurization of France；Cambridge, Mass.: Harvard University Press, 1988）；Donna Haraway，《谦逊见证者@第二千年：女性人 © 遇见癌鼠 ™——女性主义与技术科学》（Modest-Witness@SecondMillennium.FemaleMan© Meets OncoMouse™: Feminism and Technoscience；New York: Routledge, 1997）。

32. Emile Durkheim, *The Rules of Sociological Method* (New York: Free Press, 1938), xlvi. On the question of nonhuman agency and knowing the protagonists in advance see Bruno Latour, *The Pasteurization of France* (Cambridge, Mass.: Harvard University Press, 1988); and Donna Haraway, *Modest-Witness@SecondMillennium. FemaleMan© Meets OncoMouseTM: Feminism and Technoscience* (New York: Routledge, 1997).

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↑33. Karl Marx，《资本论：政治经济学批判》第 1 卷《资本主义生产过程》（Capital: A Critique of Political Economy, vol. 1, The Process of Capitalist Production；New York: Modern Library, 1906），第 170 页。

33. Karl Marx, *Capital: A Critique of Political Economy,* vol. 1, *The Process of Capitalist Production* (New York: Modern Library, 1906), 170.

↑34. Jacques Derrida，《马克思的幽灵：债务状况、哀悼工作与新国际》（Specters of Marx: The State of the Debt, the Work of Mourning, and the New International；New York: Routledge, 1994），第 170 页。

34. Jacques Derrida, *Specters of Marx: The State of the Debt, the Work of Mourning, and the New International* (New York: Routledge, 1994), 170.

↑35. 我对阿布德的叙述，主要取自维塔利斯以阿布德为个案、研究这一时期埃及商业与政治的出色修正主义著作。1955 至 1963 年，政府通过一系列国有化措施接管了阿布德的商业帝国。1961 年 10 月，政府查封 168 名“反动资本家”的财产时，没收了阿尔曼特庄园的剩余部分。阿布德于 1963 年 12 月在伦敦去世（Vitalis，《当资本家发生冲突》，第 210—214 页）。我 1997 年访问该庄园时，最后一次查封已被撤销，部分财产已经归还。

35. My account of ʿAbbud is drawn largely from Vitalis’s fine revisionist study of Egyptian business and politics in this period, based upon the case of ʿAbbud. The government took control of ʿAbbud’s business empire in a series of nationalizations between 1955 and 1963. It seized the remainder of the Armant estate in October 1961, when it sequestered the property of 168 “reactionary capitalists.” ʿAbbud died in London in December 1963 (Vitalis, *When Capitalists Collide,* 210–14). I visited the estate in 1997, by which time the last sequestration had been reversed and some of the property returned.

↑36. 在伊拉克，阿布德曾于 1913 年与威廉·威尔科克斯爵士合作修建幼发拉底河大坝。Eric Davis，《挑战殖民主义：米斯尔银行与埃及工业化，1920—1941》（Challenging Colonialism: Bank Misr and Egyptian Industrialization, 1920–1941；Princeton, N.J.: Princeton University Press, 1983），第 152—153 页。

36. In Iraq, ʿAbbud worked with Sir William Willcocks on the construction of the Euphrates Dam in 1913. Eric Davis, *Challenging Colonialism: Bank Misr and Egyptian Industrialization, 1920–1941* (Princeton, N.J.: Princeton University Press, 1983), 152–53.

↑37. 关于 20 世纪 40 年代的商业政治，另见 Eric Davis，《挑战殖民主义》；Robert L. Tignor，《埃及的国家、私人企业与经济变迁，1918—1952》（The State, Private Enterprise, and Economic Change in Egypt, 1918–1952；Princeton, N.J.: Princeton University Press, 1984）及《埃及纺织业与英国资

37. On the business politics of the 1940s see also Eric Davis, *Challenging Colonialism,* and Robert L.Tignor, *The State, Private Enterprise, and Economic Change in Egypt, 1918–1952* (Princeton, N.J.: Princeton University Press, 1984), and *Egyptian Textiles and British Capital, 1930–1956* (Cairo: American University in Cairo

本，1930—1956》（Egyptian Textiles and British Capital, 1930–1956；Cairo: American University in Cairo Press, 1989）。关于更广泛阶级政治的研究包括：ʿAbd al-ʿAzim Ramadan，《埃及的阶级斗争（1837—1952）》（Siraʿ al-tabaqat fi misr (1837–1952)；Beirut: al-Muʾassasa al-ʿArabiyya li-ʾl-Dirasat wa-ʾl-Nashr, 1978；第 75 页给出了从“福尔图纳斯伯爵”处购得的阿尔曼特庄园面积）及《埃及民族运动的发展：1937 年至 1948 年》（Tatawwur al-haraka al-wataniyya fi misr: min sanat 1937 ila sanat 1948；Cairo: al-Hayʾa al-Misriyya al-ʿAmma li-ʾl-Kitab, 1998）；ʿAsim al-Disuqi，《大农业地主及其在埃及社会中的作用》（Kibar mullak al-aradi al-ziraʿiyya wa dawruhum fi al-mujtamaʿ al-misri；Cairo: Dar al-Thaqafa al-Jadida, 1976）；Magda Baraka，《两次革命之间的埃及上层阶级，1919—1952》（The Egyptian Upper Class Between Revolutions, 1919–1952；Reading, Eng.: Ithaca Press, 1998）。

Press, 1989). Studies of the broader class politics include ʿAbd al-ʿAzim Ramadan, *Siraʿ al-tabaqat fi misr (1837–1952)* (Beirut: al-Muʾassasa al-ʿArabiyya li-ʾl-Dirasat wa-ʾl-Nashr, 1978) (who gives the size of the Armant estate, purchased from “Count Fortunas,” on p. 75) and *Tatawwur al-haraka al-wataniyya fi misr: min sanat 1937 ila sanat 1948* (Cairo: al-Hayʾa al-Misriyya al-ʿAmma li-ʾl-Kitab, 1998); ʿAsim al-Disuqi, *Kibar mullak al-aradi al-ziraʿiyya wa dawruhum fi al-mujtamaʿ al-misri* (Cairo: Dar al-Thaqafa al-Jadida, 1976); and Magda Baraka, *The Egyptian Upper Class Between Revolutions, 1919 – 1952* (Reading, Eng.: Ithaca Press, 1998).

↑38. 七十米高的伊莫比利亚大楼于 1940 年 1 月竣工，内有 38 间商铺、82 间办公室和 218 套公寓。Mercedes Volait，《埃及现代建筑与〈建筑〉杂志，1939—1958》（L’Architecture moderne en Egypte et la revue Al-ʿImara, 1939–1958），Dossiers du CEDEJ 1987，第 4 号（Cairo: Centre d’Etudes et de Documentation Economique, Juridique et Sociale, 1988），第 62—63 页；Arthur Goldschmidt Jr.，《现代埃及传记辞典》（Biographical Dictionary of Modern Egypt；Boulder, Colo.: Lynne Rienner, 2000），第 4 页。

38. The seventy-meter-tall Immobilia Building, completed in January 1940, contained 38 shops, 82 offices, and 218 apartments. Mercedes Volait, *L’Architecture moderne en Egypte et la revue Al-ʿImara,* 1939–1958, Dossiers du CEDEJ 1987, no. 4 (Cairo: Centre d’Etudes et de Documentation Economique, Juridique et Sociale, 1988), 62 – 63; Arthur Goldschmidt Jr., *Biographical Dictionary of Modern Egypt* (Boulder, Colo.: Lynne Rienner, 2000), 4.

↑39. Gallagher，《埃及的其他战争》，第 20、89、188 页。

39. Gallagher, *Egypt’s Other Wars,* 20, 89, 188.

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↑40. 被免职的部长马克拉姆 • 欧贝德，在被免职前一直是党内二号人物，也是阿布德的老对头。两个月后，他又被开除出党。接任财政部长的阿明 • 奥斯曼，是阿布德的密友。Vitalis，《当资本家发生冲突》，第 120—124 页；Yunan Labib Rizq，《华夫脱党与黑皮书》（al-Wafd wa-ʾl-kitab al-aswad；Cairo: Muʾassasat al-Dirasat al-Siyasiyya wa-ʾl-Istratijiya, 1978），第 17—122 页；Muhammad Husayn Haykal，《我的埃及政治回忆录》（Mudhakkirati fi al-siyasa al-misriyya），三卷本（Cairo: Dar al-Maʿarif, 1977–78），第 2 卷《1938 年 7 月 29 日至 1952 年 7 月 26 日：法鲁克时代》（Min 29 yuliya 1938 ila 26 yuliya 1952: ʿAhd Faruq），第 193—245 页。

40. Makram ʿUbayd, the dismissed minister and until his dismissal the number two man in the party, was an old nemesis of ʿAbbud. Two months later he was expelled from the party. His replacement as finance minister, Amin ʿUthman, was a close friend of ʿAbbud. Vitalis, *When Capitalists Collide,* 120 – 24; Yunan Labib Rizq, *al-Wafd wa-ʾl-kitab al-aswad* (Cairo: Muʾassasat al-Dirasat al-Siyasiyya wa-ʾl-Istratijiya, 1978), 17–122; Muhammad Husayn Haykal, *Mudhakkirati fi al-siyasa al-misriyya,* 3 vols. (Cairo: Dar al-Maʿarif, 1977 – 78), vol. 2, *Min 29 yuliya 1938 ila 26 yuliya 1952: ʿAhd Faruq,* 193–245.

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↑41. 基勒恩勋爵（迈尔斯 • 兰普森爵士），《日记》，私人文书收藏，牛津大学圣安东尼学院中东中心，1943 年 2 月 19 日至 3 月 30 日条目；Vitalis，《当资本家发生冲突》，第 122 页。

41. Lord Killearn (Sir Miles Lampson), *Diaries,* Private Papers Collection, Middle East Centre, St. Antony’s College, Oxford, entry dated Feb. 19 – Mar. 30, 1943; Vitalis, *When Capitalists Collide,* 122.

↑42. Vitalis，《当资本家发生冲突》，第 121 页。

42. Vitalis, *When Capitalists Collide,* 121.

↑43. 同书，第 126 页。

43. Ibid., 126.

↑44. 同书，第 119—128 页；Ramadan，《埃及的阶级斗争》，第 108—110 页。

44. Ibid., 119–28; Ramadan, *Siraʿ al-tabaqat fi misr,* 108–10.

↑45. Robert L. Tignor，《两次大战之间埃及的民族主义、经济规划与发展项目》（“Nationalism, Economic Planning, and Development Projects in Interwar Egypt”），International Journal of African Historical Studies 10，第 2 期（1977）：第 185—208 页，考察了两次大战之间埃及围绕灌溉和水电计划控制权展开的斗争。

45. Robert L. Tignor, “Nationalism, Economic Planning, and Development Projects in Interwar Egypt,” *International Journal of African Historical Studies* 10, no. 2 (1977): 185–208, examines the struggles over the control of irrigation and hydroelectric schemes in interwar Egypt.

↑46. 有一批重要的人类学文献打开了商品概念，考察商品如何参与经济学所界定范围之外更广泛的流转。见 Arjun Appadurai，《导论：商品与价值政治》（“Introduction: Commodities and the Politics of Value”），载 Arjun Appadurai 编，《物的文化生命》（The Cultural Life of Things；Cambridge: Cambridge University Press, 1986）；Nicholas Thomas，《纠缠的物：太平洋地区的交换、物质文化与殖民主义》（Entangled Objects: Exchange, Material Culture, and Colonialism in the Pacific；Cambridge, Mass.: Harvard University Press, 1991）。

〔校注：底本书名为 The Cultural Life of Things，照录。剑桥大学出版社 1986 年书目及扉页列出的实际书名是 The Social Life of Things: Commodities in Cultural Perspective（《物的社会生命：文化视角下的商品》）。〕

46. There is an important anthropological literature that opens up the concept of commodity to examine its involvement in wider circulations outside those defined by economics. See Arjun Appadurai,“Introduction: Commodities and the Politics of Value,” in *The Cultural Life of Things,* ed. Arjun Appadurai (Cambridge: Cambridge University Press, 1986), and Nicholas Thomas, *Entangled Objects: Exchange, Material Culture, and Colonialism in the Pacific* (Cambridge, Mass.: Harvard University Press, 1991).

↑47. 这一论点的经典表述见 Clifford Geertz，《深描》（“Thick Description”），载《文化的解释：论文选》（The Interpretation of Cultures: Selected Essays；New York: Basic Books, 1973）。

47. A classic statement of this argument is in Clifford Geertz, “Thick Description,” in *The Interpretation of Cultures: Selected Essays* (New York: Basic Books, 1973).

↑48. Willcocks 与 Craig，《埃及灌溉》，第 2 卷，第 677—678 页。威尔科克斯在 Sir William Willcocks，《美索不达米亚的灌溉》（Irrigation of Mesopotamia；London: E. & F. N. Spon, 1917）中扩展了这一论点，将盆地灌溉系统的优势，同拦河坝和蓄水库的新方法作了比较。

48. Willcocks and Craig, *Egyptian Irrigation,* 2:677–78. Willcocks extended this argument in Sir William Willcocks, *Irrigation of Mesopotamia* (London: E. & F. N. Spon, 1917), where he compared the advantages of the system of basin irrigation to the new methods of barrages and storage reservoirs.

↑49. Emil Ludwig，《尼罗河：一条河流的生命历程》（Der Nil: Lebenslauf Eines Stromes；Amsterdam: Querido Verlag, 1935–36）；英译本《尼罗河：埃及传奇之河的宏大故事——六千年的惊心历史》（The Nile: The Mighty Story of Egypt’s Fabulous River—6,000 Years of Thrilling History），Mary H. Lindsay 译（New York: Viking Press, 1937），第 7 页。

49. Emil Ludwig, *Der Nil: Lebenslauf Eines Stromes* (Amsterdam: Querido Verlag, 1935–36), English translation, *The Nile: The Mighty Story of Egypt's Fabulous River—6,000 Years of Thrilling History,* trans. Mary H. Lindsay (New York: Viking Press, 1937), 7.

↑50. Willcocks 与 Craig，《埃及灌溉》，第 2 卷，第 744—745 页。

50. Willcocks and Craig, *Egyptian Irrigation,* 2:744–45.

↑51. 关于研究行动中发生的科学，而非其结果，见 Latour，《法国的巴斯德化》及《潘多拉的希望：论科学研究的实在性》（Pandora’s Hope: Essays on the Reality of Science Studies；Cambridge, Mass.: Harvard University Press, 1999）。比起拉图尔的“事实拜物”（factish；《潘多拉的希望》，第 274 页），我更倾向于使用从德里达《马克思的幽灵》第 170 页借来的“人为制成的”（artifactual）。拉图尔在《我们从未现代过》中对德里达著作的简略否定，令人失望。关于自然 / 文化的区分，见 Jacques Derrida，《论文字学》（Of Grammatology），Gayatri Chakravorty Spivak 译（Baltimore, Md.: Johns Hopkins University Press, 1976），第 46—48 页。

51. On studying science as it happens in action rather than its outcome, see Latour, *Pasteurization of France* and *Pandora's Hope: Essays on the Reality of Science Studies* (Cambridge, Mass.: Harvard University Press, 1999). Rather than Latour’s term “factish” (*Pandora's Hope,* 274), I prefer “artifactual,” borrowed from Derrida, *Specters of Marx,* 170. Latour’s brief dismissal of Derrida’s work in *We Have Never Been Modern* is disappointing. On the nature/culture distinction see Jacques Derrida, *Of Grammatology,* trans. Gayatri Chakravorty Spivak (Baltimore, Md.: Johns Hopkins University Press, 1976), 46–48.

↑52. Willcocks 与 Craig，《埃及灌溉》，第 2 卷，第 718—758 页。引文出自第 738 页。

52. Willcocks and Craig, *Egyptian Irrigation,* 2:718–58. The quotation is from 738.

↑53. Robert Mabro，《埃及经济，1952—1972》（The Egyptian Economy, 1952–1972；Oxford: Clarendon Press, 1974），第 86 页。

53. Robert Mabro, *The Egyptian Economy, 1952–1972* (Oxford: Clarendon Press, 1974), 86.

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↑54. 同书，第 89 页。关于计算大坝影响日益困难的问题，见世界水坝委员会，《水坝与发展》。

54. Ibid., 89. On the increasing difficulty in calculating the impact of dams, see World Commission on Dams, *Dams and Development.*

↑55. 关于科学的殖民生产，见 Gyan Prakash，《另一种理性：科学与现代印度的想象》（Another Reason:

55. On the colonial production of science, see Gyan Prakash, *Another Reason: Science and the Imagination*

Science and the Imagination of Modern India; Princeton, N.J.: Princeton University Press, 1999）。

†56. R. B. Buckley，《灌溉袖珍手册》（Irrigation Pocket-Book; London: Spon, 1913），转引自 Willcocks 与 Craig，《埃及灌溉》，第 152—153 页。

†57. 阿兰达蒂 • 洛伊在《更大的共同利益》（“The Greater Common Good”）中讨论了大坝剥夺地方知识的力量，该文收入《生活的代价》（The Cost of Living），可见 http://www.narmada.org 。

†58. A. B. De Guerville，《新埃及》（New Egypt; London: William Heinemann, 1905），第 224—227 页。

†59. Tignor，《两次大战之间埃及的民族主义、经济规划与发展项目》。另见 John Waterbury，《尼罗河谷的水政治》（Hydropolitics of the Nile Valley; Syracuse, N.Y.: Syracuse University Press, 1979）。

†60. 威尔科克斯已近七十岁，获免入狱，并被允许返回他出生、成长和学习工程的孟加拉，担任政府灌溉顾问。Goldschmidt，《传记辞典》，第 225 页。

†61. Hussein Kamel Selim，《埃及农业发展二十年》（Twenty Years of Agricultural Development in Egypt; Cairo: Ministry of Finance, Egypt, 1940），第 66—67 页。作者是福阿德一世大学商学院副院长。

†62. Al-Ahram，1942 年 12 月 1 日，转引自 Gallagher，《埃及的其他战争》，第 26 页。

†63. Mirrit Butrus Ghali，《明日的政策：政治、经济与社会纲领》（Siyasat al-ghad: barnamij siyasi wa-iqtisadi wa-ijtimaʿi; Cairo: Matbaʿat al-Risala, 1938），第 145 页；英译本 The Policy of Tomorrow，Ismail R. el-Faruqi 译（Washington, D.C.: American Council of Learned Societies, 1953），第 102 页。20 世纪 30 年代末出版的一系列讨论埃及乡村问题的著作，没有讨论土地改革。例如 Henry Habib Ayrout，《农民的风俗与习惯》（Moeurs et coutumes des fellahs; Paris: Payot, 1938），阿拉伯文版题为《农民》（al-Fallahun），Muhammad Ghallab 译（Cairo: Matbaʿat al-Kawthar, 1943）；ʿAʾisha ʿAbd al-Rahman（笔名 Bint al-Shatiʾ），《埃及乡村》（al-Rif al-misri; Cairo: Maktabat al-Wafd, 1936）；Hafiz ʿAfifi，《政治边缘：我们的一些民族问题》（ʿAla hamish al-siyasa: baʿd masaʾilna al-qawmiyya; Cairo: Dar al-Kutub al-Misriyya, 1938）。另见 Gabriel Baer，《现代埃及的土地所有制，1800—1950》（Landownership in Modern Egypt, 1800–1950; Oxford: Oxford University Press, 1962），第 201—222 页，以及本书下文第 4 章。

†64. Warriner，《中东的土地与贫困》，第 49 页。

†65. Gallagher，《埃及的其他战争》，第 40—55 页。

†66. Ihsan ʿAbd al-Qaddus，Ruz al-Yusuf，1944 年 3 月 2 日；相关讨论见 Gallagher，《埃及的其他战争》，第 49—50 页。

†67. Baer，《现代埃及的土地所有制》，第 210—215 页。1944 年的法律草案经参议院——立法机关上院——社会事务委员会修改，将拟议上限提高到

of Modern India (Princeton, N.J.: Princeton University Press, 1999).

56. R. B. Buckley, *Irrigation Pocket-Book* (London: Spon, 1913), quoted in Willcocks and Craig, *Egyptian Irrigation,* 152–53.

57. Arundhati Roy discusses the power of dams to take away local knowledge in “The Greater Common Good,” in *The Cost of Living,* available at http://www.narmada.org.

58. A. B. De Guerville, *New Egypt* (London: William Heinemann, 1905), 224–27.

59. Tignor, “Nationalism, Economic Planning, and Development Projects.” See also John Waterbury, *Hydropolitics of the Nile Valley* (Syracuse, N.Y.: Syracuse University Press, 1979).

60. Approaching seventy, Willcocks was spared imprisonment and allowed to return to Bengal, where he had been born and raised and had studied engineering, to work as a government advisor on irrigation. Goldschmidt, *Biographical Dictionary,* 225.

61. Hussein Kamel Selim, *Twenty Years of Agricultural Development in Egypt* (Cairo: Ministry of Finance, Egypt, 1940), 66–67. The author was vice-dean, Faculty of Commerce, Fuʾad I University.

62. *Al-Ahram,* Dec. 1, 1942, cited in Gallagher, *Egypt's Other Wars,* 26.

63. Mirrit Butrus Ghali, *Siyasat al-ghad: barnamij siyasi wa-iqtisadi wa-ijtimaʿi* (Cairo: Matbaʿat al-Risala, 1938), 145, English translation *The Policy of Tomorrow,* trans. Ismail R. el-Faruqi (Washington, D.C.: American Council of Learned Societies, 1953), 102. There is no discussion of land reform in a series of books published in the late 1930s addressing the problems of rural Egypt, such as Henry Habib Ayrout, *Moeurs et coutumes des fellahs* (Paris: Payot, 1938), published in Arabic as *al-Fallahun,* trans. Muhammad Ghallab (Cairo: Matbaʿat al-Kawthar, 1943); ʿAʾisha ʿAbd al-Rahman (pseudonym Bint al-Shatiʾ), *al-Rif al-misri* (Cairo: Maktabat al-Wafd, 1936); Hafiz ʿAfifi, *ʿAla hamish al-siyasa: baʿd masaʾilna al-qawmiyya* (Cairo: Dar al-Kutub al-Misriyya, 1938). See also Gabriel Baer, *Landownership in Modern Egypt, 1800–1950* (Oxford: Oxford University Press, 1962), 201–22, and chapter 4, below.

64. Warriner, *Land and Poverty in the Middle East,* 49.

65. Gallagher, *Egypt's Other Wars,* 40–55.

66. Ihsan ʿAbd al-Qaddus, *Ruz al-Yusuf,* Mar. 2, 1944, discussed in Gallagher, *Egypt's Other Wars,* 49–50.

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67. Baer, *Landownership in Modern Egypt,* 210–15. The 1944 draft law was amended by the Social Affairs Committee of the Senate (the upper house of the legisla-

一百费丹；这正是 Mirrit Butrus Ghali 那份影响甚大的土地改革蓝图《农业改革》（al-Islah al-ziraʿi；Cairo: Jamaʿat al-Nahda al-Qawmiyya, 1945）所倡导的上限。他认为，现有超过一百费丹的庄园，会在两三代之内因继承而分割。本章所用 acre 一词，指埃及亩，即费丹（feddan）。

ture) to raise the proposed ceiling to one hundred acres, the limit advocated in Mirrit Butrus Ghali's influential blueprint for land reform *al-Islah al-ziraʿi* (Cairo: Jamaʿat al-Nahda al-Qawmiyya, 1945). He argued that existing estates larger than one hundred acres would be broken up by inheritance within two or three generations. The term acre in this chapter refers to an Egyptian acre, or feddan.

†68. ʿAsim al-Disuqi，《大农业地主》，第 306—312 页。在议会之外，社会主义左翼认为，这些土地改革提案不过是资产阶级限制土地垄断的努力；土地垄断正在推高物价，进而推高工资，并降低消费品消费。不过，他们也认为，提案有一个有益作用：让人民看到，私有财产并非神圣不可侵犯。Ahmad Sadiq Saʿd 发表于 al-Fajr al-jadid 的文章，1945 年 7 月 16 日和 8 月 16 日；相关讨论见 Rifʿat al-Saʿid，《埃及左翼报刊，1925—1948》（al-Sihafa al-yasariyya fi misr, 1925–1948；Cairo: Maktabat Madbuli, 1977），第 130—131 页。

68. ʿAsim al-Disuqi, *Kibar mullak al-aradi al-ziraʿiyya,* 306–12. Outside parliament, the socialist left saw these land reform proposals as simply an effort by the bourgeoisie to limit the monopolization of land, which was forcing up prices, and therefore wages, and reducing the consumption of consumer goods. Nevertheless, it was argued, the proposals had the useful effect of revealing to the people that private property was not something sacred. Ahmad Sadiq Saʿd, writing in *al-Fajr al-jadid,* July 16, 1945, and Aug. 16, 1945, discussed in Rifʿat al-Saʿid, *al-Sihafa al-yasariyya fi misr, 1925–1948* (Cairo: Maktabat Madbuli, 1977), 130–31. p. 311

†69. 1948 年 3 月约 575 户获得土地，1951 年 2 月为 597 户，1951 年 5 月为 240 户。美国国家档案馆，第 59 档案群，国务院中央档案，埃及 1950—1954 年，874.16/2-1951，美国驻开罗使馆致国务院，1951 年 2 月 19 日，《向农民授予土地》；以及 874.16/5-1851，美国驻开罗使馆致国务院，1951 年 5 月 18 日，《在卡夫尔萨阿德分配土地》。缩微版（University Publications of America, 1985），下文简称 USRG 59。

69. Some 575 families received land in March 1948, 597 in February 1951, and 240 in May 1951. United States National Archives, Record Group 59, Department of State, Central Files, Egypt 1950–54, 874.16/2–1951, U.S. Embassy Cairo to State, Feb. 19, 1951, Presentation of Land to Peasants, and 874.16/5–1851, U.S. Embassy Cairo to State, May 18, 1951, Distribution of Land at Kafr Saad. Microform (University Publications of America, 1985), hereafter referred to as USRG 59.

†70. Vitalis，《当资本家发生冲突》，第 179—180 页。关于位于 Kesma 的香水工厂，见 USRG 59，1950–54，874.395/10–352，美国驻开罗使馆致国务院，1952 年 10 月 3 日。关于造纸厂，见 USRG 59，1950–54，874.392/4–2252，国务院，谈话备忘录，《在埃及兴建纸制品工厂》；及 USRG 59，1950–54，874.392/8–454，1954 年 8 月 4 日。该造纸厂拟与 W. R. Grace & Co. 合作，在纳贾哈马迪建设，采用格雷斯公司秘鲁子公司开发的新工艺，用甘蔗渣——甘蔗榨汁后的残渣——生产牛皮纸和新闻纸。此前，格雷斯公司已为甘蔗渣找到另一种用途，即在炸药制造中用它稀释硝酸铵。

70. Vitalis, *When Capitalists Collide,* 179–80. On the perfume venture, a plant at Kesma, see USRG 59, 1950–54, 874.395/10–352, U.S. Embassy Cairo to State, Oct. 3, 1952. On the paper factory, see USRG 59, 1950–54, 874.392/4–2252, Department of State, Memorandum of a conversation, Construction of paper products plant in Egypt; and USRG 59, 1950–54, 874.392/8–454, Aug. 4, 1954. The paper plant was to be built at Najʿ Hamadi in collaboration with W. R. Grace & Co. using a new procedure developed by a Grace subsidiary in Peru to make kraft paper and newsprint from sugarcane bagasse, the refuse from crushed sugarcane. Grace had previously found another use for the bagasse, using it to dilute the ammonium nitrate in the manufacture of explosives.

†71. USRG 59，1950–54，874.3972/11–851，卡弗里致国务院，1951 年 11 月 8 日，附 1951 年 11 月 6 日一等秘书 Randall S. Williams 与苏伊士肥料厂 Clark Davis 会面的备忘录。关于战后的发展政治，见 Arturo Escobar，《遭遇发展：第三世界的形成与解体》（Encountering Development: The Making and Unmaking of the Third World；Princeton, N.J.: Princeton University Press, 1995）；James Ferguson，《反政治机器：莱索托的“发展”、去政治化与官僚权力》（The Anti-Politics Machine: “Development,” Depoliticization, and Bureaucratic Power in Lesotho；Cambridge: Cambridge University Press, 1990）；Akhil Gupta，《后殖民的发展：现代印度形成中的农业》（Postcolonial Developments: Agriculture in the Making of Modern India；Durham, N.C.: Duke University Press, 1998）；以及

71. USRG 59, 1950–54, 874.3972/11–851, Caffery to State, Nov. 8, 1951, enclosing memorandum of meeting between Randall S.Williams, First Secretary, and Clark Davis of Suez Fertilizer Works, Nov. 6, 1951. On the postwar politics of development see Arturo Escobar, *Encountering Development: The Making and Unmaking of the Third World* (Princeton, N.J.: Princeton University Press, 1995); James Ferguson, *The Anti-Politics Machine: "Development," Depoliticization, and Bureaucratic Power in Lesotho* (Cambridge: Cambridge University Press, 1990); Akhil Gupta, *Postcolonial Developments: Agriculture in the Making of Modern India* (Durham, N.C.: Duke University Press, 1998); and three collections of essays: Jonathan Crush, ed., *Power of Development* (London: Routledge, 1995), Frederick p. 311–312

三本文集：Jonathan Crush 编，《发展的权力》（Power of Development；London: Routledge, 1995），Frederick Cooper 与 Randall Packard 编，《国际发展与社会科学：知识的历史与政治论集》（International Development and the Social Sciences: Essays on the History and Politics of Knowledge；Berkeley: University of California Press, 1997），Wolfgang Sachs 编，《发展辞典：知识与权力指南》（The Development Dictionary: A Guide to Knowledge and Power；London: Zed Press, 1992）。

Cooper and Randall Packard, eds., *International Development and the Social Sciences: Essays on the History and Politics of Knowledge* (Berkeley: University of California Press, 1997), and Wolfgang Sachs, ed., *The Development Dictionary: A Guide to Knowledge and Power* (London: Zed Press, 1992).

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†72. USRG 59，1950–54，874.3971/1–951，Salmawy 公司总裁 Mohamed Salmawy 致国务院埃及事务处的信；USRG 59，1950–54，874.00-TA/8–2552，美国驻开罗使馆致国务院，《技术合作署埃及项目 1953 及 1954 财政年度的计划与预算》，第 3 页；USRG 59，1950–54，874.00-TA/9–652，美国驻开罗使馆致国务院，《1953 财政年度项目需求的优先次序》。

72. USRG 59, 1950–54, 874.3971/1–951, Letter from Mohamed Salmawy, President, Salmawy Co., to Egyptian Desk, Department of State; USRG 59, 1950–54, 874.00-TA/8–2552, American Embassy Cairo to Department of State, Program and Budget, TCA-Egypt, for Fiscal Years 1953 and 1954, 3; USRG 59, 1950–54, 874.00-TA/9–652, American Embassy Cairo to Department of State, Priority of Project Needs for FY 1953.

†73. USRG 59，1950–54，874.00-TA/11–2351，美国驻开罗使馆致开罗技术合作署，《埃及技术援助计划：1952 和 1953 财政年度预算》，第 3—4 页。

73. USRG 59, 1950–54, 874.00-TA/11–2351, American Embassy Cairo to Technical Cooperation Administration, Cairo, A Plan for Technical Assistance for Egypt. Budgets for FY 1952 and FY 1953, 3–4.

†74. USRG 59，1950–54，874.00-TA/10–852，美国驻开罗使馆致国务院，《埃及的第四点计划》。

74. USRG 59, 1950–54, 874.00-TA/10–852, American Embassy Cairo to Department of State, The Point IV Program in Egypt.

†75. USRG 59，1950–54，874.00-TA/5–1253，美国驻开罗使馆致国务院，《1953 年 4 月 1 日至 30 日第四点计划活动报告》，第 5—6、18 页；USRG 59，1950–54，874.00-TA/7–2053，美国驻开罗使馆致国务院，《亚瑟·D. 利特尔公司目前在埃及遇到的困难》。

75. USRG 59, 1950–54, 874.00-TA/5–1253, American Embassy Cairo to Department of State, Report of Point IV Activities from April 1 through April 30, 1953, 5–6, 18; USRG 59, 1950–54, 874.00-TA/7–2053, American Embassy Cairo to Department of State, Current Difficulties in Egypt of Arthur D. Little, Inc.

†76. USRG 59，1950–54，874.3972/11–851，卡弗里致国务院，1951 年 11 月 8 日，附 1951 年 11 月 6 日一等秘书 Randall S. Williams 与苏伊士肥料厂 Clark Davis 会面的备忘录。阿布德此前已前往华盛顿，寻求重新安排肥料厂美国贷款的还款计划，把财务困难归咎于 1949 年埃及镑贬值。USRG 59，1950–54，874.39/1–1950，华盛顿国务院，《与阿布德帕夏及化学建筑公司波特将军会面的备忘录》，1950 年 1 月 19 日。

76. USRG 59, 1950–54, 874.3972/11–851, Caffery to State, Nov. 8, 1951, enclosing memorandum of meeting between Randall S.Williams, First Secretary, and Clark Davis of Suez Fertilizer Works, Nov. 6, 1951. ʿAbbud had already traveled to Washington to seek a rescheduling of his U.S. loan repayments for the fertilizer plant, blaming his financial difficulties on the 1949 devaluation of the Egyptian pound. USRG 59, 1950–54, 874.39/1–1950, Department of State, Washington, memorandum of meeting with Abboud Pasha and General Porter of Chemical Construction Company, Jan. 19, 1950.

†77. James Scott，《国家的视角：那些试图改善人类状况的计划何以失败》（Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed；New Haven, Conn.: Yale University Press, 1998），考察了许多我在本章及后续章节所探讨的同类技术政治。虽然我赞赏斯科特的许多论点，但我的分析在重要方面与他不同。尤其是，斯科特关注现代国家如何误用科学的力量，并把这种误用同恰当的科学区分开来；我关注的则是何种社会与政治实践，同时生产了科学的力量和现代国家的力量。

77. James Scott, *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed* (New Haven, Conn.: Yale University Press, 1998), examines many of the same kinds of techno-politics that I explore in this and subsequent chapters. While I admire many of Scott's arguments, my own analysis differs from his in important ways. In particular, whereas Scott is concerned with the way modern states have misused the powers of science, and distinguishes this misuse from proper science, I am concerned with the kinds of social and political practice that produce simultaneously the powers of science and the powers of modern states.

†78. Ramadan，《埃及的阶级斗争》，第 213 页；Salib Sami，《萨利卜·萨米帕夏回忆录，1891—1952》（Dhikriyyat Salib Basha Sami, 1891–1952），Sami Abu al-Nur 编（Cairo: Maktabat Madbuli, 1999）。

78. Ramadan, *Siraʿ al-tabaqat fi misr,* 213; Salib Sami, *Dhikriyyat Salib Basha Sami, 1891–1952,* ed. Sami Abu al-Nur (Cairo: Maktabat Madbuli, 1999).

↑79. 我对 1952 年事件的叙述，主要依据 Joel Gordon，《纳赛尔的受祝福运动：埃及自由军官与七月革命》（Nasser’s Blessed Movement: Egypt’s Free Officers and the July Revolution；Oxford: Oxford University Press, 1992），第 14—57 页。ʿAbd al-ʿAzim Ramadan 也指出，军官们夺权之前没有考虑过土地改革问题，见《纳赛尔与三月危机》（ʿAbd al-Nasir wa-azmat maris；Cairo: Ruz al-Yusuf, 1976），第 14—18 页。

79. My account of the events of 1952 is based largely on Joel Gordon, *Nasser’s Blessed Movement: Egypt’s Free Officers and the July Revolution* (Oxford: Oxford University Press, 1992), 14–57. ʿAbd al-ʿAzim Ramadan also says that the army officers had not considered the question of land reform before seizing power: *ʿAbd al-Nasir wa-azmat maris* (Cairo: Ruz al-Yusuf, 1976), 14–18.

↑80. 1952 年土地改革法规定，“任何人不得拥有超过 200 费丹的土地”，但允许地主通过向两个子女各赠予最多 50 费丹，再保留 100 费丹。超过 300 费丹上限的土地，须由政府在五年内征收，再以每份二至五费丹卖给小农。地价定为土地税的七十倍，分三十年支付，这还不到市场价值的一半。五年的实施期限，使地主有时间通过私人出售或转让给其他亲属来规避法律。Doreen Warriner，《中东的土地改革与发展：埃及、叙利亚和伊拉克研究》（Land Reform and Development in the Middle East: A Study of Egypt, Syria, and Iraq），第 2 版（London: Oxford University Press, 1962），第 31—35 页。另见本书下文第 5、7 章。

80.The 1952 agrarian reform law laid down that “no person may own more than 200 acres of land,” but allowed owners to keep another 100 acres by giving up to 50 acres to each of two children. Land over the 300-acre limit was to be requisitioned by the government over five years and sold to small farmers in holdings of two to five acres. The price of the land, to be paid over thirty years, was fixed at seventy times the land tax, which represented less than half its market value. The five-year implementation period gave owners time to evade the law through private sales or transfers to other relatives. Doreen Warriner, *Land Reform and Development in the Middle East: A Study of Egypt, Syria, and Iraq,* 2nd ed. (London: Oxford University Press, 1962), 31–35. See also chapters 5 and 7 below.

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↑81. Vitalis，《当资本家发生冲突》，第 172—214 页。

81. Vitalis, *When Capitalists Collide,* 172–214.

↑82. 工厂之所以不经济，是因为美国政府担心本国科学家和工程师无法掌握德国新近开发的高温高压哈伯—博施固氮技术，便同美国氰胺公司签约，采用耗电量大得多的氰胺法，在马斯尔肖尔斯建造肥料厂。

82. The plant was uneconomical because the U.S. government, afraid that American scientists and engineers would not be able to handle the high-pressure and high-temperature Haber-Bosch nitrogen-fixing technology recently developed in Germany, contracted with the American Cyanamid Corporation to build the fertilizer plant at Muscle Shoals using the cyanamid process, which consumed much greater amounts of electricity.

↑83. Thomas P. Hughes，《权力网络：西方社会的电气化，1880—1930》（Networks of Power: Electrification in Western Society, 1880–1930；Baltimore, Md.: Johns Hopkins University Press, 1983），第 286—287、293—295 页。另见 Preston J. Hubbard，《田纳西河流域管理局的起源：马斯尔肖尔斯争论，1920—1932》（Origins of the TVA: The Muscle Shoals Controversy, 1920–1932；New York: Norton, 1961）。

83. Thomas P. Hughes, *Networks of Power: Electrification in Western Society, 1880–1930* (Baltimore, Md.: Johns Hopkins University Press, 1983), 286–87, 293–95. See also Preston J. Hubbard, *Origins of the TVA: The Muscle Shoals Controversy, 1920–1932* (New York: Norton, 1961).

↑84. Feliks Bochenski 与 William Diamond，《中东的“田纳西河流域管理局”》（“TVA’s in the Middle East”），Middle East Journal 4，第 1 期（1950）：第 52—82 页，所引见第 55 页。

84. Feliks Bochenski and William Diamond, “TVA’s in the Middle East,” *Middle East Journal* 4, no. 1 (1950): 52–82, at 55.

↑85. 至少 113000 人因高坝而迁离原居地，海岸线每年退缩五至八米，土地退化与盐碱化持续。到 1989 年，灌溉面积并未超过建坝之前。World Rivers Review，2000 年 2 月，第 5 页，可见 http://www.irn.org 。公共工程部 1920 年发表的尼罗河开发蓝图曾主张，三角洲沿海湖泊的二十万费丹不应开垦种植，因为其渔获更有价值。埃及政府公共工程部，《尼罗河控制》（Nile Control；Cairo, 1920），转引自 Tignor，《民族主义、经济规划与发展项目》，第 191 页。另见 Tom Little，《阿斯旺高坝》（High Dam at Aswan；New York: John Day, 1965）。

85. At least 113,000 people were displaced by the High Dam, five to eight meters of coastline were lost every year, and degradation of the land and salination continued. By 1989 the irrigated area was no greater than before the dam. *World Rivers Review,* Feb. 2000, 5, available at http://www.irn.org. The blueprint for Nile development published by the Ministry of Public Works in 1920 had argued that 200,000 acres of coastal lakes in the delta should not be brought under cultivation because their yield in fish was more valuable. Government of Egypt, Ministry of Public Works, *Nile Control* (Cairo,

1920), cited Tignor, "Nationalism, Economic Planning, and Development Projects," 191. See also Tom Little, *High Dam at Aswan* (New York: John Day, 1965).

↑86. Marx，《资本论》，第 1 卷，第 198 页。马克思在随后的段落中明确指出，这只是对人类意向性的初步说明。他接着探讨，现代性的机械化世界如何把意识从个体那里移走，并使其在个体面前显现为一种居于现代技术机器中的智识。相关讨论见 Moishe Postone，《时间、劳动与社会统治》（Time, Labor, and Social Domination；Cambridge: Cambridge University Press, 1993）。

86. Marx, *Capital,* 1:198. Marx makes it clear in passages that follow that this is only a preliminary account of human intentionality. He goes on to explore how the mechanized world of modernity removes consciousness from individuals and makes it appear to them as an intelligence residing in the machinery of modern technology. For a discussion, see Moishe Postone, *Time, Labor, and Social Domination* (Cambridge: Cambridge University Press, 1993).

↑87. Timothy Mitchell，《殖民埃及》（Colonising Egypt；Berkeley: University of California Press, 1991），及《现代性的舞台》（"The Stage of Modernity"），载 Timothy Mitchell 编，《现代性的问题》（Questions of Modernity；Minneapolis: University of Minnesota Press, 2000），第 1—34 页。

87. Timothy Mitchell, *Colonising Egypt* (Berkeley: University of California Press, 1991), and "The Stage of Modernity," in *Questions of Modernity,* ed. Timothy Mitchell (Minneapolis: University of Minnesota Press, 2000), 1–34.

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↑88. 1918 年，J. R. Geigy S.A. 与瑞士另外两家化学公司汽巴和山德士联合，组成巴塞尔利益共同体（Interessengemeinschaft Basel，即巴塞尔联合集团），与法本竞争。1929—1939 年，这两个卡特尔又与英国、法国化学公司联合，组成四方卡特尔（Quadrapartite Cartel），一直延续到第二次世界大战。

88. In 1918 J. R. Geigy S.A. joined with Switzerland's two other chemical companies, Ciba and Sandoz, to form the Interessengemeinschaft Basel (Basel Syndicate) to compete with I. G. Farben. In 1929–39 the two cartels joined with British and French chemical firms to form the Quadrapartite Cartel, which lasted until World War II.

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↑89. G. Fischer 教授，《1948 年诺贝尔生理学或医学奖颁奖词》（"Presentation Speech, Nobel Prize in Physiology or Medicine, 1948"），可见 http://www.nobel.se/laureates/medicine-1948-press.html 。

89. Professor G. Fischer,"Presentation Speech, Nobel Prize in Physiology or Medicine, 1948," available at http://www.nobel.se/laureates/medicine-1948-press.html.

〔校注：原注中的网址按底本保留，跨行空白已合并。旧网址的保留不表示目前仍可访问；全部外部核验的范围及链接见 work/external_checks_ch01.md。〕

↑90. Harrison，《蚊子、疟疾与人》，第 218—227 页。

90. Harrison, *Mosquitoes, Malaria and Man,* 218–27.

↑91. Gallagher，《埃及的其他战争》，第 198 页注 64。

91. Gallagher, *Egypt's Other Wars,* 198 n. 64.

↑92. 见 Fischer，《颁奖词》。

92. See Fischer, "Presentation Speech."

↑93. Fischer，《颁奖词》。皮特—格雷迪试验室以其发明者命名，相关描述可见 http://www.clorox.com/science/labs/insect_lab.html 。

93. Fischer, "Presentation Speech." A description of the Peet-Grady chamber, named after its inventors, is available at http://www.clorox.com/science/labs/insect_lab.html.

↑94. 关于以这种方式分析科学，见 Bruno Latour，《法国的巴斯德化》。

94. For the analysis of science in these terms, see Bruno Latour, *The Pasteurization of France.*

↑95. Harrison，《蚊子、疟疾与人》，第 219、222、223 页。

95. Harrison, *Mosquitoes, Malaria and Man,* 219, 222, 223.

↑96. 同书，第 218—227 页。

96. Ibid., 218–27.

↑97. 同上。

97. Ibid.

↑98. Titus Bradley，《疟疾：历史与分布》（"Malaria: History and Distribution"），可见 http://www.micro.msb.le.ac.uk/224/Bradley/history.html 。

98. Titus Bradley, "Malaria: History and Distribution," available at http://www.micro.msb.le.ac.uk/224/Bradley/history.html.

↑99. 选择 2007 年，是因为世界上当时仍在生产 DDT 的最大生产国之一墨西哥，已把这一年定为停止制造和使用该杀虫剂的期限。

99. The year 2007 was chosen because it was the date that Mexico, one of the world's largest remaining DDT producers, had set to stop manufacturing and using the pesticide.

↑100. 世界自然基金会加拿大分会与美国分会，《解决 DDT 困境：保护生物多样性与人类健康》（Resolving the DDT Dilemma: Protecting Biodiversity and Human Health；Toronto and Washington, D.C.: WWF Canada and WWF U.S., 1998）。20 世纪 90 年代，国际卫生机构开始认真推广蚊虫管理中 DDT 的替代方案。值得注意的是，所用方法——带有人体气味和杀虫剂的诱饵布、浸渍合成除虫菊药剂的蚊帐，以及取代集中化军事式运动的地方性社区参与项目——都试图利用一个事实：蚊子是一种依赖人类社区的社会性昆虫。

100. World Wildlife Fund Canada and World Wildlife Fund U.S., *Resolving the DDT Dilemma: Protecting Biodiversity and Human Health* (Toronto and Washington, D.C.: WWF Canada and WWF U.S., 1998). In the 1990s, international healthcare agencies began to seriously promote alternatives to DDT for the management of mosquitoes. Interestingly, the methods employed—cloths baited with human odor and insecticide, bed nets impregnated with synthetic pyrethrum, and locally based, community-participant projects in place of centralized, military-style campaigns—tried to take advantage of the fact that the mosquito is a social insect, dependent upon human communities.

↑101. USRG 59，1950–54，874.3972/10–651，开罗致国务院，1951 年 8 月 28 日，《化学肥料——埃及》。

101. USRG 59, 1950–54, 874.3972/10–651, Cairo to State, Aug. 28, 1951, Chemical Fertilizers—Egypt.

↑102. 1954 年，英国化学巨头 ICI 的子公司帝国化学工业（埃及）有限公司，以及 E. I. du Pont de Nemours & Co. 旗下 Gresselli Chemicals 部门的当地代理商 Salmawy & Co.，进口 DDT 浓缩物及其他化学品，每年生产约七千公吨以 DDT 为基础的棉田粉剂和其他杀虫剂。Salmawy 公司的办公室设在开罗阿布德的伊莫比利亚大楼，它也进口了那些直升机。USRG 59，1950–54，874.3971/1–951，Salmawy 公司总裁 Mohamed Salmawy 致国务院埃及事务处的信；USRG 59，1950–54，874.397/11–1754，美国驻开罗使馆商务助理 Nestor Lardicos 致国务院，1954 年 11 月 17 日，《埃及害虫防治产品调查》。

102. In 1954 Imperial Chemical Industries (Egypt) Ltd., a subsidiary of the British chemical giant ICI, and Salmawy & Co., the local agent for the Gresselli Chemicals division of E. I. du Pont de Nemours & Co., were importing DDT concentrate and other chemicals to produce about seven thousand metric tons a year of DDT-based cotton dust and other pesticides. Salmawy, whose offices were in ʿAbbud's Immobilia building in Cairo, also imported the helicopters. USRG 59, 1950–54, 874.3971/1–951, Letter from Mohamed Salmawy, President, Salmawy Co., to Egyptian Desk, Dept of State; USRG 59, 1950–54, 874.397/11–1754, Nestor Lardicos, Commercial Assistant, U.S. Embassy Cairo to State, Nov. 17, 1954, Survey of Pest Control Products, Egypt.

↑103. Vitalis，《当资本家发生冲突》，第 178 页。

103. Vitalis, *When Capitalists Collide,* 178.

↑104. USRG 59，1950–54，874.3971/5–2851，美国驻开罗使馆三等秘书 Myles Standish III 致国务院，《DDT 工厂建设》；USRG 59，1950–54，874.395/10–352，美国驻开罗使馆致国务院，1952 年 10 月 3 日；USRG 59，1950–54，874.397/6–1351，美国驻开罗使馆致国务院，1951 年 6 月 13 日。

104. USRG 59, 1950–54, 874.3971/5–2851, Myles Standish III, 3rd Secretary, U.S. Embassy Cairo to State, Construction of DDT Plant; USRG 59, 1950–54, 874.395/10–352, U.S. Embassy Cairo to State, Oct. 3, 1952; USRG 59, 1950–54, 874.397/6–1351, U.S. Embassy Cairo to State, June 13, 1951.

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↑105. 1954 年的报告称，工厂正在建设，计划 1956 年投产。世卫组织提供四名技术人员，联合国儿童基金会提供 25 万美元购买机器。USRG 59，1950–54，874.397/11–1754，美国驻开罗使馆商务助理 Nestor Lardicos 致国务院，1954 年 11 月 17 日，《埃及害虫防治产品调查》；USRG 59，1950–54，874.3971/8–1152，美国驻开罗大使 Jefferson Caffery 致国务院，《世卫组织、联合国儿童基金会与埃及政府关于兴建 DDT 工厂的协议》。

105. In 1954 it was reported that the plant was under construction and scheduled to begin production in 1956. The WHO provided four technicians and UNICEF provided $250,000 to purchase machinery. USRG 59, 1950–54, 874.397/11–1754, Nestor Lardicos, Commercial Assistant, U.S. Embassy Cairo to State, Nov. 17, 1954, Survey of Pest Control Products, Egypt; USRG 59, 1950–54, 874.3971/8–1152, Jefferson Caffery, Ambassador, U.S. Embassy Cairo to State, Agreement between WHO and UNICEF and the GOE to build DDT plant.

↑106. USRG 59，1950–54，874.397/11–1754，美国驻开罗使馆商务助理 Nestor Lardicos 致国务院，1954 年 11 月 17 日，《埃及害虫防治产品调查》。

106. USRG 59, 1950–54, 874.397/11–1754, Nestor Lardicos, Commercial Assistant, U.S. Embassy Cairo to State, Nov. 17, 1954, Survey of Pest Control Products, Egypt.